

# 2110211 Introduction to Data Structure

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## Lecture 1

### C++ Intro

#### C++ Workflow

Source Code (\*.cpp) → Compiler (gcc) ผ่าน Code::Blocks → Executable File (\*.exe)

#### Hello World

```
#include <iostream>

int main() {
    std::cout << "Hello, CP!" << std::endl;
    return 0;
}
```

### C++ from Java & Python

#### ตัวแปร

- ต้องประกาศ
- ต้องระบุประเภท
- ตัวแปร โดยปกติเป็น “กล่อง” ที่เก็บข้อมูล ไม่ใช่ของที่ “อ้างอิง” ข้อมูล

```
#include <iostream>

using namespace std;

int main() {
    int x;
    x = 10;
    bool y = true;
    cout << y << endl;
    cout << (x+20) << endl;
    // x = y; // ไม่ได้
}
```

### C++, if, for, while, function

#### If

- ต้องมีวงเล็บ ตรงเงื่อนไข
- ใช้ {} เป็นตัวระบุ suite (ใน c++ เรียก block)
- ไม่มี elif ต้องใช้ else if

```
int main() {
    int age;
    cout << "Please enter your age:";
    cin >> age;
    if ( age < 5 ) {
        cout << "You are a kid!\n";
    } else if ( age < 100 ) {
        cout << "You are not old!\n";
    } else {
        cout << "You live long!\n";
    }
    return 0;
}
```

**For loop**

- เหมือน Java
- ประกอบด้วย 3 ส่วน
  - initial
  - condition
  - iteration

```
#include <iostream>

using namespace std;

int main() {
    for (int i=0; i < 10; i++) { // i = i + 1
        cout << "i = " << i << endl;
    }
    return 0;
}
```

**While loop**

```
int main() {
    int x = 0;
    while ( x < 10 ) {
        cout << "x = " << x << endl;
        x++;
    }
    return 0;
}
```

```
int main() {
    int x = 20;
    do {
        x--;
        cout << x << endl;
    } while ( x > 10 )
}
```

**Function**

- Pass by value vs pass by reference

```
void pass_by_value(int x) {
    cout << "X is " << x << endl;
    x = 30;
}
```

```
void pass_by_reference(int &x) {
    cout << "X is " << x << endl;
    x = 40;
}
```

```
int main() {
    cout << "Pass by Value, direct" << endl;
    pass_by_value(10); // X is 10
    cout << endl;

    int x = 20;
    cout << "Pass by value, variable" << endl;
}
```

```

pass_by_value(x); // X is 20
cout << "outside PbV function x = " << x << endl; // 20
cout << endl;

cout << "Pass by reference" << endl;
pass_by_reference(x); // X is 20
cout << "outside PbR function x = " << x << endl; // 40

// the following line cannot be compiled
// because we need reference
// pass_by_reference(20);
}

```

## Lecture 2

### Array & Vector

#### Array

- Two types: Static and Dynamic
- Raw Data
- Fast random access
- Static = size cannot be changes
- Dynamic = can change size

```

#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
    vector<int> a = {2,3};
    vector<bool> b = {true,false,true};

    a.push_back(10);
    a.push_back(20);

    a.insert( a.begin(), 99);
    a.insert( a.end(), -5);

    sort(a.begin(),a.end());

    cout << "a is ";
    for (size_t i = 0; i < a.size(); i++) {
        cout << a[i] << " ";
    }
    cout << endl;

    cout << "b is ";
    for (size_t i = 0; i < b.size(); i++) {
        cout << b[i] << " ";
    }
    cout << endl;
}

```

## Word Count Problem

```
#include <iostream>
#include <fstream>
#include <string>
#include <vector>
#include <set>
#include <algorithm>
#include "Tokenizer.h"

using namespace std;

void printUniqueWords1(string filename);
void printUniqueWords2(string filename);
void printUniqueWords3(string filename);
void printUniqueWords4(string filename);
void printWord(string filename);

int main() {
    string filename = "test.txt";
    printWord(filename);
    printUniqueWords1(filename);
    printUniqueWords2(filename);
    printUniqueWords3(filename);
    printUniqueWords4(filename);
    return 0;
}

bool search(string words[], int n, string w) {
    for (int i = 0; i < n; i++) {
        if (words[i] == w) return true;
        n++;
    }
    return false;
}

void printWord(string filename) {
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        n++;
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

void printUniqueWords1(string filename) { // wastes too much memory
    string words[100000];
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) words[n++] = token;
    }
}
```

```

}
tokenizer.close();
cout << "A total of " << n << " words" << endl;
}

```

## Dynamic Array

```

void printUniqueWords2(string filename) { // expandable array; wastes time
    int cap = 1;
    string *words;
    words = new string[cap];
    int n = 0;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) {
            if (n == cap) {
                string *a = new string[2*cap];
                for (int i=0; i<n; i++) a[i] = words[i];
                delete[] words;
                words = a;
                cap *= 2;
            }
            words[n++] = token;
        }
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

```

## Word Count by Vector & Set

### Vector

```

void printUniqueWords3(string filename) { // slow find time
    vector<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (words.end() == find(words.begin(), words.end(), token))
            words.push_back(token);
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

```

### Set

```

void printUniqueWords4(string filename) { // quick find existence
    set<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        // if (words.end() == find(words.begin(), words.end(), token)) // slow find
        if (words.end() == word.find(token)) // if can't find then return word.end()
            words.insert(token);
    }
    tokenizer.close();
}

```

```
    cout << "A total of " << n << " words" << endl;
}
```

## Lecture 3

### std::pair

#### Basic

```
#include <iostream>

int main() {
    std::pair<int,float> x;
    std::pair<int,float> y;

    x.first = 10;
    x.second = 20.65;

    std::cout << x.first << " "; // 10
    std::cout << x.second << std::endl; // 20.65

    y = x;

    std::cout << y.first << std::endl; // 10
    std::cout << y.second + 20 << std::cout; // 40.65
}
```

Pair reserves 2 spaces in RAM for first and second elements. When you use `y = x`, it copies values of x to y. You need to build template by specifying data types in a pair.

#### Initialize

```
#include <iostream>

int main() {
    // default constructor
    std::pair<std::string,bool> p; // default of that type: {"", false}
    std::cout << "default [" << p.first << "] [" << p.second << "]" << std::endl;

    // initialize by { }
    std::pair<std::string,bool> p1 = { "somchai", true };

    // create pair without specifying type by "make_pair"
    std::pair<bool,int> p2;
    p2 = std::make_pair(false,10);

    std::pair<bool,int> p3(p2); // must be same type p3 = p2

    // more complex pair
    std::pair< std::pair<float,int>, std::string > p4 = { {20.5, -3}, "abc"};
    std::cout << p4.first.first << " " << p4.first.second << " " << p4.second <<
std::endl; // 20.5 -3 abc
}
```

#### Array inside Pair

```
#include <iostream>
```

```
int main() {
    std::pair<std::string*,std::pair<int, float>> p;

    p.first = new std::string[5];
    p.first[0] = "a";
    p.first[1] = "b";
    p.first[2] = "c";

    p.second = std::make_pair(10,1.5);

    std::cout << p.first[0] << " " << p.second.first << " " << p.second.second;
}
```

## Template

- Template allows “same code, different data types”
- `std::pair` is a “class template”
  - In generic term, pair is defined as `pair<T1, T2>`
- To use `std::pair`, we must supply “Type information” to the template what T1 and T2 should be.

## STD and Namespace

- The “Fullname” of `cout` is `std::cout`
- `std` is a namespace
- We need to use fullname
  - Too lazy? use “using” keywords

```
#include <iostream>

// this tells C++ that when we say cout, we mean std::cout;
using std::cout;

int main() {
    // we still need to use std::endl
    cout << "Yes" << std::endl;
}

#include <iostream>

// this tells C++ that when we say something that is does not understand, C++ should
try to use std as its namespace

// this is VERY BAD PRACTICE in real world.
// 10/10 not recommend
// ... but it's ok for this class
using namespace std;

int main() {
    cout << "Yes" << endl;
}
```

## std::vector

A linear storage of a single data type

### Basic

```
#include <iostream>
#include <vector>
```

```
using namespace std;

int main() {
    vector<int> v1;
    cout << "Size of v1 is " << v1.size() << endl; // 0

    vector<int> v2 = {2,3,4};

    cout << v2[1] << endl; // 3
    v1 = v2;
    v1[0] = 20;

    cout << v1[0] << ", " << v1[1] << v1[2] << endl; // 20, 3, 4

    v1.push_back(99);
    cout << v1.size() << endl;
}
```

- Vector starts with empty element
- Use size() to get the number of element
- Use empty() to check if a vector has any element

### Initialize

- With specific size
- With specific size and starting value

```
#include <iostream>
#include <vector>

using namespace std;

void print_vector(vector<float> v) {
    for (size_t i = 0; i < v.size(); i++) {
        cout << v[i] << " ";
    }
    cout << endl;
}

int main() {
    vector<float> v1(10); // size = 10
    print_vector(v1);

    vector<float> v2(5,3.55); // size = 5, default = 3.55
    print_vector(v2);

    vector<float> v3(v2);
    print_vector(v3);
}
```

## Vector Access & Overflow Problem

### Access

```
#include <iostream>
#include <vector>

using namespace std;
```

```

int main() {
    vector<float> v1(2);
    vector<float> v2(2);

    cout << "-- v1 --" << endl;
    for (int i = 0; i < 7; i++) {
        v1[i] = i; // this automatically assigns value
        cout << i << ": " << v1[i] << endl; // print nonsense if out of range
    }
    cout << "-- v2 --" << endl;
    for (int i = 0; i < 7; i++) {
        cout << i << ": " << v2[i] << endl; // print "4 5 6" due to buffer overflow (take
space of v1)
    }
    cout << "using at" << endl;
    v2.at(1) = 99;
    // this will cause exception 'std::out_of_range'
    for (int i = 0; i < 7; i++) {
        cout << ": " << v2.at(i) << endl;
    }
}

```

- Operator [] won't check for 'out-of-range'
  - Reading, writing beyond size is undefined
  - Might crash
  - Grader will give 'x'
- at () will check bound
  - But slower

You should be careful of using spaces, it can access illegal memories and cause weird behaviors.

## Modifying Vector

### Resizing

- Resize change the size
  - Enlarge will fill with default

```

#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ":";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v(3,10);
    print(v); // 10,10,10,
    v.resize(6);
    print(v); // 10,10,10,0,0,0,
    v.resize(1);
    print(v); // 10,
    v.resize(7);
}

```

```
    print(v); // 10,0,0,0,0,0,0
}
```

### Modify

- pop\_back
  - Erase last element
- insert(position,value)
- erase(position)
- Careful!
  - Both insert and erase position must be valid
  - If not valid, it can crash

```
#include <iostream>
#include <vector>
```

```
using namespace std;
```

```
void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}
```

```
int main() {
    vector<int> v(3,8); // v.begin() is called iterator
    v.insert( v.begin(), 1); // 1, 8, 8, 8
    v.insert( v.begin()+3, 2); // 1, 8, 8, 2, 8
    v.insert( v.end(), 3); // 1, 8, 8, 2, 8, 3
    print(v);
    v.erase(v.begin()); // 8, 8, 2, 8, 3
    v.erase(v.begin()+2); // 8, 8, 8, 3
    print(v);
    v.pop_back(); // 8, 8, 8
    print(v);
}
```

### std::find & std::sort

#### Find

- find(a,b,c)
  - a and b are position (iterator)
  - find c from a to before b
  - if not found return b
- Must #include <algorithm>

```
#include <iostream>
#include <vector>
#include <algorithm>
```

```
using namespace std;
```

```
int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    int x = 5;
    if ( find(v.begin(), v.end(), x) != v.end() ) {
```

```

    cout << "found" << endl;
} else {
    cout << "not found" << endl;
}

if (find(v.begin(), v.begin()+3, 3) != v.begin()+3) cout << "FOUND" << endl;

// How many "YES" will be printed? (CHEAT QUESTION)
if (find(v.begin(), v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin(), v.begin()+4, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin()+4, v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin()+4, v.begin()+8, x) != v.end()) cout << "YES" << endl; // YES
// First 3 can't even give v.end(), last did find so it gave that position instead
}

```

### Sort

```

#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    print(v); // 9, -1, 3, 7, 5, 2, 1, 4,
    sort(v.begin()+2, v.begin()+6);
    print(v); // 9, -1, 2, 3, 5, 7, 1, 4,
}

```

### Functions that Work with Vector

- sort
- find
- min\_element
- max\_element
- lower\_bound
- upper\_bound

Need to use *iterator*

## Lecture 4

### Vector Iterator

#### Iterator

- Iterator is a *pointer to position*
- First element is `begin()`
- The one *after* the last element is `end()`

- For insert, valid position is from `begin()` to `end()`, inclusive
- For erase, valid position is from `begin()` up to *before* `end()`
- Different vector, different iterator
  - `v1.end()` is not the same as `v2.end()`

### Iterator Arithmetic

- We can add by integer to an iterator
- We can subtract two iterators of the same type
- We can use increment (`++`) and decrement (`--`)

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int>    v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    vector<int>::iterator it1 = v1.begin() + 3; // 30
    vector<float>::iterator it2 = v2.end();      // after 2.73

    // getting value at iterator by using "*" operator
    cout << *it1 << endl;      // 30
    cout << *(it2-1) << endl;  // 2.73
    cout << *it1+2 << endl;    // 32

    // iterator arithmetics
    vector<int>::iterator it3 = it1 + 2;
    cout << "data at it3: " << *it3 << endl; // 50
    cout << "difference of it3,it1: " << (it3 - it1) << endl; // 2

    vector<float>::iterator it4 = v2.end();
    it4--;
    cout << "data of it4: " << *it4 << endl; // 2.73

    // this cannot be done
    // cout << (it2 - it1) << endl;
}
```

### Iterator All Elements by Iterator

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int>    v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    cout << "----v1----" << endl;
    auto it = v1.begin();
    while (it < v1.end()) { // should use != in other iterator types
        cout << it - v1.begin() << ": " << *it << endl;
        it++;
    }

    cout << "----v2----" << endl;
```

```

    for (auto it = v2.begin(); it < v2.end(); it++) {
        cout << it - v2.begin() << ": " << *it << endl;
    }
}

```

- We can compare iterator
  - Beware!! Iterator of some data structure does not support > or < comparator
  - But for vectors, it's ok
- We can use auto keyword to automatically define a type of a variable

## Iterator Invalidate

### Some Functions that Use Iterators

```

#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int main() {
    vector<int>    v1 = {3,0,1,2,4,-3,9,8};
    vector<float> v2 = {10.2, -4, 0.13, 3.14, 2.73};

    auto it1 = min_element(v1.begin(),v1.end());
    auto it2 = max_element(v2.begin()+2,v2.end());

    cout << *it1 << endl; // -3
    cout << *it2 << endl; // 3.14
}

```

- min\_element and max\_element get the iterator of min, max element

### New Idiom that Iterate All Elements

- Shorter syntax for loop
- Can use reference operator (&)

```

#include <iostream>
#include <vector>
#include <algorithm>
#include <string>
using namespace std;

int main() {
    vector<string> v1 = {"somchai", "somying", "somsak"};

    // range-based for loop
    for (string x : v1) {
        // x is a copy of each element in v1
        cout << x << ", ";
    }

    // using reference
    // x is THE element of v1, meaning we can modify it
    for (auto &x : v1) { x.replace(0,4,"--"); }
    for (auto &x : v1) { cout << x << " "; } // --hai --ing --ak
    cout << endl;
}

```

### Iterator Invalidation

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {10,20};

    auto = v1.end()-1; // this points to 20
    // we resize (enlarge) v1, now [it] is invalidated
    v1.resize(10);

    // this might not crash but it actually points to somewhere not in v1
    cout << *it << endl;

    // this will crash the program
    // v1.insert(it,99);
    for (auto &x: v1) {cout << x << " " ;}
}
```

- Some operation on the data structure *invalidate* existing iterators
  - For vector, this includes *addition* and *deletion* of element (including *resize*)

### Set

- Storing distinct data of same type
  - The data type must be *comparable*, i.e., we can tell if a is more or less than b
- Somewhat slow insert
- Fast look up
- Fast erase
- Iterator starts from “minimum element” and goes in increasing value direction
  - Can be used to (somewhat) fast storing

### Basic

- Notice that s does not include duplicate elements
- Also see that when we iterate, member is sorted
  - This is distinct characteristic of set

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set<int> s = {4,1,3,2,1,1,3,4};

    cout << "Size of s is " << s.size() << endl; // 4

    s.insert(10);
    s.insert(5);
    s.erase(3);

    cout << "member of s: "; // 1 2 4 5 10
    for (auto it = s.begin(); it != s.end(); it++) {
        cout << *it << " ";
    }
}
```

### Somewhat Slow Insert, Iterate but Fast Find

- We will see this in the detail around last part of this course
  - For now, please believe that
    - If there are  $N$  elements in the set
    - Insert takes time directly proportional to  $\log(N)$
    - Each `it++` or `it--` takes times directly proportional to  $\log(N)$
    - Each find takes time directly proportional to  $\log(N)$

### Set Iterator

#### Demos Comparing Vector & Set

- Insert
  - Vector is faster than set
- Find random number
  - Set is much faster than vector

#### Set Iterator

- We cannot do `s.begin() + x`
  - Because, going to the next element (which is the *successor*) in set is not as fast as vector, c++ forbids `begin() + x`
    - alternatively, you can use `auto it = s.begin(); for (int i = 0; i < s.size(); i++) {it++};`
  - We cannot *compare by > or <*
- We can still use `it++` or `it--` to go to the next or previous (*successor* or *predecessor*) or `x`

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set< pair<string,int> > s = { {"somchai",5}, {"abc",6}, {"abcd",-3}, {"somchai",-4},
                                {"z",0}, {"z",-1}, {"z",9} };

    for (auto &x : s) {
        cout << x.first << "," << x.second << endl;
    }

    cout << "-- find --" << endl;
    auto it = s.find({"z",-1});
    cout << (*it).first << "," << (*it).second << endl;
    it--;
    it--;
    cout << it->first << "," << it->second << endl;
    it++;
    cout << it->first << "," << it->second << endl;
}
```

#### Additional Function

- `set.lower_bound`
- `set.upper_bound`
- `set.count`

### std::map

Associative data structure with same property as set

## Map

- Is very similar to Python's dict in usage
- Is internally implemented as a set with "pair" data type
  - *Same properties, same limitations* as set but more convenience to use as associative data structure
- Associative (mapping) between a *Key Type* and a *Mapped Type*

## Basic

```
#include <iostream>
#include <map>
using namespace std;

int main() {
    // map between "Key Type" string and "Mapped Type" int
    map<string,int> m;
    m["somchai"] = 10;
    m["somying"] = -5;
    cout << "Size = " << m.size() << endl; // 2

    // accessing unseen key create a map with default value
    cout << "m[\"xxx\"] = " << m["xxx"] << endl; // 0
    // each element is a pair of key type and mapped type
    for (auto it = m.begin(); it != m.end(); it++) {
        cout << it->first << " is mapped to " << it->second << endl;
    } // somchai: 10, somying: -5

    // this will create mapping "abc" to 0 first and then increase it
    m["abc"]++;
    cout << "now size = " << m.size() << endl; // 4
    for (auto &x : m) {
        cout << x.first << " is mapped to " << x.second << endl;
    } // abc: 1, somchai: 10, somying: -5, xxx: 0
}
```

## Checking If Map Has This Key?

- Use find()

```
#include <iostream>
#include <map>
using namespace std;

int main() {
    map<int,string> m;
    m[1] = "somchai";
    m[99] = "nattee";

    int k = 99;
    map<int,string>::iterator it;
    if ((it = m.find(k)) != m.end()) {
        cout << "Key " << it->first << " is mapped to " << it->second << endl;
    } else {
        cout << "Key " << k << " does not exist in m." << endl;
    }

    // this is not the correct way to check if key exists why??
    if (m[k] != "") {
        cout << "exists" << endl;
    }
}
```

```

    } else {
        cout << "does not exists" << endl;
    }
}

```

### Requirement of std::set and std::map

- Set data type and map key type must be comparable
  - We must be able to compare *order* of two elements
- Type that we can use directly
  - int, bool, float, string, double, char, ... and most of other numerical data type
  - Pair can also be used if both the types of first and second are comparable
    - Pair compare first then second

### Practice Reading C++ Docs

- Both map and set has *insert* and *erase* function
- What is the return value of both function of each data structure?
  - For `set<int> s`, can we do `s.erase(20)`
  - For `map<string, bool> m`, can we do `m.erase("Somchai")??`
- If we wish to erase element from index 3 to index 4096 in a vector
  - Is there any function from vector that we can easily use?

### Pair-Sum Problem

- Pair Sum
  - Given an array of integers, our task is to find whether there exists a pair of elements in the array such that their summation equal to  $X$
- Input:
  - Array of integers (our main array); this is a large array
  - $M$  values of  $X$ , for each value  $X$ , we have to determine if a pair whose sum equal to  $X$  exists.
- Output:
  - For each value of  $X$ , print "YES" if we found such pair; print "NO" otherwise

### Answer

```

#include <iostream>
#include <array>
#include <algorithm>
#include <cmath>

using namespace std;

bool check(int a[], int n, int X);

int main() {
    int a[15] = {323, 232, 138, 230, 437, 176, 127, 111, 329, 1, 296, 12, 3, 2, 6};
    int n = sizeof(a) / sizeof(a[0]);
    int X = 10000;

    cout << check(a,n,X);
}

bool check(int a[], int n, int X) {
    for (int i=0; i<(pow(2,n)); i++) {
        int t = X;
        for (int j=0; j<n; j++) {
            if (i / ((int) pow(2,j+1)) % 2 == 0) {
                t -= a[j];
            }
        }
    }
}

```

```

    }

    if (0 == t) {
        cout << "YES" << endl;
        return true;
    }
    if (0 > t) {break;}
}
}
cout << "NO" << endl;
return false;
} // My stupid solution where I brute force every possible combinations

```

## Lecture 5

### Stack

การเพิ่ม / ลบข้อมูลในกองซ้อน

- ข้อมูล เข้าหลัง ออกก่อน (Last\_In First-Out)

กองซ้อน : stack

```

bool          empty();
unsigned      size();
T             top();           // ดูบนสุด
void          push(T element); // ใส่ข้อมูล
void          pop();           // เอาออก

```

ข้อควรระวัง

- ใช้ #include <stack>
- top() ตอนไม่มีข้อมูล → พัง
- pop() ตอนไม่มีข้อมูล → พัง
- stack ไม่มี iterator
  - ▶ ดูได้เฉพาะข้อมูลที่เพิ่งใส่เข้าไปล่าสุด (top)
  - ▶ ไม่มี begin(), end() ให้ใช้
  - ▶ ถ้าอยากมาดูข้อมูลทุกตัวใน stack ต้องจำใจ pop ข้อมูลออกมา

ตัวอย่างการใช้งาน Stack

- การตรวจสอบโครงสร้างแบบซ้อนกัน เช่น การใส่วงเล็บ () {} [] ...
- การจัดเก็บตัวแปรและการทำ function calls
- การประมวลผลนิพจน์ทางคณิตศาสตร์ postfix → Discrete Math
- การทำ undo/redo
- การค้นคำตอบแบบ depth-first search
- ...

### Parentheses Checking

การตรวจสอบการใส่วงเล็บ

- ถูก : ( { ( ) [ { } ] } )
- ผิด : เปิดปิดไม่ตรงกัน ( { ] )
- ผิด : มีปิดมากเกินไป ( { ( ) } ) ) }
- ผิด : มีเปิดมากเกินไป ( { ( ) }
- วิธีทำ

- อ่านมาทีละตัว
- ถ้าเป็นวงเล็บเปิด ให้ push ลง stack
- ถ้าเป็นวงเล็บปิด ให้ pop จาก stack มาตรวจสอบว่าเป็นวงเล็บเปิดที่ตรงกันกับวงเล็บปิดหรือไม่
- เมื่อใดที่อยาก pop ถ้า isEmpty แสดงว่า ปิดมีมากไป
- เมื่ออ่านเสร็จหมด stack ยังมีข้อมูล แสดงว่า เปิดมีมากไป

## Function Call Stack

### Using Stack in Java Virtual Machine

- jvm uses java stack to store
  - state of method calls
  - parameters and local variables
  - temporary memory for computation (operand stack)

```
public class StackFrame {
    public static void main(String[] args) {
        a(3,2);
        b(5);
    }
    static void a(int x, int y) {
        int z = x/y;
        b(z);
    }
    static void b(int x) {
        ++x;
    }
}
```

We make a stack frame of function calls, each element consists of arguments and returning argument of the function. Function main contains args and RA JVM and calls a, a will be added to stack and contains z, y, x, and RA 04:, then it calls b. After b is finished, it will come back to line 9 and popped itself. It will do a and go to line 4, and so on. Arguments in each functions are separated variables.

```
#include <iostream>

using namespace std;

void test(int x) {
    if (x > 0) {
        int y = x-1;
        cout << "I am test(" << x << "). continue... " << endl;
        test(y);
        cout << "now x is " << x << endl;
    } else {
        cout << "I am test(0). stop"
    }
}

int main() {
    test(4);
    return 0;
}
```

### Output

```
I am test(4). continue...
I am test(3). continue...
```

```

I am test(2). continue...
I am test(1). continue...
I am test(0). stop
now x is 1
now x is 2
now x is 3
now x is 4

```

## Postfix Evaluation

### Infix and Postfix Expressions

- infix (เติมกลาง)
  - $a + b * c / d - 2$ ,  $(a + b) * c / (d - 2)$
  - use order of operations
  - use parentheses
- postfix (เติมหลัง)
  - $a b c * d / + 2 -$ ,  $a b + c * d 2 - /$
  - order is from left to right
  - parentheses may be omitted

```

1  2  +  3  *  4  1  -  /
      3  3  *  4  1  -  /
          9  4  1  -  /
              9          3  /
                          3

```

### Calculate Postfix Expressions with Stack

- Value of  $2 3 + 4 5 - 6 * + ?$
- Use stack to help
- Solution
  - Check each element in postfix from left to right
  - If it is operand, push stack
  - If it is operator, pop elements from stack as many as operator needs, then push the result
  - After complete, answer will be at the top of stack

## Convert Infix to Postfix

### Use Stack to Convert

- input : infix expression
- output : postfix expression
- Steps
  - Read each element in infix
  - If it is operand, move it to the back of output
  - If it is operator
    - May pop operator to the end of output
    - push operator to stack
  - If all infix is read
    - pop all operators to the end of input

```

string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) { // Read each element
        char token = infix[i];
        if (token เป็น operand) {

```

```

    postfix += token; // if operand, add to output
} else {
    while ( ??? ) {
        postfix += s.top(); // if operator, pop stack to result,
        s.pop();
    }
    s.push(token); // then push new operator
}
}
while (!s.empty()) {postfix += s.top(); s.pop();}
return postfix;
}

```

### Priority of Operators

input 2 \* 3 + 4

| read | output    | stack |
|------|-----------|-------|
| 2    |           |       |
| *    | 2         |       |
| 3    | 2         | *     |
| +    | 2 3       | *     |
| 4    | 2 3 *     | +     |
|      | 2 3 * 4   | +     |
|      | 2 3 * 4 + |       |

(\* comes before and is more important than +)

input 2 + 3 \* 4

| read | output    | stack |
|------|-----------|-------|
| 2    |           |       |
| +    | 2         |       |
| 3    | 2         | +     |
| *    | 2 3       | +     |
| 4    | 2 3       | * +   |
|      | 2 3 4     | * +   |
|      | 2 3 4 *   |       |
|      | 2 3 4 * + |       |

### Code to Compare Priority

```

string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) {
        char token = infix[i];
        if (priority.find(token) != priority.end()) {
            postfix += token;
        } else {
            int p = priority[token]; // Get priority of the new operator
            while ( !s.empty() && priority[s.top()] >= p ) { // lazy evaluation; left first
                postfix += s.top(); // pop if top operator is not less important than new op
                s.pop();
            }
            s.push(token);
        }
    }
    while (!s.empty()) {postfix += s.top(); s.pop();}
}

```

```
    return postfix;
}
```

## Lazy Evaluation

### Setting Priority of Operators

- $\wedge$  represents power 7
- $\wedge$  before  $+$   $-$   $*$   $/$
- $*$   $/$  before  $+$   $-$
- $*$  same as  $/$  5
- $+$  same as  $-$  3

```
map<char,int> priority =
    {{'+',3},{'-',3},{ '*',5},{ '/',5},{ '^',7}};
```

### Complications with Parentheses

- In parentheses is like sub-expression
- Open parentheses always push (very high priority)
- Close parentheses is very low priority
- If found, pop until sees open parentheses

```
input    2    *    (    3    +    4    )
```

| read | output    | stack           |
|------|-----------|-----------------|
| 2    |           |                 |
| *    | 2         |                 |
| (    | 2         | *               |
| 3    | 2         | ( *             |
| +    | 2 3       | ( *             |
| 4    | 2 3       | + ( *           |
| )    | 2 3 4     | + ( * (pop all) |
|      | 2 3 4 + * |                 |

```
int p = priority[token];
while ( !s.empty() && priority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
s.push(token);

int p = outpriority[token];
while ( !s.empty() && inpriority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
if (token == ')') s.pop(); else s.push(token);
```

### Priority of Operators with Parentheses

|               | + | - | * | / | $\wedge$ | ( | ) |
|---------------|---|---|---|---|----------|---|---|
| In expression | 3 | 3 | 5 | 5 | 7        | 9 | 1 |
| In stack      | 3 | 3 | 5 | 5 | 7        | 0 |   |

### Right to Left Operator

|               | + | - | * | / | $\wedge$ | ( | ) |
|---------------|---|---|---|---|----------|---|---|
| In expression | 2 | 2 | 4 | 4 | 8        | 9 | 1 |

|          |   |   |   |   |   |   |
|----------|---|---|---|---|---|---|
| In stack | 3 | 3 | 5 | 5 | 7 | 0 |
|----------|---|---|---|---|---|---|

**Note** Power in expression is more important than in stack

input    2    +    3    -    4    ^    5    ^    6

| read | output            | stack |
|------|-------------------|-------|
| 2    |                   |       |
| +    | 2                 |       |
| 3    | 2                 | +     |
| -    | 2 3               | +     |
| 4    | 2 3 +             | -     |
| ^    | 2 3 + 4           | ^ -   |
| 5    | 2 3 + 4           | ^ -   |
| ^    | 2 3 + 4 5         | ^ -   |
| 6    | 2 3 + 4 5         | ^ ^ - |
|      | 2 3 + 4 5 6       | ^ ^ - |
|      | 2 3 + 4 5 6 ^ ^ - |       |

### Conclusion

- Implementations of stack to solve problems
- Main operations: push / pop / top
- Make stack with array
- If reserve enough space, all runs is  $\Theta(1)$ ; constant time

## Lecture 6

### Queue

- Just like a normal queue in real life.
  - First-in-First-Out data structure
  - One way in (back of queue), one way out (front of queue)
  - push = add data to the back of the queue
  - pop = remove data from the head of the queue

### Basic

```
#include <iostream>
#include <queue>
#include <vector>

using namespace std;

int main() {
    queue<int> q;
    q.push(1);
    q.push(2);
    q.push(3);
    while (q.empty() == false) {
        cout << q.front() << endl;
        q.pop();
    }
    cout << "-- example 2 --" << endl;
    queue<vector<int>> q2;
    vector<int> v1 = {1,2,3};
    vector<int> v2 = {99,88,-1};
```

```

q2.push( v1 );
q2.push( v2 );
cout << q2.back()[1] << endl;
cout << q2.front().size() << endl;
auto x = q2.front();
q2.pop();
cout << x[0] << endl;
}

size_t    q.size()
bool      q.empty()
void      q.push(T data)
void      q.pop()
T         q.front()
T         q.back()

```

### Limitation

- Same limitation as stack
  - No iterator
    - No `begin()`, `end()`
    - Can only access front and back of the queue
    - If we wish to access all members, we have to pop it all
  - Do not call `front()`, `back()`, `pop()` when the queue is empty

## Radix Sort

Queue Application: Fast sorting with no comparison

### Overview

- Put all data in an array
- For each digit  $X$ , from LSD to MSD
  - PUT TO QUEUE step: Sort by digit  $X$  by putting all of data from the array into  $B$  queues
    - $B$  is the base of the number
    - For example, for a base 10 number, we will have Queue[0] to Queue[9]
    - Put into the queue labelled with that digit
  - GET FROM QUEUE step: Start from queue 0 to queue  $B - 1$ , remove data from the queue and put back to the array

### Example

Input 115 15 42 305 21 8 463

#### Round 1 (digit 0), from queue

| (0) | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|     |     |     |     |     | 305 |     |     |     |     |
|     |     |     |     |     | 15  |     |     |     |     |
|     | 21  | 42  | 463 |     | 115 |     |     | 8   |     |

⇒ 21 42 463 115 15 305 8

Data is now sorted by the last digits, because we pop from queue 0 to 9

#### Round 2 (digit 1), from queue

| (0) | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 8   | 15  |     |     |     |     |     |     |     |     |
| 305 | 115 | 21  |     | 42  |     | 463 |     |     |     |

⇒ 305 8 115 15 21 42 463

Data is now sorted by last two digits, because we goes from queue 0 to 9, which is grouped by digit 1 and the data in each queue is sorted by the last digit.

### Round 3 (digit 2), from queue

| (0) | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 42  |     |     |     |     |     |     |     |     |     |
| 21  |     |     |     |     |     |     |     |     |     |
| 15  |     |     |     |     |     |     |     |     |     |
| 8   | 115 |     | 305 | 463 |     |     |     |     |     |

⇒ 8 15 21 42 115 305 463

### Code

```
#define base 10

int getDigit(int v, int k) {
    // return the kth digit of v (MSD is digit 0)
    int i;
    for (int i=0; i<k; i++) v /= base;
    return v % base;
}

// d = number of digits
void radixSort(vector<int> &data, int d) {
    queue<int> q[base];
    for (int k=0; k<d; k++) {
        for (auto &x : data) {
            q[getDigit(x,k)].push(x);
            for (int i=0, j=0; i<base; i++) {
                while(!q[i].empty()) {
                    data[j++] = q[i].front(); q[i].pop();
                }
            }
        }
    }
}
```

### Breadth First Search

Queue Application: Gotta generate 'em all

#### The Problem

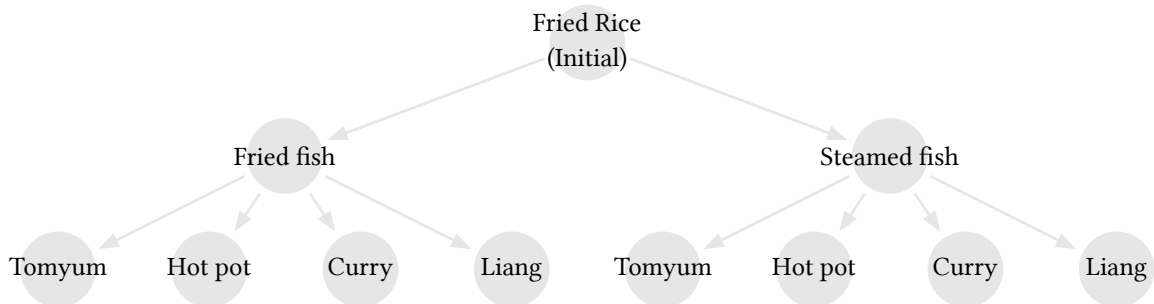
- Given a positive integer  $X$
- Start with a number 1, find a sequence of arithmetics operations, either “ $\times 3$ ”, or “ $/ 2$ ” that makes 1 into  $X$ 
  - the  $/ 2$  is integer division, e.g.,  $5 / 3 = 1$  (not 1.6666)
  - The sequence must be as short as possible
- Example
  - $10 = 1 \times 3 \times 3 \times 3 \times 3 / 2 / 2 / 2$
  - $31 = 1 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 / 2 / 2 / 2 / 2 / 2 \times 3 \times 3 / 2$

#### The Idea

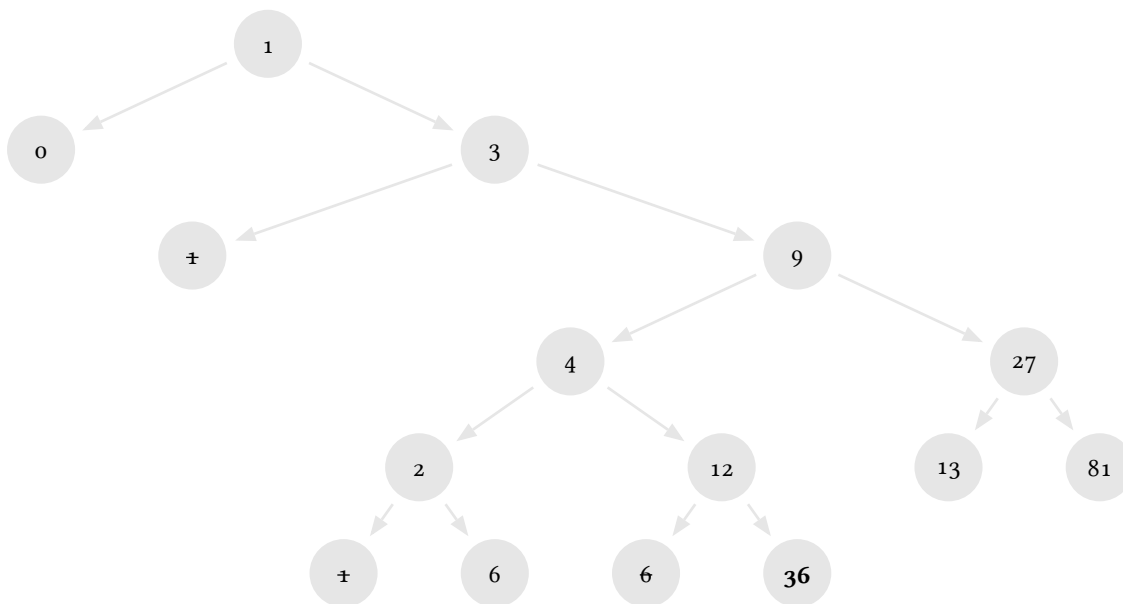
- Generate all possible sequences
  - Start with length 1, 2, 3, ... until we find one that gives  $X$
- This is called an *exhaustive search* algorithm
  - Systematically enumerate all possible somethings

**Tree Structure (Search Tree)**

- A structure to illustrate *search* algorithm
- Divide into steps
  - Start with *initial solution*
  - For each possible outcome (called a *state*) of each step, *generate all* proper possible *next step*
    - Also, check if the current step is what we needed
- Written as a diagram of *node* and *edge*

**Enumerate****M3D2 Problem & Solution****Back to Our Problem**

- Start with 1
- Each step is either \* 3 or / 2
- Issue: might get repeated number
  - Solution: if we have found it, do not generate new step

**M3D2 Code****Code**

- `Queue` makes ordering of how we pick a state to enumerate
- From top to bottom and left to right

```

void m3d2(int target) {
    map<int, int> prev;
    queue<int> q;
  
```

```

int v = 1;
q.push(1); prev[1] = -1;
while (!q.empty()) {
    v = q.front(); q.pop();
    if (v == target) break;
    int v2 = v/2;
    int v3 = v*3;
    if (prev[v2] == 0) {q.push(v2); prev[v2] = v;}
    if (prev[v3] == 0) {q.push(v3); prev[v3] = v;}
}
if (v == target) showSolution(v, prev);
}

```

### Display Solution

|         |   |    |   |   |   |   |   |    |    |    |    |    |    |
|---------|---|----|---|---|---|---|---|----|----|----|----|----|----|
| x       | 0 | 1  | 2 | 3 | 4 | 6 | 9 | 12 | 13 | 27 | 36 | 39 | 40 |
| prev[x] | 1 | -1 | 4 | 1 | 9 | 2 | 3 | 4  | 27 | 9  | 12 | 13 | 81 |

### showSolution

```

void showSolution(int v, map<int,int>& prev) {
    string out = "";
    while (prev[v] != -1) {
        if (prev[v] * 3 == v) {
            out = "x3" + out;
        } else {
            out = "/2" + out;
        }
        v = prev[v];
    }
    out = "1" + out;
    cout << out << endl;
}

```

⇒ 36 = 1 \* 3 \* 3 / 2 \* 3 \* 3

## Lecture 7

### Priority Queue

Queue with privilege

#### Intro

- Priority Queue is ...
  - A queue with priority
  - Item with high priority is promoted to the front of the queue
    - There is *no back of the queue*
  - Priority is defined by having more value
    - Comparison, by default, is to use operator <, i.e., if item A < B is true, then B *has higher priority*
    - We can have custom comparator
- Has the *same interface* as stack

#### Example

For intuitive purpose only! This is not really how priority\_queue work internally

```

push(10)
< 10      >

```

```
push(30)
< 30 10    >
```

```
push(20)
< 30 20 10 >
```

```
pop()
< 20 10    >
```

```
push(15)
< 20 15 10 >
|
top
```

### Basic

```
size_t      q.size()
bool        q.empty()
void        q.push()
void        q.pop()
T           q.top()
* there is no q.back()
```

### Limitation

- Same limitation as stack, queue
  - *no iterator*
    - No `begin()`, `end()`
    - Use `#include <queue>`
    - Can only access `top()` of the queue
    - If we wish to access all members, we have to pop it all
  - Do not call `top()`, `pop()` when the queue is empty
- The data type must be *comparable* (similar to set and map)

### Class

#### Quick Summary

- Syntax
  - class declaration must end with `;`
  - Function definition can be outside of the class
  - Access modifier is `public:`, `private:`, `protected:`
  - Constructor is a function with the same name of the class with no return type
- Object is a variable (instantiation) of a class
  - When declared, a constructor is called

#### Example 1

```
#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
}
```

```

    string getFullName() {
        return "[" + name + " " + surname + "];"
    }
private:
    string name, surname;
};

int main() {
    Student a;
    Student b;
    a.setFullName("nattee", "niparnan");
    cout << a.getFullName() << endl;
    cout << b.getFullName() << endl;
}

```

Declaring class before writing functions

```

class Student {
public:
    void setFullName(string name, string surname);
    string getFullName();
private:
    string name, surname;
};

void Student::setFullName(string name, string surname) {
    this->name = name;
    this->surname = surname;
}

string Student::getFullName() {
    return "[" + name + " " + surname + "];"
}

```

### Output

```

[nattee niparnan]
[ ]

```

### Example 2: Constructor

```

#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    Student(float score) {gpax = score;}
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
    string getFullName() {return "[" + name + " " + surname + "];"}
    bool is1stHonor() {return gpax >= 3.6;}
private:
    string name, surname;
    float gpax;
};

```

```
int main() {
    Student a(2.95);
    a.setFullName("nattee", "niparnan");

    cout << a.getFullName() << endl;
    if (a.is1stHonor()) {cout << "YES" endl;} else {cout << "NO" << endl;}
    // Student b; // <-- cannot compile because there is no default constructor
}
```

## Operator Overloading

How C++ has a function for each operator

### Overview

- Let say we write `a + b` when `a` and `b` is an object of some classes
  - This can be considered the same as calling a function `plus(a,b)`
  - C++ allows us to write a function for many operator and use it as an operator
  - For example we can write a function `times(a,b)` and let it be used as `a * b`
- This is what we called operator overloading

Instead of writing

```
Matrix multiply(Matrix a1, Matrix a2) {
    multiply(a, multiply(b,c))
}
```

We want to write

```
Matrix a, b, c
a * (b * c)
```

### Example

```
#include <queue>
#include <iostream>
#include <string>

using namespace std;

// use operator + symbol that will be overloaded
string operator*(string & lhs, const int & rhs) {
    string result = "";
    for (int i=0; i<rhs; i++) {
        result = result + lhs;
    }
    return result;
}

int main() {
    string a = "abc ";
    cout << (a * 3) << endl; // this gives "abc abc abc "
}
```

- Function must be named operator followed by the operator that we will overload
- Some operator takes two parameters (such as `+`, `-`, `*`, `/`, `%`)
- Some takes one (such as `++`, `--`, `!`, `*`, `&`)

## Overloading less-than operator

### Using with data structure that require sorting

- We have seen several data structure that requires comparability of the data, such as set, map and priority queue
- If we want to use our class with these data structure, we need to tell them how can we use them
- There are multiple ways to achieve this
  - Let us consider operator overloading

### Overloading <

- As stated earlier, set, map, and priority\_queue use *operator <* to compare two elements
- It does not work if we overload *operator >*

```
class Student {
public:
    Student(float score, string a, string b) {
        name = a;
        surname = b;
        gpax = score;
    }
    bool is1stHonor() {return gpax >= 3.6;}
    // not good, now our data is public
    string name, surname;
    float gpax;
    // overloading <
    bool operator<(const Student & other) const {
        return gpax < other.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    cout << (a < b) << endl; // 1
    priority_queue<Student> pq;
    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // attawith
}
```

## Custom Comparator

### Why custom?

- By overloading *operator <*, we have defined default ordering of that class
- What if we need another ordering, just for this *priority\_queue* only
  - For example, *Student* is ordered by *gpax* by default
  - What if we want our *priority\_queue* to order by name instead, while keeping the *Student* default ordering elsewhere
  - Better, can we have multiple *priority\_queue* with different ordering?
- Can be done via comparator class

### Example

```
#include <iostream>
#include <string>
#include <queue>
```

```

using namespace std;

class Student {...};

// Comparator class use () overloading
class StudentByNameComparator {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        return lhs.name < rhs.name;
    }
};

class GpaxThenName {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        if (lhs.gpax == rhs.gpax)
            return lhs.name < rhs.name;
        return lhs.gpax < rhs.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    Student c(4.00, "vishnu", "kotrajaras");
    cout << (a < b) << endl; // 1
    StudentByNameComparator comp1;
    GpaxThenName comp2;

    cout << comp1(a,b) << endl; // 0; can use like function

    // Use 3 template parameters
    priority_queue<Student, // class to compare
                  vector<Student>, // vector of that class
                  StudentByNameComparator> pq(comp1); // comparator class

    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // nattee

    priority_queue<Student,
                  vector<Student>,
                  GpaxThenName> pq2(comp2);

    pq2.push(a);
    pq2.push(b);
    pq2.push(c);
    cout << pq2.top().name << endl; // vishnu
}

```

#### Another Method, lambda-function

```

#include <iostream>
#include <string>
#include <queue>
using namespace std;

```

```
int main() {
    auto compare = [](const string& lhs, const string& rhs) {
        return lhs.size() < rhs.size();
    };

    cout << "Result of compare function = " << compare("xxx","z") << endl;

    priority_queue<string,vector<string>,decltype(compare)> pq(compare);
    pq.push("somchai");
    pq.push("z");
    pq.push("abc");
    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}
```

### Output

```
Result of compare function = 0
somchai
abc
z
```

## Priority Queue Template

### Templating of priority\_queue

- priority\_queue requires 3 template parameters
- priority\_queue<T, Container = vector<T>, Compare = less<T>>
- The first one is required (which is the type of the data)
- The *second* and the *third* is optional (it has default type)
  - *Second* is the container (for now, just don't think about it)
  - *Third* is the class for comparator (the class that we use to compare)
    - This one is default to less<T>

```
#include <iostream>
#include <string>
#include <queue>
using namespace std;

int main() {
    less<int> x;    // comparator that checks a < b
    greater<int> y; // checks a > b

    int a = 10;
    int b = 3;
    cout << x(a,b) << endl; // 0
    cout << y(a,b) << endl; // 1
}
```

### Using Comparator for set and map

- To use custom class with set and map, we need to do the same thing, let set and map know how to sort the data
  - Either make default ordering (overload <) in the custom class
  - Or use custom comparator when declare
- For set, the declaration is set<T, Compare = less<T>>
- For map, the declaration is map<Key, T, Compare = less<Key>>

**Assignment**

- Is any of `vector<int>`, `set<int>`, `map<int, string>`, `queue<bool>`, `stack<vector<int>>` comparable?
  - For any class that is “YES”, how it is ordered?
  - For example, if `vector<int>` is comparable, how `{1, 2, 3}` is compared to `{1, 2, 3, 4}` or `{2, 3, 4}`

**Answer**

- `vector<int>` compare each element from the left; `{2, 3, 4} > {1, 2, 3, 4} > {1, 2, 3}`
- `set<int>` compare the maximum element () of each set, if equal then look the next max
- `map<int, string>` compare by Key set, then compare max of value set
- `queue<bool>` is not comparable
- `stack<vector<int>>` is not comparable

**Summary****Data Structure Summary**

| Data Structure                       | Pro                                                                                                                                               | Cons                                                                                                                                                                                                                                         | Remark                                                                                                                                                                                    |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>pair&lt;T1, T2&gt;</code>      | Nothings... It just a pair of two data type                                                                                                       |                                                                                                                                                                                                                                              |                                                                                                                                                                                           |
| <code>vector&lt;T&gt;</code>         | <ul style="list-style-type: none"> <li>• Fast access [ ]</li> <li>• Fast append <math>\mathcal{O}(1)</math></li> </ul>                            | <ul style="list-style-type: none"> <li>• Slow find <math>\mathcal{O}(N)</math></li> <li>• Slow insert, Slow Erase <math>\mathcal{O}(N)</math></li> </ul>                                                                                     |                                                                                                                                                                                           |
| <code>set&lt;T&gt;</code>            | <ul style="list-style-type: none"> <li>• Fast find <math>\mathcal{O}(\log(N))</math></li> <li>• Item is sorted</li> </ul>                         | <ul style="list-style-type: none"> <li>• Slower to just append data than vector, stack, queue <math>\mathcal{O}(\log(N))</math></li> <li>• Iterate is also slow</li> <li>• Takes lots of memory <math>\mathcal{O}(\log(N))</math></li> </ul> | Require comparator                                                                                                                                                                        |
| <code>map&lt;Key, T&gt;</code>       |                                                                                                                                                   |                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Associative data type</li> <li>• Also require comparator of Key_Type</li> </ul>                                                                  |
| <code>stack&lt;T&gt;</code>          | <ul style="list-style-type: none"> <li>• Very fast push, pop <math>\mathcal{O}(1)</math></li> </ul>                                               | <ul style="list-style-type: none"> <li>• Very limited functionality but has special uses</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>• No iterator</li> <li>• Order of data coming out depends on something (stack, queue depends on WHEN it is pushed, pq depends on value)</li> </ul> |
| <code>queue&lt;T&gt;</code>          |                                                                                                                                                   |                                                                                                                                                                                                                                              |                                                                                                                                                                                           |
| <code>priority_queue&lt;T&gt;</code> | <ul style="list-style-type: none"> <li>• Fast get max</li> <li>• Fast delete max</li> <li>• Data is sorted</li> <li>• Memory efficient</li> </ul> | <ul style="list-style-type: none"> <li>• Slower to just append data than vector, stack, queue</li> <li>• Very limited functionality</li> </ul>                                                                                               | <ul style="list-style-type: none"> <li>• pq requires comparator</li> </ul>                                                                                                                |

**More Data Structure**

- C++ has more data structure not really covered right now
  - `list` is a vector with faster input / erase but does not have fast access
  - `unordered_set`, `unordered_map` are set and map that the data is not sorted but is much faster
  - `deque` is a queue that can push, pop at both ends
  - `multiset`, `multimap` are set and map that allows duplicated entries

**Lecture 8****Create Your Own CP::pair****Intro**

- We start writing our own data structure
  - In our own *namespace* (CP)
- We start with *pair*
  - Good introduction on writing a class
  - Some C++ feature we need to use

- const
- pass-by-value, pass-by-ref
- Header file

### What should we write in a class

- Class name and namespace
- Member *variables* which define what data we want to store in the class
- Member *functions* which define behavior of the class and how member variable is manipulated
  - Include lots of special function such as *constructor*, *destructor*, *overloading*

### CP:pair version 0.1 (minimal version)

- Templating
  - Parameter of a code
- No member functions
  - When we declare a pair, T1 and T2 must be declared as a type

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int,string> p1; // In p1, T1 is int, T2 is string
    CP::pair<bool,int> p2;  // In p2, T1 is bool, T2 is int
}
```

### Capabilities

- Can hold 2 pieces of data
- Can use template
- Works with operator =
- Has copy constructor
- Cannot check equal
- Cannot check less than

```
#include <iostream>
#include <string>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int,string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int,string> a(p1); // copy ctor
    p2 = p1;
```

```
    cout << p2.first << "," << p2.second << endl;
}
```

```
/*
if (p1 == p2) { // won't compile
    cout << "yes" << endl;
}
```

```
if (p1 < a) { // won't compile
    cout << "yes" << endl;
}
*/
```

### CP::pair version 0.2 (comparator overload)

```
#include <iostream>
#include <string>
#include <set>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // operator
        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        bool operator<(const pair<T1,T2>& other) const { // operator< must be const
            return ((first < other.first) ||
                    (first == other.first && second < other.second));
        }
    };
}

int main() {
    CP::pair<int,string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int,string> a(p1); // copy ctor
    p2 = p1;
    cout << p2.first << "," << p2.second << endl; // 20, somchai

    if (p1 == p2) {cout << "yes" << endl;} // yes
    if (p1 < a) {cout << "yes" << endl;}
    set<CP::pair<int,int>> s;
    s.insert({1,2});
    // Now we can compare and use less than (and also set, map, priority queue)
    cout << s.begin()->second << endl; // 2
}
```

## Const Member

### Const Parameter in Function

```
void test(int &x, const int& y) {
    x++;           // Okay; doable
    cout << y << endl; // Okay, we only read y
    for (int i=0; i<y; i++) { // Okay, too
        cout << i << endl;
    }
    y--; // ERROR: we promised NOT to change y
}
```

- Declared by putting a keyword const as a prefix
- Const parameter must not be modified inside the function
- Why? So that we know that the function does not modify the data
  - Especially when we pass-by-reference

### Difference of Pass-by-reference && Pass-by-value

#### Pass-by-value

- The *arguments* can be either constants or variables
- Modifying *parameters* (variable inside the function) does not change the *argument's* value
- The *argument's* value is copied to the parameter
  - **SLOWER!!!** Because we have to copy

#### Pass-by-reference

- The *arguments* must be variables
- The *parameters* are the argument's variables
  - Modify the *parameter* also modify the *argument's* variable
- **Faster**

### Const Member Function

```
class ccc {
public:
    int a,b;

    void inspect() const { // This function promises NOT to change anything
        if (a < b) cout << "yes" << endl; // Okay

        // b += 20; // <-- NOT OKAY
    }

    void mutate() { // This function might change something
        if (a < b) a += 10; // Okay
    }
};

void test2(ccc& changeable, const ccc& unchangeable) {
    changeable.inspect();
    changeable.mutate();
    unchangeable.inspect();
    unchangeable.mutate(); // ERROR: attempt to change unchangeable object
}
```

- Declared by putting a keyword const after the function declaration
- Const member function cannot modify any member data
  - Also cannot call any other function that is not const

- Why? So that we know that the function does not modify its data

## Constructor & List Initialization

### Custom Constructor

- In STL spec of `std::pair`, it has custom constructor called *initialization constructor*

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // Custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) {
            first = a;
            second = b;
        }
        // When we have a constructor, a default constructor is not auto-generated

        // Operator
        bool operator==(const pair<T1,T2> &other) {...}
        bool operator<(const pair<T1,T2>& other) {...}
    };
}

int main() {
    CP::pair<int, bool> p(10,false);
    CP::pair<string, int> q("abc",42), r("",0);

    cout << (q < r) << endl; // 0
    priority_queue<CP::pair<string,int>> pq;
    pq.push(r);
    pq.push(q);
    cout << pq.top().first << endl; // abc

    CP::pair<string, int> x(q);
    CP::pair<string, int> y = x;

    // All below cannot be compiled
    // CP::pair<string, int> w;
    // vector<CP::pair<int, int>> v(10);
}
```

### Initialization List

- Instead of writing a code to assign a value to each member, we can use *initialization list*
  - A little bit shorter code
  - Also little bit faster
  - Only way to init *const* member

```
namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;
```

```

        // custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

### Default Constructor

- A constructor used when we simply declare an object
- Auto-generated by init all members with its default constructor
- Won't be auto-gen if we have any other constructor

```

namespace CP {
    template <typename T1, typename T2>
    class pair {
        T1 first;
        T2 second;

        // Constructor
        pair() : first(), second() { }
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

## Include & Header file

### Include File

- Usually, our data structure (which is written as a class) will be used by several programs in several files.
  - It is better NOT TO copy our data structure code into each file
- Rather, put our code in a file and include it where it is needed
  - Better if we want to change it
  - Better compilation
- Introducing ".h" files

### C++ header file (.h) and #include

- To put a content of one file into another file, we use #include "filename" keyword
- C++ will simply put the content of *filename* into where we #include it
- #include has more benefit
  - Separation of *declaration* (what it is) and *definition* (how it works)
  - Not really explored in this class
- We usually use .h for a file that we will include but that is not a rule, we can use other extension

### Problem with include

#### a.h

```

class a {
    int m1, m2;
};

```

#### b.h

```

#include "a.h"

```

```

class b {
    a x;
};

```

#### c.h

#### main.cpp (expanded)

```

class a {
    int m1, m2;
};

```

```

class b {
    a x;
};

```

```

// There are multiple copies of class a
class a {
    int m1, m2;
};

```

```
#include "a.h"
```

```
class c {
    a y;
};
```

#### main.cpp

```
#include "b.h"
#include "c.h"
```

```
int main() {
    b b1;
    c c1;
}
```

#### Fix problem

##### a.h

```
#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif
```

##### b.h

```
#include "a.h"
```

```
class b {
    a x;
};
```

##### c.h

```
#include "a.h"
```

```
class c {
    a y;
};
```

```
class c {
    a y;
};
```

```
int main() {
    b b1;
    c c1;
}
```

#### main.cpp (expanded)

```
#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif

class b {
    a x;
};

// class a here is taken out of
// compilation because A_H is defined
#ifndef A_H
// #define A_H
// class a {
//     int m1, m2;
// };
#endif

class c {
    a y;
};

int main() {
    b b1;
    c c1;
}
```

#### Summary

##### Major

- **Templating**: allow user to write a code that works with different data type
- **Constructor**: a function called when we create a variable
  - Default constructor
- **Operator overloading**: for less-than and equality
- pass-by-ref vs pass-by-value

##### Minor

- Header file
- const
- List initialization

### Exercise (no grader)

1. Modify operator< so that it compare *second* before *first*
2. Modify operator< so that when we call sort(v.begin(), v.end()) where w is a vector of our pair, it is sorted from *Max* to *Min*
3. Write operator!= and operator>=

### Final version CP.h

```
#ifndef _CP_PAIR_INCLUDED_
#define _CP_PAIR_INCLUDED_

namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;

        pair() : first(), second() {}

        pair(const T1 &a, const T2 &b) : first(a), second(b) { }

        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        // 1.
        bool operator<(const pair<T1,T2>& other) const {
            return ((second < other.second) ||
                    (second == other.second && first < other.first));
        }

        // 2.
        bool operator<(const pair<T1,T2>& other) const {
            return ((first > other.first) ||
                    (first == other.first && second > other.second));
        }

        // 3.
        bool operator!=(const pair<T1,T2>& other) const {
            return !(*this == other);
        }

        bool operator>=(const pair<T1,T2>& other) const {
            return !(*this < other);
        }

    };
}
#endif
```

## Lecture 9

### CP::vector

Our first “real” data structure

#### Intro

- Now we will create more complex data structure `CP::vector`
- It can store variable length array
  - Implemented as a dynamic array
- Can be accessed by operator `[]`
  - Additional operator to be overloaded
- We also have to create our own iterator
  - Implemented as a pointer

#### Key Idea

- Vector stored 3 things (3 member data)
  - `mData`: A dynamic array, large enough space to store current data and might have reserved space
  - `mSize`: Number of data stored
  - `mCap`: Size of the dynamic array (maybe large than `mSize`)
- If the dynamic array is full and more data is being added, we create a new dynamic array and relocate data to the new array
  - This is called `expand`
  - Each `expansion` takes very long time
- Dilemma
  - Large reserve = less often relocation but use more memory
  - Small reserve = more frequent relocation but less memory

#### Example

```
mSize  mCap      mData
  4      5    [10, 3, 1, 5, ]
```

```
push_back(9)
```

```
  5      5    [10, 3, 1, 5, 9]
```

```
push_back(4)
```

```
  6     10    [10, 3, 1, 5, 9, 4, , , , ]
```

#### How much reserve should we have?

- Whenever we need to relocate, we `double` the capacity that we currently have
- If we start with a vector with zero size and continuously add data one by one, for example by `push_back`
  - Each expansion will take time `equal` to `the number of data` when we relocate (this is very slow)
- By doubling the size every time we expand, we can show that on average, `each addition of a data` (such as `push_back`, `insert`) `takes constant time!!!`

### Pointer

How to implement a dynamic array (and also iterator)

#### Pointer & Memory

- Each variable is some block in computer memory
  - Programming language just map our `variable name` to that `block of memory`
  - Programming language works with the address of that block
- Pointer variable is a variable that stor `address of memory`

- ▶ Pointer needs type i.e., address of int is not the same as address of bool
- ▶ We can use operator & to ask for the **address** of a **variable**
- ▶ We can use operator \* to ask for the **data** of an **address**

### Example

```
int x, y;    // this is normal variable x and y
int *a;     // this is int pointer variable a
int *b;     // another int pointer

x = 10;     // say x is at address 3267
a = &x;     // a points to x
b = a;      // b also points to x
*b = 30;    // change the address value to 30
y = *b;     // set value of y to 30 (hard copy)

cout << &x << endl; // 0x7ffee22885ac
cout << &y << endl; // 0x7ffee22885a8 +4
cout << &a << endl; // 0x7ffee22885a0 +8
cout << &b << endl; // 0x7ffee2288598 +8
cout << sizeof(int) << endl; // 4 bytes
cout << sizeof(*int) << endl; // 8 bytes
```

### Pointer Arithmetic

- Pointer can be added, subtracted by integer
  - ▶ It **moves the address by the size of the type** of the pointer
  - ▶ For example, when X is an int (which is 4 bytes) X + 10 result in an address 40 bytes away from X
- Two pointers of the same type can be subtracted
  - ▶ The result is the address **difference divided by size of the type** of the pointer

```
int main() {
    bool x, y, z;
    x = 1; y = 2; z = 3;
    bool *a, *b;

    a = &x;
    b = a+2;

    cout << "&x = " << &x << endl; // &x = 0x6afed4
    cout << "&y = " << &y << endl; // &y = 0x6afed8
    cout << "&z = " << &z << endl; // &z = 0x6afedc
    cout << " a = " << a << endl; // a = 0x6afed4
    cout << " b = " << b << endl; // b = 0x6afedc
    cout << b-a << endl; // 2
}
```

### Dynamic Array

- **Dynamic Array** variable of type T is a **pointer** to the **starting address** of consecutive block of type T
- Let A be a dynamic array of int
  - ▶ A[x] refer to the  $x^{\text{th}}$  block starting from A
- Static array works in the same way, that's why accessing A[x] is very fast for an array
  - ▶ It just refers to the address A[x] is  $*(A + x * \text{sizeof}(T))$

### new and delete operator

- For a datatype T, new T *allocates* a block with a size of T and then call a *constructor* of T, return the address of that memory

- For a datatype T, `new T[n]` allocates n blocks of T and then call a *constructor* of T in each block, return the address of the first block
  - For example, we use `new int [10]` to create a dynamic int array of 10 elements
- For a pointer X, `delete X` calls the *destructor* and then *de-allocates* memory pointed by X
- For a dynamic array X, `delete []` calls the *destructor* of all blocks in the dynamic array and *de-allocates* the memory allocated by X

### Example

```
class test {
public:
    // constructor
    test() : data() {cout << "created" << endl;}
    // destructor
    ~test() {cout << data << " destroyed " << endl;}
    int data;
};

int main() {
    test *a, *b;    // pointers of test

    a = new test;    // created new test pointed by a
    a->data = 10;    // (*a).data = 10
    cout << a->data << endl;    // 10
    delete a;        // destroyed

    b = new test[4]; // created 4x
    b[0].data = 10;
    b[1].data = 20;
    b[2].data = 30;
    b[3].data = 40;
    delete [] b;     // deleted 4x
}
```

### Memory Leak

- For everything, this is created by new, we must call delete on it
- If you do not, that memory is not deleted until all memory is used up

```
#include <iostream>
using namespace std;

void leaked() {
    int *a;
    a = new int[2000];
}

int main() {
    for (int i=0; i<1000000; i++) {
        int *a;
        cout << i << endl;
        leaked();
    }
}

// terminate called after throwing an instance of 'std::bad_alloc'
```

### Smart Pointer (NOT A SUBJECT OF THIS COURSE)

- A better way is to use C++ smart pointer `shared_ptr(T)`
- Smart pointer is a pointer that can delete itself when it go out of scope

- Similar concept to [Java Garbage Collection](#)
- Still possible to have memory leak

## vector.h

### Version 0.1

- Start with vector that can do push\_back, pop\_back, and []
- Also with custom constructor

```
namespace CP {
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {...}
    void expand(size_t capacity) {...}
    void ensureCapacity(size_t capacity) {...}
public:
    vector() {...}
    vector(size_t capacity) {...}
    ~vector() {...}
    // access
    T& at(int index) {...}
    T& operator[](int index) {...}
    // modifier
    void push_back(const T& element) {...}
    void pop_back() {...}
};
}
```

### Basic Constructor

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

public:
    vector() {
        int cap = 1;
        mData = new T[cap]();
        mCap = cap;
        mSize = 0;
    }

    vector(size_t cap) {
        mData = new T[cap]();
        mCap = cap;
        mSize = cap;
    }
}
```

```
vector<int> v;
```

```
vector<int> w(5);
```

|       |      |       |  |       |      |             |
|-------|------|-------|--|-------|------|-------------|
| mSize | mCap | mData |  | mSize | mCap | mData       |
| 0     | 1    | [ ]   |  | 5     | 5    | [0,0,0,0,0] |

### Destructor

- Since we new mData, we have to delete it
  - Or face a memory leak problem

```
~vector() {
    delete [] mData;
}
};
```

### Object Life Cycle

- Normal object
  - Object is created (constructor called) when declared
  - Object is destroyed (destructor called) when go out of scope
- Object created by new (both new T or new T[]) Object is created when new Object is destroyed when delete

```
class test {
public:
    // constructor
    test() : data() {cout << "created" << endl;}
    // destructor
    ~test() {cout << data << " destroyed " << endl;}
    int data;
};

int main() {
    cout << "-- Life cycle --" << endl;
    cout << "- normal cycle -" << endl;
    test u;
    u.data = 99;
    for (int i=0; i<5; i++) {
        test t;
        t.data = i*10;
    }
}
```

### Output

```
-- Life cycle --
- normal cycle -
created      <-- u
created
0 destroyed
created
10 destroyed
created
20 destroyed
created
30 destroyed
created
40 destroyed
99 destroyed <-- u
```

### Accessing Data

- The return type is T& which is a reference
- Same deal as pass-by-reference, this is called **return-by-reference**
  - So we can do `v[i] = 30` or `v[i]++`
- What is returned is actually that variable
- Also notice the difference between `at()` and `operator[]`

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {
        if (n < 0 || (size_t)n >= mSize) {
            throw std::out_of_range("index out of range");
        }
    }

public:
    T& at(int index) {
        rangeCheck(index);
        return mData[index];
    }

    T& operator[](int index) {
        return mData[index]; // return the variable
    }
};
```

### Add, remove data

- `push_back` first check if we have reserved space
  - If not, we expand
- Then, the data is put to `mData[mSize]`
- Removing Data is done by just reduce the size

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    void expand(size_t capacity) {
        T *arr = new T[capacity](); // create a new dynamic array
        for (size_t i = 0; i < mSize; i++) {
            arr[i] = mData[i]; // move all data
        }
        delete [] mData;
        mData = arr; // delete old data and point to new one
        mCap = capacity;
    }

    void ensureCapacity(size_t capacity) {
        if (capacity > mCap) {
            size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap; // double the size
            expand(s);
        }
    }
};
```

```

    }
}
public:
    void push_back(const T& element) {
        ensureCapacity(mSize+1);
        mData[mSize++] = element; // add size and put element
    }
    void pop_back() {
        mSize--;
    }
};

```

## Vector Constructors

### Problem of vo.1

- Copy constructor and assignment operator is incorrect
  - It is auto generate to copy all variables (but not the data it points to)
- Rule of three in c++
  - Consider destructor, copy constructor, assignment operator
  - If any of them is written in the code, we mostly need all of them
- Since c++11, it's rule of four an a half

```

int main() {
    CP::vector<int> w(5);

    for (int i=0; i<5; i++) w[i] = i*10;
    CP::vector<int> x(w); // this creates shallow copy of w
    CP::vector<int> y = w;
    x[3] = -1; // changes both w and y
    cout << y[3] << endl; // -1
    cout << w[3] << endl; // -1
}

```

### vo.2, add small access functions

- empty and size also exists in other data structure
- size\_t is non-negative integer type

```

bool empty() const {
    return mSize == 0;
}

size_t size() const {
    return mSize;
}

size_t capacity() const {
    return mCap;
}

```

### vo.2 adding copy constructor & assignment operator

```

// copy constructor
vector(const vector<T>& a) {
    mData = new T[a.capacity()]();
    mCap = a.capacity(); // copies only value
    mSize = a.size();
    for (size_t i=0; i<a.size(); i++) {
        mData[i] = a[i];
    }
}

```

```

    }
}

// copy assignment operator
vector<T>& operator=(vector<T> &other) {
    // protect against self-destruct
    if (mData != other.mData) {
        // delete current data
        delete [] mData;
        // copy the new data
        mData = new T[other.capacity()]();
        mCap = other.capacity();
        mSize = other.size();
        for (size_t i=0; i<a.size(); i++) {
            mData[i] = a[i];
        }
    }
}

```

### Copy-and-swap

- Utilize written copy-constructor and destructor
- Shorter code

```

// copy assignment operator using copy-and-swap idiom
vector<T>& operator=(vector<T> other) { // notice the pass-by-value!!!
    // other is copy-constructed which will be destruct at the end of this scope
    // we swap the content of this class to the other class and let it be destructed
    using std::swap;
    swap(this->mSize, other.mSize);
    swap(this->mCap, other.mCap);
    swap(this->mData, other.mData);
    return *this;
}

```

### Iterator, Insert, Erase, & Analysis

#### vo.3 Iterator and typedef keyword

```

template <typename T>
class vector {

protected:
    T *mData;
    size_t mCap;
    size_t mSize;
protected:
    typedef T* iterator; // let iterator be type name of T pointer

    // iterator
    iterator begin() {
        return &mData[0];
    }

    iterator end() {
        return begin()+mSize;
    }
}

```

- See that pointer works just like how `std::vector::iterator` works
- In fact, iterator is actually a pointer
- typedef keyword allows us to map a type name
  - `CP::vector<int>::iterator` is `int*`
  - `CP::vector<bool>::iterator` is `bool*`

### insert

- `push_back` actually call `insert(end(), element)`
- Question: why we need `pos`?

```
iterator insert(iterator it, const T& element) {
    size_t pos = it - begin();
    ensureCapacity(mSize+1);
    for (size_t i = mSize; i > pos; i--) {
        mData[i] = mData[i-1];
    }
    mData[pos] = element;
    mSize++;
    return begin()+pos;
}

void push_back(const T& element) {
    insert(end(), element);
}
```

### erase

- See that both `insert` and `erase` also change `mSize`

```
void erase(iterator it) {
    while((it+1)!=end()) {
        *it = *(it+1);
        it++;
    }
    mSize--;
}
```

### Exercise

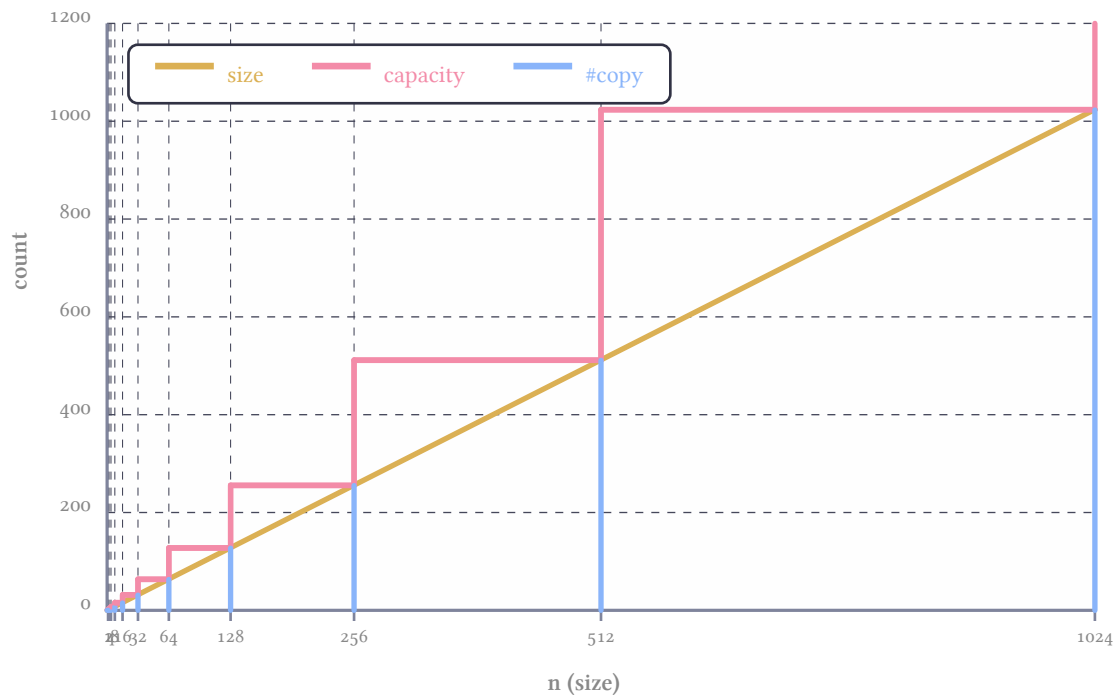
- Read the following function and see how it works in `vector.h`
  - `resize`, `clear`
  - non-stl function
    - `insert_by_pos`
    - `erase_by_pos`
    - `erase_by_value`
    - `contains`
    - `index_of`
- Read in [here](#)

### Analysis of how many data is copied by `push_back`

- When full, `push_back` have to move all data to a new dynamic array
- `ensureCapacity` double the size

### Size and Capa & Copy Count

- How much copy we need?



## Lecture 10

### Stack

#### Intro

- Now we will create less complex data structure `CP::stack`
- Just like a vector without iterator, insert, erase, resize, at end operator[]
  - Add `top()` which is just a shorthand of looking at the last element
- That's it, really

#### Key Idea

- The data is stored in the same way as a vector
  - The first element of `mData` is the bottom of stack while the last element is the top of the stack
- We just take `vector.h` and remove unnecessary function

```

| | stack
| |
| A | mSize mCap mData
| X | 4 5 [A, B, X, A]
| B |
| A |
|__|

```

#### stack.h

```

namespace CP {
    template <typename T>
    class stack {
    protected:
        T *mData;
        size_t mCap;
        size_t mSize;
        void expand(size_t capacity) {...}
        void ensureCapacity(size_t capacity) {...}
    };
}

```

```

public:
    // constructor
    stack(const stack<T>& a) {...}
    stack() {...}
    stack<T>& operator= {...}
    ~stack() {...}
    // capacity operation
    bool empty() const {...}
    size_t size() const {...}
    // access
    const T& top() const {
        return mData[mSize-1];
    }
    // modifier
    void push(const T& operator) {...} // This is push_back
    void pop() {...} // This is pop_back
};
}

```

### Speed of each operation

- All read operation always take constant time
  - `size()`, `top()` simply return something that is directly accessible
- All modify operation also take constant time
  - `push()` is constant on average (same as `push_back` of vector)
  - `pop()` is always constant

### Stack by Vector

- Instead of writing our own function, there is another way to write a stack
- We simply use `vector` as our sole data member
- Benefit: `code reuse`
- Drawback: `almost none` except that we need one more layer of function call

```

namespace CP {
    template <typename T>
    class stack {
    protected:
        vector<T> v;
    public:
        // default constructor
        stack() : v() { }
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size(); }
        // access
        const T& top() const { return v.back(); }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { v.pop_back(); }
    };
}

```

## Lecture 11

### CP::queue

Will the circle be unbroken?

#### Intro

- Queue, unlike stack, require more sophisticated technique to achieve fast performance
- We start by writing a simple class that just work (slowly)
- Then we try to improve it

#### Key Idea

- Just like stack, we will use the same format as vector, using dynamic array to store data
- However we have to somehow manage how we works with `front()` and `back()` of the queue

#### vo.1 simple implementation of the queue

- To illustrate this idea, we will use a **vector** as our data member
- `push(e)` is simply `v.push_back(e)`, this is fast
- `front()` is `v[0]`, `back()` is `v[v.size()-1]`, this is also fast
- `pop()` is `v.erase(v.begin())`, this is slow (always proportional to `v.size()`)
  - Unlike `std::queue` which has very fast `pop()`

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        std::vector<T> v;
    public:
        // default constructor
        queue() : v() { }
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size(); }
        // access
        const T& front() const { return v[0]; }
        const T& back() const { return v[v.size()-1]; }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { v.erase(v.begin()); }
    };
}
```

#### vo.1 example

```
CP::queue<int> q;      q  v
                        mSize  mCap  mData
                        0      1      ---->  [ ]

q.push(10);            q  v
                        mSize  mCap  mData
                        1      1      ---->  [10]

q.push(20);            q  v
                        mSize  mCap  mData
                        2      2      ---->  [10, 20]

q.push(30);            q  v
                        mSize  mCap  mData
                        3      4      ---->  [10, 20, 30, ]

q.pop();               q  v
                        /      /
```

```

mSize  mCap  mData    /    /
  2      4      ----> [20, 30, , ]

```

### vo.2 faster queue

- Add more data member mFront, initialized as 0
- push(e) is simply v.push\_back(e), this is fast
- front() is v[mFront], back() is v[v.size()-1], this is also fast
- pop() is mFront++, this is fast
  - However, we don't really remove anything when pop

```
#include <vector>
```

```

namespace CP {
    template <typename T>
    class queue {
    protected:
        std::vector<T> v;
        int mFront;
    public:
        // constructor
        queue() : v(), mFront() {}
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size() - mFront; }
        // access
        const T& front() const { return v[mFront]; }
        const T& back() { return v[v.size()-1]; }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { mFront++; }
    };
}

```

### vo.2 example

```

CP::queue<int> q;      q      v
                        mFront mSize mCap mData
                        0      0      1      ----> [ ]

q.push(10);            q      v
                        mFront mSize mCap mData
                        0      1      1      ----> [10]

q.push(20);            q      v
                        mFront mSize mCap mData
                        0      2      2      ----> [10, 20]

q.push(30);            q      v
                        mFront mSize mCap mData
                        0      3      4      ----> [10, 20, 30, ]

q.pop();               q      v
                        mFront mSize mCap mData
                        1      3      4      ----> [10, 20, 30, ]

```

### Problem with vo.2

- Fast but **use too many space**
- Queue grows according to **how many time push is called**
  - regardless of how many pop is called
- The data stored in the vector can be **much larger** than the actual data in the queue
- Does not really work in real world

```
for (int i=0; i<1000000; i++) {
    q.push(i);
    q.pop();
}
std::cout << q.size() << std::endl;
```

## Circular Queue

### Final Idea

| mFront      | mSize | mCap | mData | *                         |
|-------------|-------|------|-------|---------------------------|
| 2           | 2     | 4    | ----> | [ , , 30, 40]             |
| q.push(50); |       |      |       |                           |
| mFront      | mSize | mCap | mData | *                         |
| 2           | 3     | 4    | ----> | [50, , 30, 40]            |
| q.push(60); |       |      |       |                           |
| mFront      | mSize | mCap | mData | *                         |
| 2           | 4     | 4    | ----> | [50, 60, 30, 40]          |
| q.pop();    |       |      |       |                           |
| mFront      | mSize | mCap | mData | *                         |
| 3           | 3     | 4    | ----> | [50, 60, , 40]            |
| q.push(99); |       |      |       |                           |
| mFront      | mSize | mCap | mData | *                         |
| 3           | 4     | 4    | ----> | [50, 60, 99, 40]          |
| q.push(1);  |       |      |       |                           |
| mFront      | mSize | mCap | mData | * \ \ \                   |
| 0           | 5     | 4    | ----> | [40, 50, 60, 99, 1, , , ] |

- We take vo.2 and **reuse** the area at the beginning of mData
  - Expand when necessary
  - Re-arrange when expand

### Circular Queue

- We can think of mData to be circular
  - End of the last element of the mData is connected to the first element
- Consider  $i^{\text{th}}$  element
  - the next element is  $(i+1) \% \text{mCap}$
  - The previous element is  $(i-1+\text{mCap}) \% \text{mCap}$
  - Next k element is  $(i+k) \% \text{mCap}$

## Circular Queue Implementation

### queue.h

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        T *mData;
        size_t mCap;
        size_t mSize;
        size_t mFront;
        void expand(size_t capacity) {...}
        void ensureCapacity(size_t capacity) {...}
    public:
        // constructor - almost the same but have to take care of mFront
        queue(const queue<T>& a) {...}
        queue() {...}
    };
}
```

```

    queue<T>& operator=(queue<T> other) {...}
    ~queue() {...}
    // capacity function - same as vector
    bool empty() const {...}
    size_t size() const {...}
    // access - circular queue implementation
    const T& front() const {...}
    const T& back() const {...}
    // modifier
    void push(const T& element) {...}
    void pop() {...}
};
}

```

### Ctor, Dtor, copy

```

template <typename T>
class queue {
protected:
    T *mData;  size_t mCap;  size_t mSize;  size_t mFront;
public:
    // default constructor
    queue() : mData(new T[1]()), mCap(1),
             mSize(0), mFront(0) { }
    // copy constructor
    queue(const queue<T>& a) : mData(new T[a.mCap]()), mCap(a.mCap),
                             mSize(a.mSize), mFront(a.mFront) {
        for (size_t i = 0; i < a.mCap; i++) {
            mData[i] = a.mData[i];
        }
    }
    // copy assignment operator
    queue<T>& operator=(queue<T> other) {
        using std::swap;
        swap(mSize, other.mSize);
        swap(mCap, other.mCap);
        swap(mData, other.mData);
        swap(mFront, other.mFront);
        return *this;
    }
    ~queue() {
        delete [] mData;
    }
};

```

### front(), back(), pop()

```

template <typename T>
class queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
public:
    // access
    const T& front() const {
        return mData[mFront];
    }
};

```

```

    }
    const T& back() const {
        return mData[(mFront + mSize - 1) % mCap];
    }
    // modify
    void pop() {
        mFront = (mFront + 1) % mCap;
        mSize--;
    }
};

```

- back = mFront + mSize - 1
  - Also circular (by % mCap)
- pop = move mFront by 1
  - Also circular
  - Also change size

### push, expand

- push add data to (mFront + mSize) % mCap
  - The space just after back()
- Expand re-pack the mData so that mFront is 0
- ensureCapacity is the same

```

template <typename T>
class queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
    void expand(size_t capacity) {
        T *arr = new T[capacity]();
        for (size_t i = 0; i < mSize; i++) {
            arr[i] = mData[(mFront + i) % mCap];
        }
        delete [] mData;
        mData = arr;
        mCap = capacity;
        mFront = 0;
    }
    void ensureCapacity(size_t capacity) {
        if (capacity > mCap) {
            size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
            expand(s);
        }
    }
public:
    void push(const T& element) {
        ensureCapacity(mSize+1);
        mData[(mFront + mSize) % mCap] = element;
        mSize++;
    }
};

```

### Analysis

- All access, modification is fast (constant time)

- Space is re-used
  - It is not shrunk when mSize reduce
  - Space is not more than double of maximum mSize during its lifetime

### Queue Exercise

- We implement circular queue by maintain mFront and use circular logic ( $\% \text{ mCap}$ ) to calculate the position of back of the queue
  - Can we maintain mBack instead?
  - Can we maintain both mFront and mBack but not mSize?
- How about mCap, if we know mFront, mSize, mBack, can we calculate mCap?

| mFront | mSize | mBack | front()   | back()                     | size()  |
|--------|-------|-------|-----------|----------------------------|---------|
| YES    | YES   | No    | v[mFront] | v[(mFront + mSize) % mCap] | mSize() |
| No     | YES   | YES   | ????      | v[mBack]                   | mSize   |
| YES    | No    | YES   | v[mFront] | v[mBack]                   | ????    |

### Answer

- Use  $\text{front}() = v[(\text{mCap} + \text{mBack} - \text{mSize}) \% \text{mCap}]$
- Use  $\text{size}() = (\text{mCap} + \text{mBack} - \text{mFront}) \% \text{mCap}$ , but this will break when  $\text{mBack} == \text{mFront}$  because it can't determine whether the queue is empty or full.
- Impossible, mCap can be scaled to any size larger than the current elements.

### Now, meet deque

- Can you modify queue to include
  - `push_front()`, add to the front of the queue
  - `pop_back()`, remove from back of the queue
- All operation should still be constant time

### Answer

```
void push_front(const T& element) {
    ensureCapacity(mSize+1);
    mFront = (mCap + mFront - 1) % mCap;
    mData[mFront] = element;
    mSize++;
}
void pop_back() {
    mSize--;
}
```

## Lecture 12

### Complexity Analysis

How fast is our code?

#### Overview

- Introducing a **measurement** of efficiency of a program
- Why we need measurement
- How we measure
  - How to analyze our code to get a measurement
- What is measurement
  - **Asymptotic Notation**

### Key Idea

- We need a useful way to describe efficiency of our code
  - Useful = able to be **easily use to predict** how much resource (time / memory) that our program will need
  - Useful = **not overly complex** in analysis
  - Need to balance between usefulness and complexity
- Ultimately, we introduce a **class of efficiency** that says how our code use resource with respect to size of data
  - Focus on **on growth of resource usage**

### Preview

```
int find_max(vector<int> v) {
    int m = v[0];
    for (size_t i = 0; i < v.size(); i++)
        if (v[i] > m)
            m = v[i];
    return m;
}
```

- This code takes time **directly proportional** to **the size of the data**
  - Size  $N$  takes time  $T$
  - Size  $5N$  should take time  $5T$

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = 0; j < v.size(); j++)
            if (i != j && v[i] + v[j] == k)
                count++;
    return count/2;
}
```

- This code takes time **directly proportional** to **the square of the size of the data**
  - Size  $N$  takes time  $T$
  - Size  $5N$  should take time  $25T$

### Why don't we use real world clock?

- Ultimately, we want to know **how long our program takes** to do each operation
  - Use in design, how much resource we need
  - Help use **choose appropriate data structure**
- **Real world clock measurement is possible** but has **many drawback**
  - System dependency
  - Too complex (we have to build our system)
  - Too specific

### Dependency of System

- Same program in different system = different time
  - CPU
  - RAM
  - Compiler
  - Operating parameter
    - Heat? Other running program?

### How to measure

- Counting instruction
  - Depend on code only
- Dependency on size of data
  - Focus on large data

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = 0; j < v.size(); j++)
            if (i != j && v[i] + v[j] == k)
                count++;
    return count/2;
}
```

// 1  
 // 2 + n + n ( n = v.size() )  
 // (2 + n + n) \* n  
 // 2 \* n \* n  
 // 1 \* ??? ( <= (d) )  
 // 1

```
}
```

```
// Total = 1 + 2+n+n + (2+n+n)n + 2n^2 + n^2 + 1
//          = 4 + 4n + 5n^2
```

### Time per instruction

- In reality, different CPU instructions use different time
- Same instruction but different CPU also use different number of cycle
- **However, we just ignore it**
  - For now

### Focusing on large data

- In many cases, the size of the data we are working with will affect the time our code use
- Large data usually mean longer time
- What matter is when the data is large

### Growth rate simplifies analysis

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = i+1; j < v.size(); j++)
            if (v[i] + v[j] == k)
                count++;
    return count;
}
```

// i=0: 1 + n-0 + n-1  
 // i=1: 1 + n-1 + n-2  
 // <-- i=2: 1 + n-2 + n-3  
 // . . .  
 // i=n-1: 1 + 1 + 0  
 // sum =  $\Sigma(1+2n-2i-1) = n^2 + n$

```
// Total = 1 + 2+n+n + n^2+n + n^2 + n^2 + 1
//          = 4 + 3n + 3n^2
```

### Small Detail

| n      | 5n <sup>2</sup> + 4n + 4 |          | 3n <sup>2</sup> + 3n + 4 |          | n <sup>2</sup>         |        |
|--------|--------------------------|----------|--------------------------|----------|------------------------|--------|
|        | raw                      | growth   | raw                      | growth   | raw                    | growth |
| 10     | 544                      |          | 334                      |          | 100                    |        |
| 20     | 2084                     | 3.830 88 | 1264                     | 3.784 43 | 400                    | 4      |
| 40     | 8164                     | 3.917 47 | 4924                     | 3.895 57 | 1600                   | 4      |
| 80     | 32 324                   | 3.959 33 | 19 444                   | 3.948 82 | 6400                   | 4      |
| 160    | 128 644                  | 3.979 83 | 77 284                   | 3.9747   | 25 600                 | 4      |
| 320    | 513 284                  | 3.989 96 | 308 164                  | 3.987 42 | 102 400                | 4      |
| 640    | 2 050 564                | 3.994 99 | 1 230 724                | 3.993 73 | 409 600                | 4      |
| 1280   | 8 197 124                | 3.9975   | 4 919 044                | 3.996 87 | 1 638 400              | 4      |
| 2560   | 32 778 244               | 3.998 75 | 19 668 484               | 3.998 44 | 6 553 600              | 4      |
| 5120   | 1.31 × 10 <sup>8</sup>   | 3.999 37 | 78 658 564               | 3.999 22 | 26 214 400             | 4      |
| 10 240 | 5.24 × 10 <sup>8</sup>   | 3.999 69 | 3.15 × 10 <sup>8</sup>   | 3.999 61 | 1.05 × 10 <sup>8</sup> | 4      |
| 20 480 | 2.1 × 10 <sup>9</sup>    | 3.999 84 | 1.26 × 10 <sup>9</sup>   | 3.9998   | 4.19 × 10 <sup>8</sup> | 4      |

When counting instruction, it is usually OK to focus on **most executed line**

## Measurement by Growth Rate

### What?

- Growth rate = how much resource usage growth with respect to change of input
  - Resource usage = number of instruction used
  - Input = size of data
- Emphasizes long term trend

### Why?

- System independent
  - The result can be used to predict behavior on any system
- Focus on change of resource usage with respect to size of input
- Can regard small detail
  - Simple to calculate
  - Applicable in real world

## Asymtotic Notation

Classification of growth rate

### Overview

- Formally, it is a **set of function** having the **growth rate** related to something
- The definition focus **on growth of the function** while **disregard small detail**
- Also provide some workaround on dependency of value of input

### What is?

- A kind notation written as  $O(f(n))$ 
  - $O$  can be one of  $\mathcal{O}, \Theta, \Omega, \sigma, \omega$
  - $f(n)$  is some expression
- Example  $\mathcal{O}(n)$  or  $\Theta(n^2 + 3)$  or  $\omega(n^2 \log(n))$
- Usage
  - “This code is  $\mathcal{O}(n)$ ” (read as Big-Oh of n)
  - “This function takes time in  $\Theta(\log(n))$ ” (read as Big-theta of log n)
  - “Time complexity of this program is  $\mathcal{O}(n^2)$ ”

### Meaning

- “A is  $\Theta(f(x))$ ” means the **growth rate of A is equal** to the **growth rate of  $f(x)$**
- “B is  $\mathcal{O}(f(x))$ ” means the **growth rate of B is less than or equal** to the **growth rate of  $f(x)$**
- For  $\Omega, \sigma, \omega$ , (**Big-Omega**, **little-oh**, **little-omega**), we won't use it for now but the meaning is similar (which are **more than or equal**, **less than**, **more than**, respectively)
- Conventionally, we usually use  **$N$  for the size of the data**

### Usage

- Let  **$f(n)$**  be the number of instruction needed by **code A** when the **size of data is  $n$**
- We will calculate asymptotic notation that  **$f(n)$**  is a member of
  - Find  $g(n)$  such that  $f(n)$  is  $\mathcal{O}(g(n))$  or  $\Theta(g(n))$  (or  $\Omega(g(n))$  or ...)
- Let's say we have analyze that  **$f(n)$  is  $\mathcal{O}(g(n))$** 
  - We now understand that the growth rate of instruction required by  $n$  grows slower or the same as how  $g(n)$  grow

### Comparing growth rate of $f(n)$ and $g(n)$

- The relation of growth rate of  $f(n)$  and  $g(n)$  depends on the value of  $f(n)/g(n)$  when  $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \begin{cases} 0 & \text{if } f(n) \text{ grows slower than } g(n) \\ c & \text{if } f(n) \text{ grows similarly to } g(n) \\ \infty & \text{if } f(n) \text{ grows faster than } g(n) \end{cases}$$

### Example

- $f(n) = 4 + 3n + 4n^2$

- $g(n) = n^2$

$$\lim_{n \rightarrow \infty} \frac{4n^2 + 3n + 4}{n^2} = \lim_{n \rightarrow \infty} \left( 4 + \frac{3}{n} + \frac{4}{n^2} \right) = 4$$

- Hence  $f(n)$  grows similarly to  $g(n)$ 
  - Therefore,  $f(n) = \Theta(n^2)$

#### Another Example

- $f(n) = 0.00005n^2$
- $g(n) = 100000n$

$$\lim_{n \rightarrow \infty} \frac{0.00005n^2}{100000n} = \lim_{n \rightarrow \infty} 10^{-10}n = \infty$$

- Hence  $f(n)$  grows faster than  $g(n)$  (also means  $g(n)$  grows slower than  $f(n)$ )
  - Therefore  $g(n) = \mathcal{O}(n^2)$
  - and  $f(n) = \Omega(100000n)$

#### Dependency on the value of input

- Consider `vector::insert(iterator it, T value)`
- The time it takes depends on both the size of the vector and the value of it
  - Larger size  $\rightarrow$  more time
  - Closer to `end()`  $\rightarrow$  less time

#### Big-O describes upper bound

- `vector::insert` is  $\mathcal{O}(n)$ 
  - Its growth rate does not exceed  $n$
  - There are cases that it may grow less than  $n$  (insert at `end()`)
- This is very useful in real world
  - Knowing maximum load
  - Not overly complex in analysis

#### Big-O Example

- Find is  $\mathcal{O}(n)$ 
  - At worst, it can't find value and a & b points to `begin()` & `end()`
    - This case, find growth rate is  $n$
  - At best, it always find value at the first position (a)
    - This case, find growth as 1

```
bool find(iterator a, iterator b, T value) {
    while (a < b) {
        if (*a == value)
            return true;
        a++;
    }
    return false;
}
```

#### L'Hôpital's Rule

$$\lim_{n \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{n \rightarrow \infty} \frac{f'(x)}{g'(x)}$$

If  $f(x), g(x)$  must be diffable,  $g(x)$  is non-zero,  $\lim_{n \rightarrow \infty} f'/g'$  exists

- Can help
- $F(n) = \log n$

- $g(n) = n^{0.5}$

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{\log(n)}{\sqrt{n}} &= \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(10)\sqrt{n}} \\
 &= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{\ln(n)}{\sqrt{n}} \\
 &= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{2\sqrt{n}}} \\
 &= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{2}{\sqrt{n}} \\
 &= 0
 \end{aligned}$$

### Exercise

- $f(n) = \lg^c n$
- $g(n) = n^k$
- We know that  $c > 0, k > 0$
- Does  $f(n)$  grow slower than  $g(n)$ ?

Show that  $\lg^c n$  is  $\mathcal{O}(n)$ ;

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{\lg^c n}{n} &= \lim_{n \rightarrow \infty} \frac{c \lg^{c-1} n}{n} \\
 &= \lim_{n \rightarrow \infty} \frac{c(c-1) \lg^{c-2} n}{n} \\
 &\vdots \\
 &= \lim_{n \rightarrow \infty} \frac{c!}{n} \\
 &= 0
 \end{aligned}$$

Since  $n$  is  $\mathcal{o}(n^k)$  for  $k > 1$ , therefore  $\lg^c n$  is  $\mathcal{o}(n^k)$ .

## Big Theta

### Big-Theta is tight bound

- `std::count` always go through entire array
- Regardless of the value in the array, it always perform `if (*first == value)`

```

size_t count(iterator first, iterator last, const T& value) {
    size_t ret = 0;           // 1
    for (; first != last; ++first) { // n + n
        if (*first == value)    // 1 * n
            ret++;              // 1 * ??? ( <= (d) )
    }
    return ret;               // 1
}

```

```

// Total = 1 + n+n + 1*n + 1*n + 1
//         = 2 + 4n

```

### More Example

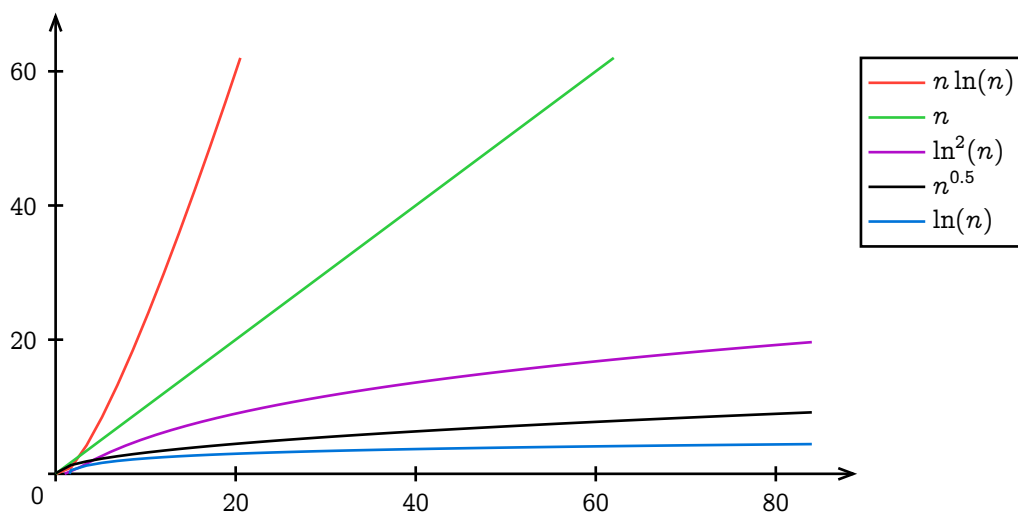
| $\Theta(1)$ | $\Theta(n)$    | $\Theta(n^2)$                            |
|-------------|----------------|------------------------------------------|
| $f(n) = 5$  | $f(n) = n$     | $f(n) = n^2$                             |
| $f(n) = 0$  | $f(n) = n + 3$ | $f(n) = \binom{n}{2} = \frac{n(n-1)}{2}$ |
| $f(n) = c$  |                | $f(n) = 400n^2 + an + b$                 |

|  |                                                 |  |
|--|-------------------------------------------------|--|
|  | $f(n) = \frac{n}{1200} + 86$ $f(n) = 40000000n$ |  |
|--|-------------------------------------------------|--|

Observation: multiplicative and addition constants in  $f(n)$  can usually be ignored, since it will be disregarded by  $\lim$ , Degree cannot.

### Well known growth rate class

|                               |               |
|-------------------------------|---------------|
| $\Theta(1)$                   | Constant      |
| $\Theta(\log(n))$             | Logarithm     |
| $\Theta(\log^c(n)), c \geq 1$ | Polylogarithm |
| $\Theta(n^a), 0 < a < 1$      | Sublinear     |
| $\Theta(n)$                   | Linear        |
| $\Theta(n \log(n))$           | Linearithmic  |
| $\Theta(n^2)$                 | Quadratic     |
| $\Theta(n^c), c \geq 1$       | Polynomial    |
| $\Theta(c^n), c > 1$          | Exponential   |
| $\Theta(n!)$                  | Factorial     |



$\ln^2(n)$  grows slower than  $n^{0.5}$ . At around  $n = 56000$ ,  $\ln^2(n)$  is less than  $n^{0.5}$ .  
 $x^4$  grows slower than  $2^n$ . At around  $n = 16$ ,  $x^4$  is less than  $2^n$ .

### Beware

- It is **wrong** to say that `vector::insert` is  $\Theta(n)$ 
  - Because there is a case that it grows slower than  $n$
- It is **wrong** to say that `vector::push_back` is  $\Theta(n)$ 
  - Because there is a case that it grows slower than  $n$
- It is **ok** to say that `std::count` is  $\mathcal{O}(n)$ 
  - Because while it always grows as  $n$ , it does not grow faster than  $n$
  - $\mathcal{O}(n)$  is upper bound
  - But it is better to say that `std::count` is  $\Theta(n)$

### How to analyze using asymptotic notation

1. Write a code
2. Calculate the function  $f(n)$  that counts the number of instruction of the code when the data is size  $n$ 
  - Usually, just focus on **most executed line**

3. Find  $g(n)$  and a notation  $X$  such that  $f(n)$  is  $X(g(n))$
- It's either **Big-O** or **Big-Theta**. If there is a case that it can grow slower than  $G(n)$ , use Big-O

### Another Example

- Let's analyze `vector::push_back`

```

iterator insert(iterator it, const T& element) {
    size_t pos = it - begin();
    ensureCapacity(mSize + 1);
    for (size_t i = mSize; i > pos; i--) {
        mData[i] = mData[i-1]; // <----- Most executed line: 1 (b)
    }
    mData[pos] = element;
    mSize++;
    return begin()+pos;
}

void push_back(const T& element) {
    insert(end(), element);
}

void expand(size_t capacity) {
    T *arr = new T[capacity]();
    for (size_t i = 0; i < mSize; i++) {
        arr[i] = mData[i]; // <----- Most executed line: 0, 0, 0, 0, 2, 4, 8, 16 (a)
    }
    delete [] mData;
    mData = arr;
    mCap = capacity;
}

void ensureCapacity(size_t capacity) {
    if (capacity > mCap) {
        size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
        expand(s);
    }
}

// (a) can be 0 to n
// (b) can also be 0 to n
// Best case 0 + 0
// Worst case n + n
// vector::push_back is O(n)

```

### Another Definition for $\mathcal{O}$ and $\Theta$

- Using set builder notation
- $\mathcal{O}(g(n)) = \{f(n) \mid \text{there exists } c > 0 \text{ and } n_0 \geq 0 \text{ such that } f(n) \leq cg(n) \text{ for } n \geq n_0\}$
- $\Theta(g(n)) = \{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1g(n) \leq f(n) \leq c_2g(n) \text{ for } n \geq n_0\}$
- The result is the same as definition using `lim`

### Summary

- We use Asymptotic Notation to describe efficiency of a program
  - Measure instruction count instead of time
  - Focus on growth rate of instruction count
- Find most frequently executed line in the code and count it
- Maps to Big-Theta if we have tight bound
- Use Big-O if we have upper bound

## Example

**Showing**  $\log_2(n!)$  is  $\Theta(n \log_2(n))$

- Will use set definition of Big-Theta  
 $\{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1 g(n) \leq f(n) \leq c_2 g(n) \text{ for } n \geq n_0\}$
- Need to find  $c_1, c_2$  and  $n_0$

**Finding**  $c_1$  and  $c_2$

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 1$$

$$n! \leq n \times n \times n \times n \times \dots \times n$$

$$\log n! \leq \log n^n \quad \boxed{\log n^n = n \log n \text{ when } n \geq 1}$$

$$\text{Found } c_2 = 1 \text{ for upper bound} \quad \boxed{\log n! \leq n \log n \text{ when } n \geq 1}$$

$$n! = n \times (n-1) \times \dots \times \left\lfloor \frac{n}{2} \right\rfloor \times \left( \left\lfloor \frac{n}{2} \right\rfloor - 1 \right) \times \dots \times 1$$

$$n! \geq \left\lfloor \frac{n}{2} \right\rfloor \times \left\lfloor \frac{n}{2} \right\rfloor \times \dots \times \left\lfloor \frac{n}{2} \right\rfloor \times 1 \times \dots \times 1$$

$$n! \geq \left( \frac{n}{2} \right)^{n/2} \quad \boxed{\log \left( \frac{n}{2} \right)^{n/2} = \frac{n}{2} \log n - \frac{n}{2}}$$

$$\log n! \geq \log \left( \frac{n}{2} \right)^{n/2} \quad \boxed{0.5n \log n - 0.5n \geq 0.4n \log n \text{ when } n \geq 32}$$

$$\text{Found } c_1 = 0.4 \text{ for lower bound} \quad \boxed{\log n! \geq 0.4n \log n \text{ when } n \geq 32}$$

$$\boxed{0.4n \log n \leq \log n! \leq n \log n \text{ when } n \geq 32, c_1 = 0.4, c_2 = 1, n_0 = 32}$$

$$\Rightarrow \boxed{\log n! \text{ is } \Theta(n \log n) \text{ and } n \log n \text{ is } \Theta(\log n!)}$$

## Lecture 13

### Priority Queue Simple Implementation

Featuring Binary Heap

#### Overview

- Simple implementation of `priority_queue`
- Quick intro to **Graph** and **Tree**
- **Binary Heap**
- `priority_queue` with **Binary Heap**

#### `priority_queue`

- Queue by value
- Max-in-First-Out

```
int main() {
    priority_queue<int> pq;
    pq.push(4);
    pq.push(20);
    pq.push(3);

    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}
```

**Vo.1, priority\_queue by vector**

- Use **vector** to store data
- push = simply push\_back
- top, pop = find the max value and return / erase
- max\_element returns iterator to max element

```
namespace CP {
    template <typename T>
    class priority_queue {
    protected:
        std::vector<T> v;
    public:
        bool empty(); { return v.empty(); }
        bool size(); { return v.size(); }

        void push(const T &e) { v.push_back(e); }

        T top() { return *std::max_element(v.begin(), v.end()); }

        void pop() { v.erase(std::max_element(v.begin(), end())); }
    };
}
```

**max\_element**

```
iterator max_element(iterator first, iterator last) {
    if (first == last) return last;
    iterator largest = first;
    ++first;
    for (; first != last; ++first)
        if (*largest < *first)
            largest = first;
    return largest;
}
```

**Vo.1 complexities**

|       |                             |
|-------|-----------------------------|
| push  | $\mathcal{O}(1)$ *amortized |
| top() | $\Theta(n)$                 |
| pop() | $\Theta(n)$                 |

**Vo.2 faster pop, top (and push??)**

- vo.1 has many drawbacks
  - Consecutive call of top is slow (it shouldn't)
  - Both pop and top works almost the same
- vo.2 focus on **slower push** while keeps **pop, top fast**
- Make the array **sorted**
  - Max will be at the back
  - Fast pop, top

```
namespace CP {
    template <typename T>
    class priority_queue {
    protected:
        std::vector<T> v;
    public:
        bool empty(); { return v.empty(); }
```

```

    bool size(); { return v.size(); }

    T& top() { return v[v.size()-1]; }

    void pop() { v.erase(v.end()-1); }

    void push(const T &e) {
        // do something
    }
};
}

```

### vo.2 push

- Maintain that the vector is sorted at every push

```

void push(const T& e) {
    v.push_back(e); // <-- O(1)
    sort(v.begin(), v.end()); // <-- O(n log(n))
}

void push(const T& e) {
    auto it = v.begin();
    while (it < v.end() && *it <= e) // <-- O(n)
        it++;
    v.insert(it, e); // <-- O(n)
} // Total of this program is O(n)

void push(const T& e) { // upper_bound is O(log(n))
    v.insert(std::upper_bound(v.begin(), v.end(), e), e); // <-- O(n)
}

```

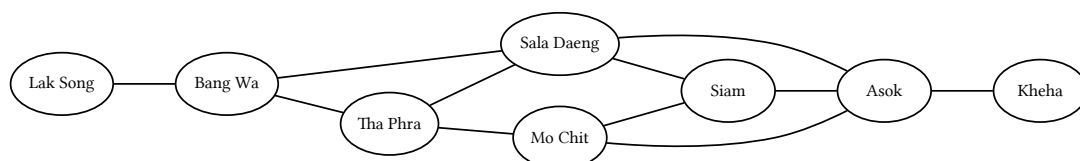
### Which one is better?

- vo.1 fast push
- vo.2 fast pop, top
- Depends on which operation we use most often
- The real version works like vo.2, we maintain some **rules** of the data that is stored in the vector such that
  - We know where max is (for fast pop)
  - Much faster push, a little bit slower pop
  - Use structure called **Binary Heap**

## Graph & Tree

### Graph

- Discrete Math Graph
- A math model that describe entities and connectivity between them



### Graph Model

- Graph consists of two things

- **Nodes (vertex, vertices)** are **things** we want to connect
- **Edges** are pairs, each pair is a **connectivity** between two nodes
- Graph  $G = (V, E)$  where  $V$  is a set of nodes and  $E$  is a set of edges

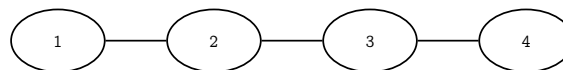
$V = \{\text{"Mo Chit", "Siam", "Asok", "Sala Daeng", "Tha Phra", "Lak Song", "Bang Wa", "Kheha"}\}$

$E =$

$\{(\text{"Mo Chit", "Asok"}), (\text{"Mo Chit", "Siam"}), (\text{"Tha Phra", "Mo Chit"}), (\text{"Sala Daeng", "Tha Phra"}), \dots\}$

### Tree

- A special kind of graph
  - Has  $N$  nodes and  $N - 1$  edges
  - Every nodes must be connected (we can start from any node and can walk through edge to reach any node)

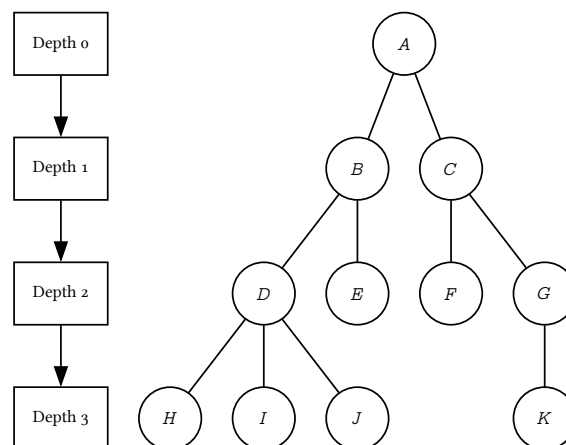


$V = \{1, 2, 3, 4\}$

$E = \{(1, 2), (2, 3), (3, 4)\}$

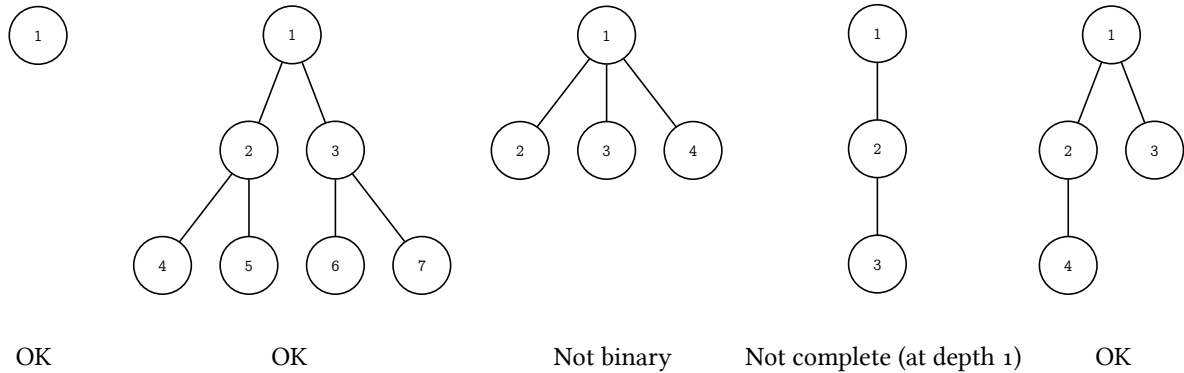
### Rooted Tree

- Tree where one node is defined as **root**
- For an edge in a rooted tree, a node that is closer to the root is called **parent** while the other node is called **child**
- **Ancestor** of node  $A$  = parent of parent ... of  $A$
- **Descendant** of node  $A$  = child of child ... of  $A$
- Root is usually drawn at the top and is considered as the **starting point**
  - Root is at level 0 (depth 0)
  - Children of root are drawn at the same level at **level 1** (depth 1)
  - Children of children of root are at **level 2** (depth 2)



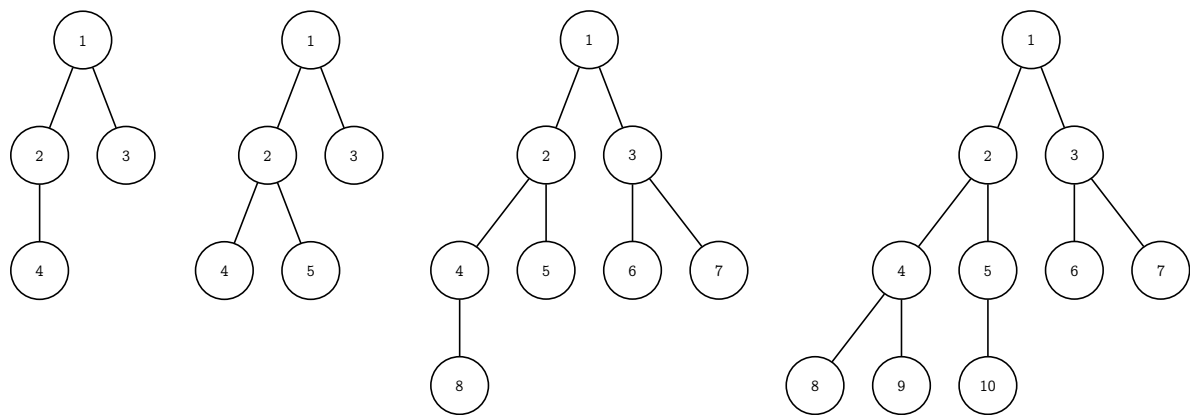
### Complete Binary Tree

- Binary Tree = a tree that every node has at most 2 children
- Complete tree = the tree must be filled with every possible node at every level (except the deepest level which must be filled as far to the left as possible)
  - Blank tree is considered a complete binary tree

**Exercise**

- Draw a Complete Binary Tree that has 4, 5, 8, 10 nodes

Hint: The answer is unique (There are exactly 1 way to draw a complete binary tree of size k)

**Special Property of a Complete Binary Tree**

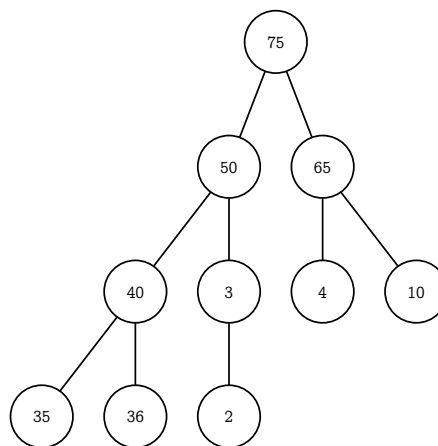
- There is exactly one way to go from any node to any node
- Maximum depth is  $\log_2 n$  where  $n$  is the number of nodes
  - Because we require completeness, and we have 2 possible children

**Binary Heap**

Use Complete Binary Tree to make priority\_queue

**Binary Heap**

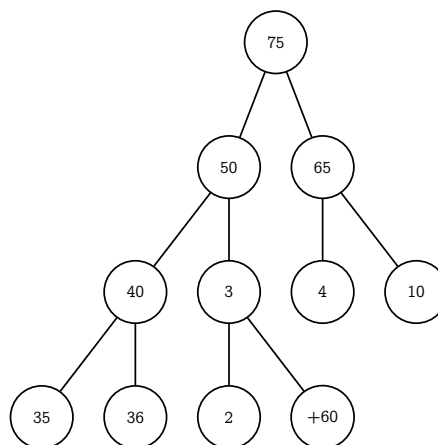
- We use **Complete Binary Tree** to store data
  - A **value** is stored at the **node**
- When data is modified (via push or pop), we must maintain these rules
  1. Tree must always be **Complete Binary Tree**
  2. For any node, its **value must be greater or equal that of its children**



### Adding data to Binary Heap

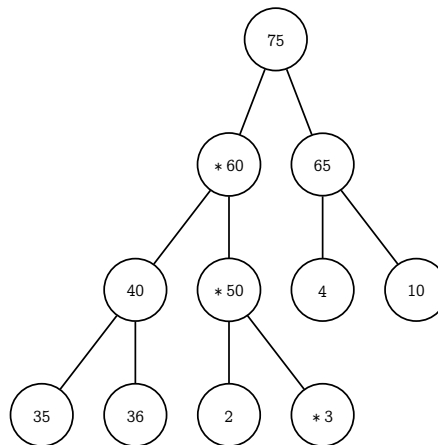
- Maintain Binary Heapness
  - structure of data
  - value of data
- Structure rules says where the new nodes should be
  - Next to **right-most child of deepest level**
  - But if we put new data there, the **value rules might be broken**
    - Fix it

push(60)



### Fix from adding a new node

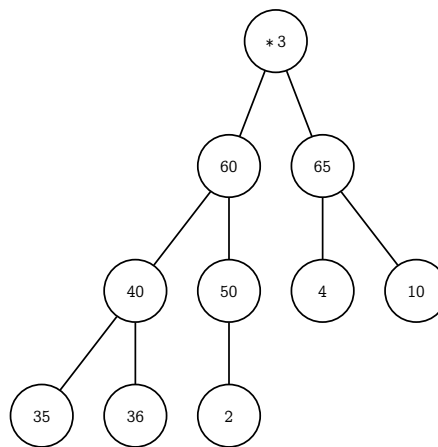
- Fix:
  - Check where we just add a node, if value rules is broken, swap with parent
  - After swap, re-check with new parent
  - Kepp doing until correct or at root



See that: after each swap, the swapped nodes does not violate with **its new children**

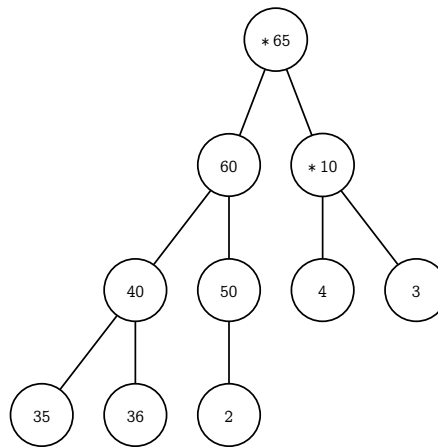
### Delete maximum data

- Similar to push, we will try to maintain structure first
- delete will remove root, find something to replace
  - Use the lowest, right-most node
- Value rules might be broken
  - Fix it



### Fix from deleting root node

- Fix:
  - Start at replaced root, if value rules is broken, swap with maximum child
  - After swap, re-check with new children
    - Beware! There is a case where we might have only one child
  - Keep doing until correct or has no children



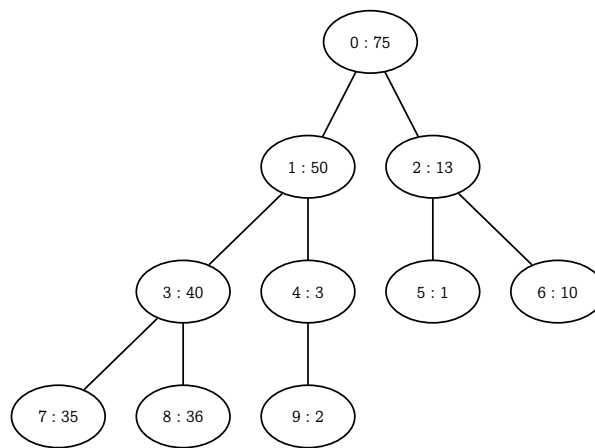
### Analysis

- How fast is push, pop
- push
  - Add to vector is  $\mathcal{O}(1)$  amortized
  - Fixing value rules is  $\mathcal{O}(h)$  where  $h$  is the maximum number of depth of the tree (we call this value tree height)
  - Notice that tree height is  $\mathcal{O}(\log(n))$
- pop
  - Fixing value rules is  $\mathcal{O}(h)$  where  $h$  is the maximum number of tree height

### Push + Fix Up

#### How to store a tree?

- Use **dynamic array**
  - Each node can be labelled from 0 to  $n-1$
- **Root** is at 0
- **Left child** of node  $i$  is at  $(i*2)+1$
- **Right child** of node  $i$  is at  $(i*2)+2$
- **Parent** of node  $i$  is at  $(i-1)/2$



Dynamic array: [75, 50, 13, 40, 3, 1, 10, 35, 36, 2]

### Layout

```

template <typename T, typename Comp = std::less<T>>
class priority_queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    Comp mLess;
    void expand(size_t capacity) {} // Fix value rules
    void fixUp(size_t idx) {}
    void fixDown(size_t idx) {}
public:
    // constructor
    priority_queue(priority_queue<T,Comp>& a);
    priority_queue(const Comp& c = Comp());
    priority_queue<T, Comp>& operator=(priority_queue<T,Comp> other);
    ~priority_queue();
    // capacity function
    bool empty() const;
    size_t size() const;
    // access
    const T& top();
    // modifier
    void push(const T& element);
    void pop();
};

```

### Constructor

- Using `list initialize`
- See that `mData` is dynamic array in the same way as vector
- `mLess` is something that is just either copied or default initialize
  - Will talk about it later

```

priority_queue(const Comp& c = Comp()):
    mData( new T[1]() ),
    mCap( 1 ),

```

```

    mSize( 0 ),
    mLess( c )
{ }

priority_queue(priority_queue<T, Comp>& a) :
    mData(new T[a.mCap]()),
    mCap(a.mCap),
    mSize(a.mSize),
    mLess(a.mLess)
{
    for (size_t i = 0; i < a.mCap; i++) {
        mData[i] = a.mData[i];
    }
}

```

### Destructor and Copy Assignment Operator

- Using standard copy-and-swap idiom

```

~priority_queue() {
    delete [] mData;
}

priority_queue<T,Comp>& operator=(priority_queue<T,Comp> other) {
    using std::swap;
    swap(mSize,other.mSize);
    swap(mCap,other.mCap);
    swap(mData,other.mData);
    swap(mLess,other.mLess);
    return *this;
}

```

### Push

- See that the **right-most child of the deepest level** is at `mData[mSize-1]` and the new node should be at `mData[mSize]`
- We do the same thing as vector's `push_back`
- Then fix the value rule

```

void expand(size_t capacity) { // Same as CP::vector
    T *arr = new T[capacity]();
    for (size_t i = 0; i < mSize; i++) {
        arr[i] = mData[i];
    }
    delete [] mData;
    mData = arr;
    mCap = capacity;
}

```

```

void push(const T& element) {
    if (mSize + 1 > mCap)
        expand(mCap * 2);
    mData[mSize] = element;
    mSize++;
    fixUp(mSize-1);
}

```

### Fix Up

- Instead of actual swap, we perform insert and find appropriate position at the same time

```
void fixUp(size_t idx) {
    T tmp = mData[idx];
    while (idx > 0) {
        size_t p = (idx - 1) / 2;
        if ( tmp < mData[p] ) break;
        mData[idx] = mData[p];
        idx = p;
    }
    mData[idx] = tmp;
}

mData p = /, idx = *, tmp = 60
0 1 2 3 4 5 6 7 8 9 10
[ 75, 50, 13, 40, /3, 1, 10, 35, 36, 2,*60]

[ 75,/50, 13, 40, *3, 1, 10, 35, 36, 2, 3]

[/75,*50, 13, 40, 50, 1, 10, 35, 36, 2, 3]

tmp = 60 < 75 = mData[p] => break;mData[idx] = tmp;
[ 75, 60, 13, 40, 50, 1, 10, 35, 36, 2, 3]
```

### mLess

- priority\_queue allows a custom comparator
- Custom comparator  $X$ 
  - We must be able to  $X(a, b)$  where  $X$  will compare  $a$  and  $b$  and return true only when  $a$  is less than  $b$
  - $X$  is a variable that implement operator()
- Initialize at constructor as variable mLess to be of type Comp in template
- Any comparison of our data (type T) must be done by mLess

```
void fixUp(size_t idx) {
    T tmp = mData[idx];
    while (idx > 0) {
        size_t p = (idx - 1) / 2;
        if ( mLess(tmp, mData[p]) ) break;
        mData[idx] = mData[p];
        idx = p;
    }
    mData[idx] = tmp;
}
```

### Pop + FixDown

#### Pop

```
void pop() {
    mData[0] = mData[mSize-1];
    mSize--;
    fixDown(0);
}

void fixDown(size_t idx) {
    T tmp = mData[idx];
    size_t c;
    while ((c = 2 * idx + 1) < mSize) {
        if ( c + 1 < mSize && mLess(mData[c], mData[c+1]) ) c++;
    }
    mData[c] = tmp;
    idx = c;
}
```

```

        if ( mLess(mData[c],tmp) ) break;
        mData[idx] = mData[c];
        idx = c;
    }
    mData[idx] = tmp;
}

```

- while loop checks if we have **at least one child**
- **c** is the index of **highest value child**
  - Must consider the case where we have only one child
- Exercise: read the rest yourself

## Fast Construction

### Construct PQ from n data

```

priority_queue(std::vector<T> &v, const Comp& c = Comp()) :
    mData( new T[v.size()]() ), mCap( v.size() ), mSize( 0 ), mLess( c ) {
    for (size_t i = 0; i < mSize; i++) push(v[i]); // n times and each log(n)
}

```

$$\begin{aligned}
 \text{Total} &\leq \lfloor \log_2 1 \rfloor + \lfloor \log_2 2 \rfloor + \dots + \lfloor \log_2 n \rfloor \\
 &\leq \lfloor \log_2 (1 \times 2 \times 3 \times \dots \times n) \rfloor = \lfloor \log_2 n! \rfloor
 \end{aligned}$$

$\lfloor \log_2 n! \rfloor \text{ is } \mathcal{O}(n \log n)$

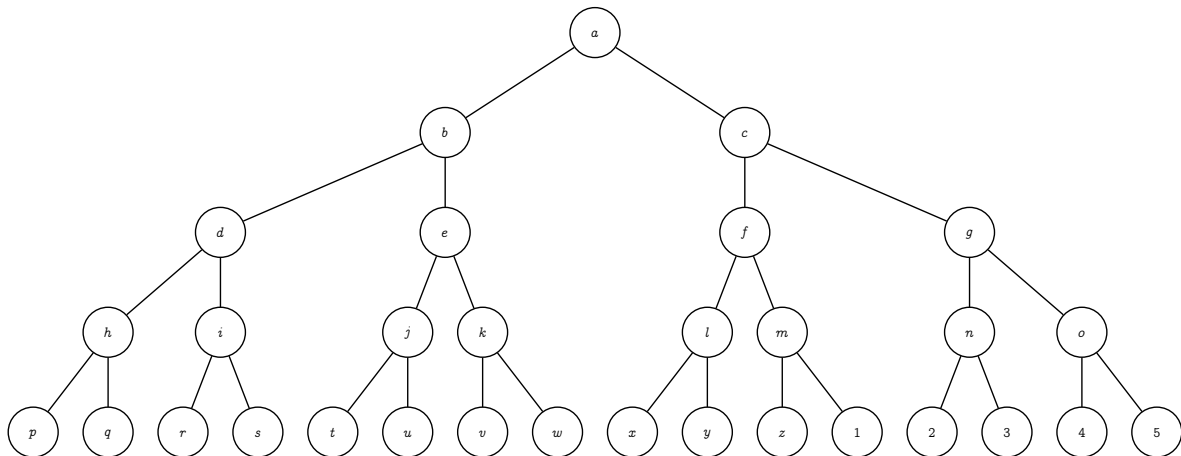
### Better Method

```

priority_queue(std::vector<T> &v, const Comp& c = Comp()) :
    mData( new T[v.size()]() ), mCap( v.size() ), mSize( 0 ), mLess( c )
{
    for (size_t i = 0; i < mSize; i++) mData[i] = v[i];
    for (int i = mSize/2-1; i > 0; i++) fixDown(i);
}

```

- Consider each node to be a binary heap
- Fix down from back to front

**How fast?**

- Binary Tree Property:
  - There are at most  $2^k$  nodes at depth  $k$
  - For a tree of height  $h$ , at depth  $k$ , fix down needed at most  $h - k$  iterations

Total nodes = 31

Tree height =  $\log_2(31) = 4$

Depth:  $k$ , Nodes:  $2^k$ , Max fix:  $h - k$   
 Total =  $\sum_{k=0}^h k 2^{h-k}$

| Depth | Nodes | Max fix per node |
|-------|-------|------------------|
| 0     | 1     | 4                |
| 1     | 2     | 3                |
| 2     | 4     | 2                |
| 3     | 8     | 1                |
| 4     | 16    | 0                |

$$\sum_{k=0}^h k 2^{h-k} = 2^h \sum_{k=0}^h k 2^{-k} < 2^h \underbrace{\sum_{k=0}^{\infty} k 2^{-k}}_{\text{This is 2}} = 2^{h+1} = 2^{\log_2 n + 1} = \mathcal{O}(n)$$

**Exercise**

If we use different number of children per node, say 4-ary heap, with same logic still holds.

1. How to keep data in dynamic array?
2. Write the new functions `fixUp` and `fixDown`
3. Is it faster?? Consider any  $n$ -ary heap

Will do this part later

**Lecture 14****Linked List**

Faster insert and erase but slower access

**Overview**

- Vector takes  $\mathcal{O}(n)$  in insert / erase at position specified by an iterator
  - But it is very fast to access any member,  $\mathcal{O}(1)$

- List gives better performance on insert / erase as  $\mathcal{O}(1)$ , providing that we have iterator to the position of insertion / erase
- Achieved by storing data into a **node** and each **node** use a **pointer** to identify the **next element**
  - Access to any elements is  $\mathcal{O}(n)$ , if we don't have an iterator to that element
- **Use more memory** than vector

## List vs Vector

| List                                                                                                                                                                                                                                                                                                                                                              | Vector                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Allocate each data separately               <ul style="list-style-type: none"> <li>▸ Each data points to where is the next data</li> <li>▸ Very fast <b>insert</b> / <b>erase</b> (just change some pointer)</li> <li>▸ Very slow access because we don't know where <math>k</math>-th element is</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>• Allocate data as a consecutive block               <ul style="list-style-type: none"> <li>▸ Very fast to access any element</li> <li>▸ Very slow <b>insert</b> / <b>erase</b> requires every element after point of insertion</li> </ul> </li> </ul> |

**vo.1 node**

- Simple object that stores a **data** and a **link to another node**
- **NULL** is a special value for any pointer that points to nowhere
  - Draw as a ground

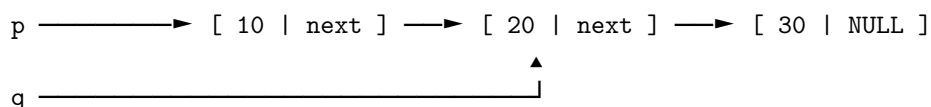
```
template <typename T>
class node {
public:
    T data;
    node *next;
    node() :
        data( T() ), next( NULL ) { }

    node(const T& data, node* next) {
        data( T(data) ), next( next ) { }
    }
};
```

### Pointer to Node

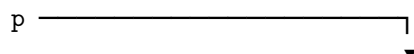
- Working with linked list needs a pointer to a node

```
int main() {
    CP::node<int> *p = NULL;
    p = new CP::node<int>(10, NULL);
    CP::node<int> *q;
    q = new CP::node<int>(20, NULL);
    p->next = q;
    q->next = new CP::node<int>(30, NULL);
}
```



What is the result of?

```
p = p->next->next; // p points to [30, NULL]
q = p;             // q points to [30, NULL]
q = q->next;        // q points to NULL
```



[ 10 ]  $\longrightarrow$  [ 20 ]  $\longrightarrow$  [ 30 | NULL ]

q  $\longrightarrow$  NULL

### vo.1 Simple (Singly) Linked List

```
template <typename T>
class list {
protected:
    class node {
    public:
        T data;
        node *next;
        node() :
            data( T() ), next( NULL ) { }

        node(const T& data, node* next) {
            data( T(data) ), next( next ) { }
        }
    };
protected:
    node *mFirst;
    size_t mSize;
public:
    list() : mFirst( NULL ), mSize( 0 ) { }
    ~list() { clear(); }
    // ... more function
};
```

- Node is a inner class

### Insertion

Linked List

```
mSize mFirst      node
[ 4, ] -> [20, next] -> [ 3, next] -> [16, next] -> [77, NULL]

d -> [99, NULL]
```

x  
▼

- To insert a value 99 before value 16, let x be a pointer that point to the node of 16
  1. Create a new node d containing a data to be inserted
  2. Change pointer of a node before x (which points to x) to point to d instead
  3. Make pointer of d points to x

```
CP::node<int> *x = mFirst->next->next;
CP::node<int> *d = new CP::node<int>(99, NULL);
mFirst->next->next = d;
d->next = x;
```

```
mSize mFirst      node
[ 4, ] -> [20, next] -> [ 3, d ] -> [16, next] -> [77, NULL]

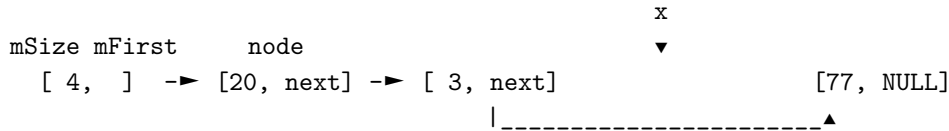
                        ▼      ▲
                        d --> [99, x] ---|
```

### Erase

- To delete a value of 16, in the node pointed by x

1. Change pointer of **a node before x** (which points to x) to point to the node that x points to instead
2. Don't forget to delete x

```
CP::node<int> *x = mFirst->next->next;
mFirst->next->next = x->next;
delete x;
```



## Doubly Linked List

### Problem with SLL

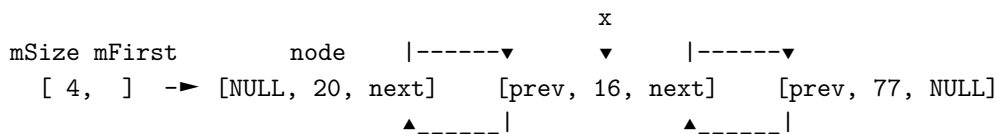
- erase / insert at iterator X is hard
  - If we have an iterator point to X, we **cannot** easily go to the **node before X**
  - Cannot go backward
    - Need to start from mFirst and move on
- Adding data to the end takes long time
  - We have to get iterator that points to the last element (which is  $\mathcal{O}(n)$ )

### Finding node before X

```
node *p = mFirst;
while (p != NULL && p->next != x) p = p->next;
```

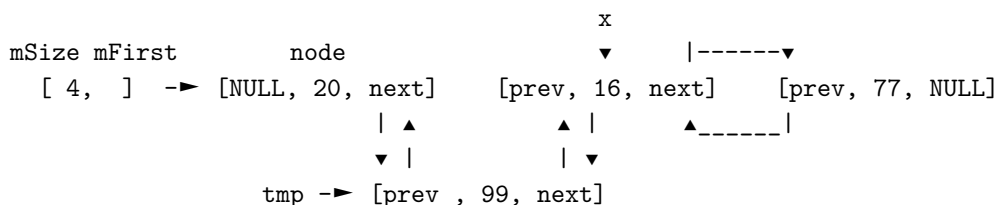
### vo.2 Doubly Linked List

- Each node has 2 pointers
  - next and prev
- Can now move forward and backward
- Now, if X is a pointer to a node, we can easily go to **node before X** and then erase or insert



### Insert in Doubly Linked List

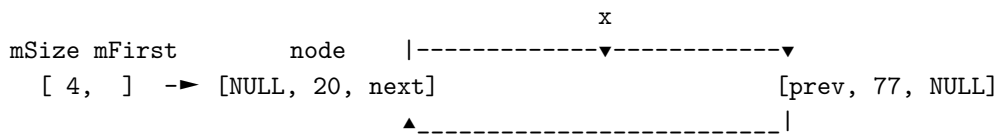
```
CP::node<int> *tmp =
    new CP::node<int> (99, NULL, NULL);
tmp->next = x;
tmp->prev = x->prev;
x->prev->next = tmp;
x->prev = tmp;
```



### Erase in Doubly List

```
x->prev->next = x->next;
x->next->prev = x->prev;
delete x;
```

Note: When we refer to a pointer,  $x$  is the variable itself, but the value of  $x$  is the destination address it is pointing to. Say  $x$  holds a node containing  $[x \rightarrow \text{prev}, 16, x \rightarrow \text{next}]$ . In this set up,  $x \rightarrow \text{prev} \rightarrow \text{next}$  represents the next pointer inside the node before  $x$ . If we write  $x \rightarrow \text{prev} \rightarrow \text{next} = x \rightarrow \text{next}$ ; it means that the previous node's next pointer now stores the destination of the node after  $x$ . In other words, it successfully skips  $x$  from the sequence.



## Circular Linked List

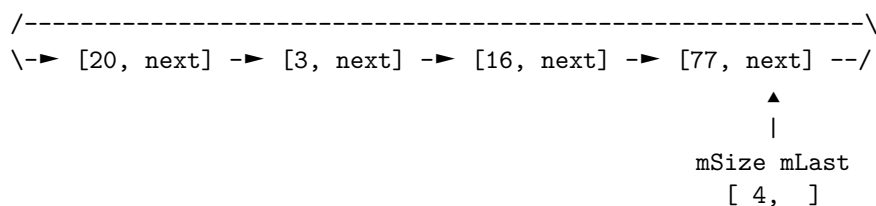
### Problem Solved

- erase / insert at iterator  $X$  is now **easy**
- But, adding data at the end (`push_back`) is still hard
  - Need to get  $X$  to point to the last element
  - `push_back` is popular in real world
  - Right now we have only `push_back` (fast addition to the first)
- Also some minor issue about the **code cleanliness**
  - insert / erase the first / last node

### vo.3 Circular Linked List

- Use `mLast` instead of `mFirst`
- Fast access to both first and last element

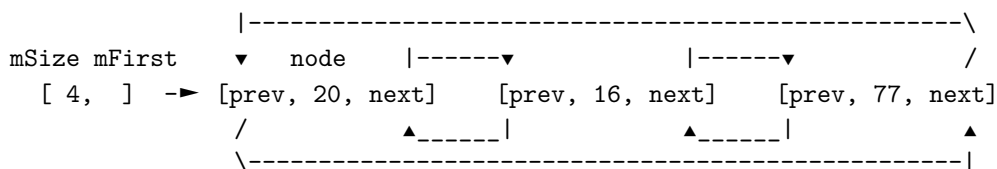
```
CP::node<int> *first = mLast->next;
```



### vo.4 Circular Doubly Linked List

- Special linking to last element
  - Fast access to both first and last element
  - Can now easily insert at the end

```
CP::node<int> *last = mFirst->prev;
```



## Linked List with Header

### Minor Problem

- Code is **not really clean** because of the first element (or the last element)
- Consider insert at the first node and insert at the second node of Doubly Linked List
  - Assume we have a pointer to that node

```
// assume thta s points to the second node
CP::node<int> *tmp =
```

```

    new CP::node<int>(99, NULL, NULL);
tmp->next = s;
tmp->prev = s->prev;
s->prev->next = tmp;
s->prev = tmp;

// assume that f points to the first node
CP::node<int> *tmp =
    new CP::node<int>(99, NULL, NULL);
tmp->next = f;
tmp->prev = f->prev;
mFirst = tmp; // Because first node is different
f->prev = tmp;

```

### Special First Node Problem

- First node (and last node) is **different from other nodes** in both singly or doubly linked list
- For circular singly and circular doubly, each node look the same **but we also have to adjust mFirst** (or mLast)
- This affects both insert and erase
- Also need to deal **when mFirst is NULL** (when mSize == 0)

```

// circular doubly linked list
void push_front(const T& e) {
    if (mSize == 0) {
        mFirst = new node(e, NULL, NULL);
        mFirst->next = mFirst;
        mFirst->prev = mFirst;
    } else {
        node* tmp =
            new node(e, mFirst->prev, mFirst);
        mFirst->prev->next = tmp;
        mFirst->prev = tmp;
        mFirst = tmp;
    }
}

```

### Another Example

- Remove for circular doubly linked list
- Remove is to find and then erase

```

// circular doubly linked list
void remove(const &T e) {
    node *p = mFirst;
    for (size_t i = 0; i < mSize; i++, p = p->next) {
        if (p->data == e) {
            p->next->prev = p->prev;
            p->prev->next = p->next;
            if (p == mFirst) {
                mFirst = p->next;
            }
            delete p;
            mSize--;
            break;
        }
    }
    if (mSize == 0) mFirst = NULL;
}

```

### Linked List with Header

- Add a special node that will not be used to stored data

### Simpler Code with Header

```
void push_front(const T& e) {
    node* f = mFirst->next;
    node* tmp = new node(e, f, mFirst);
    mFirst->next = tmp;
    f->prev = tmp;
}

void remove(const T& e) {
    node *p = mFirst->next;
    while (p != mHeader && p->data != e)
        p = p->next;
    if (p != mHeader) {
        p->next->prev = p->prev;
        p->prev->next = p->next;
        mSize--;
    }
}
```

- Header simplifies code
  - Because mFirst always points to the header (mFirst never is NULL)

### Variant Summary

- **Circular** makes accessing first and last element fast
- **Doubly** makes accessing previous element fast
  - Also making erase at node p easy if we have pointer to p
  - Need more space for prev pointer
- **Header** makes code simpler
  - Need more space for header node

### Final Version

- CP::list is “circular doubly linked list with header”
  - **Simple code** for insert / erase
  - Use **most space** (two pointers per node, need header node)
  - Can push\_back, pop\_back, push\_front, pop\_front
- Also need custom iterator class
  - **Iterator** just store a pointer to a node
  - We cannot directly use a pointer to a node (node\*) because we need to override some operator (--, ++, and something else)

### CP::list

#### Layout

- **Inner class** is a class inside another class
  - **Inner class** can access any members of **outer class**
  - **Outer class** cannot access protected or private of the **inner class**
- **Friend class** allows other class to access

```
template <typename T>
class list {
protected:
    class node {
        friend class list;
    };
};
```

```

    public:
        T data;
        node *prev, *next;
        // some functions
};
class list_iterator {
    friend class list;
protected:
    node* ptr;
public:
    // some functions && operators
};
public:
    typedef list_iterator iterator;
protected:
    node *mHeader; // pointer to a header node
    size_t mSize;
public:
    // functions
};

```

### Doubly Linked List Node

```

class node {
    friend class list;
public:
    T data;
    node *prev;
    node *next;

    node() :
        data( T() ), prev( this ), next( this ) { }
    node(const T& data, node* prev, node* next) :
        data( T(data) ), prev( prev ), next( next ) { }
};

```

### Constructor

```

// default constructor
list() : mHeader( new node() ), mSize( 0 ) { }

// copy constructor
list(list<T>& a) : mHeader( new node() ), mSize( 0 ) {
    for (iterator it = a.begin(); it != a.end(); it++) {
        push_back(*it);
    }
}

list<T>& operator=(list<T> other) {
    using std::swap;
    swap(this->mHeader, other.mHeader);
    swap(this->mSize, other.mSize);
    return *this;
}

~this() {
    clear();
}

```

```

    delete mHeader;
}

```

### Small functions

```

// capacity function
bool empty() const { return mSize == 0; }
size_t size() const { return mSize; }

// access
T& front() { return mHeader->next->data; }
T& back() { return mHeader->prev->data; }

// modifier
void push_back(const T& element) {
    insert(end(), element);
}
void push_front(const T& element) {
    insert(begin(), element);
}
void pop_back() {
    erase(iterator(mHeader->prev));
}
void pop_front() {
    erase(begin());
}

```

- Task is delegated to insert and erase
- Need iterator

### Iterator

```

class list_iterator {
    friend class list;
protected:
    node* ptr;
public:

    list_iterator() : ptr( NULL ) { }
    list_iterator(node *a) : ptr(a) { }

    list_iterator& operator++() {
        ptr = ptr->next;
        return (*this);
    }

    list_iterator& operator--() {
        ptr = ptr->prev;
        return (*this);
    }
}

```

- Has custom constructor that takes node pointer

```

list_iterator& operator++(int) {
    list_iterator tmp(*this);
    operator++;
    return tmp;
}

```

```
list_iterator& operator--(int) {
    list_iterator tmp(*this);
    operator--();
    return tmp;
}
```

- operator++() is an operator for ++it
- operator++(int) is a syntax for it++
- operator++(int) delegates to operator++()
- Same for operator--()

```
T& operator*() { return ptr->data; }
T* operator->() { return &(ptr->data); }
```

```
bool operator==(const list_iterator& other) {
    return other.ptr == ptr;
}
```

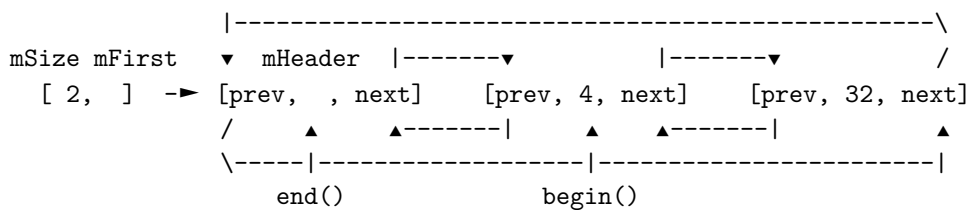
```
bool operator!=(const list_iterator& other) {
    return other.ptr != ptr;
}
};
```

### Other small functions

```
iterator begin() {
    return iterator(mHeader->next);
}
```

```
iterator end() {
    return iterator(mHeader);
}
```

```
void clear() {
    while (mSize > 0) erase(begin());
}
```



### Insert & Erase

```
iterator insert(iterator it, const T& element) {
    node *n = new node(element, it.ptr->prev, it.ptr);
    it.ptr->prev->next = n;
    it.ptr->prev = n;
    mSize++;
    return iterator(n);
}
```

```
iterator erase(iterator it) {
    iterator tmp(it.ptr->next);
    it.ptr->prev->next = it.ptr->next;
    it.ptr->next->prev = it.ptr->prev;
    delete it.ptr;
}
```

```

    mSize--;
    return tmp;
}

```

- Header make insert / erase very simple

## Lecture 15

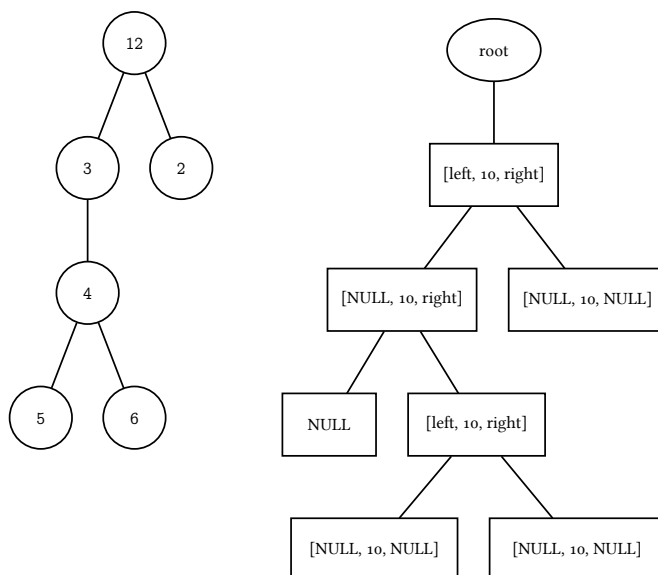
### Binary Tree

Practicing Pointer & Recursive

#### Overview

- This is a basic for the next data structure, Binary Search and AVL Tree
- Focus on using Node and Pointer
- Focus on using recursive programming
- Some applications using just Binary Tree
- There is no data in std that is Binary Tree

#### Binary Tree & Node



- A rooted tree where each node have at most two children
- Tree Node is very similar to a linked list node

```

node
    left data right
    NULL 10 NULL

class node {
public:
    ValueT data;
    node *left, *right;
    node() :
        data( ValueT() ), left( NULL ), right( NULL ) { }
    node(const ValueT& data, node* left, node* right) :
        data( data ), left( left ), right( right ) { }
};

```

**Node with parent link**

```

class node {
public:
    ValueT data;
    node *left, *right, *parent;
    node() :
        data( ValueT() ), left( NULL ), right( NULL ), parent( NULL ) { }
    node(const ValueT& data, node* left, node* right, node* parent) :
        data( data ), left( left ), right( right ), parent( parent ) { }
};

```

- Sometimes, we need a link to parent
- Root is the only node that parent is NULL

**String Encoding****Huffman Coding: Example Application of Tree**

- David Huffman proposed this as his term project in Robert Fano's class (co-worker of Claude Shannon) which beats Shannon-Fano encoding
- Encoding = associate meaning to a representation
- ASCII Code
  - Fix length encoding
  - Each char = 8 bits

**Variable Length Encoding**

*Never gonna give you up*

*Never gonna let you down*

*Never gonna run around and desert you*

16 different characters

Fix-length needs  $4 \times 86 = 344$  bits

Variable Length need 327 bits

| n    | e    | o    | u    | r    | a    | v    | g    | d    | y    | t    | w    | s    | p    | l    | i    |
|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 14   | 11   | 9    | 7    | 7    | 6    | 5    | 5    | 5    | 4    | 3    | 2    | 2    | 2    | 2    | 2    |
| 0000 | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |
| 11   | 010  | 011  | 0001 | 0011 | 0000 | 1011 | 1010 | 1000 | 0010 | 1001 | 1001 | 0010 | 0010 | 1001 | 0010 |
|      |      |      |      |      |      |      |      |      | 1    | 1    | 01   | 001  | 01   | 00   | 000  |

Encoding "never"

Fix-length      00000001011000010100

Variable-length   1101010110100011

**Huffman Coding****Problem Statement**

- Input: a string
- Output: encoding of each character in the string such that
  - The total length of encoding is minimum
  - The encoding of each character is *not ambiguous*
    - Any character encoding is not a prefix of any other character

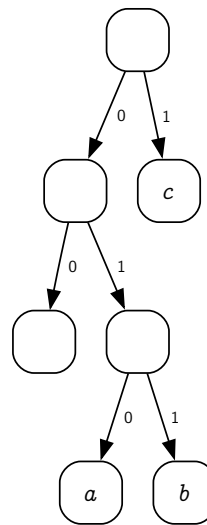
**Tree Encoding**

- Using a **tree** to represent encoding
- Each character is represented at **leaf nodes**
  - **Leaf node** is a node without children
- Encode by start at the root and **walk toward leaf nodes**
  - The path gives the encoding
  - Going to left child equal to 0
  - Going to right child equal to 1
- Guaranteed to be non-ambiguous

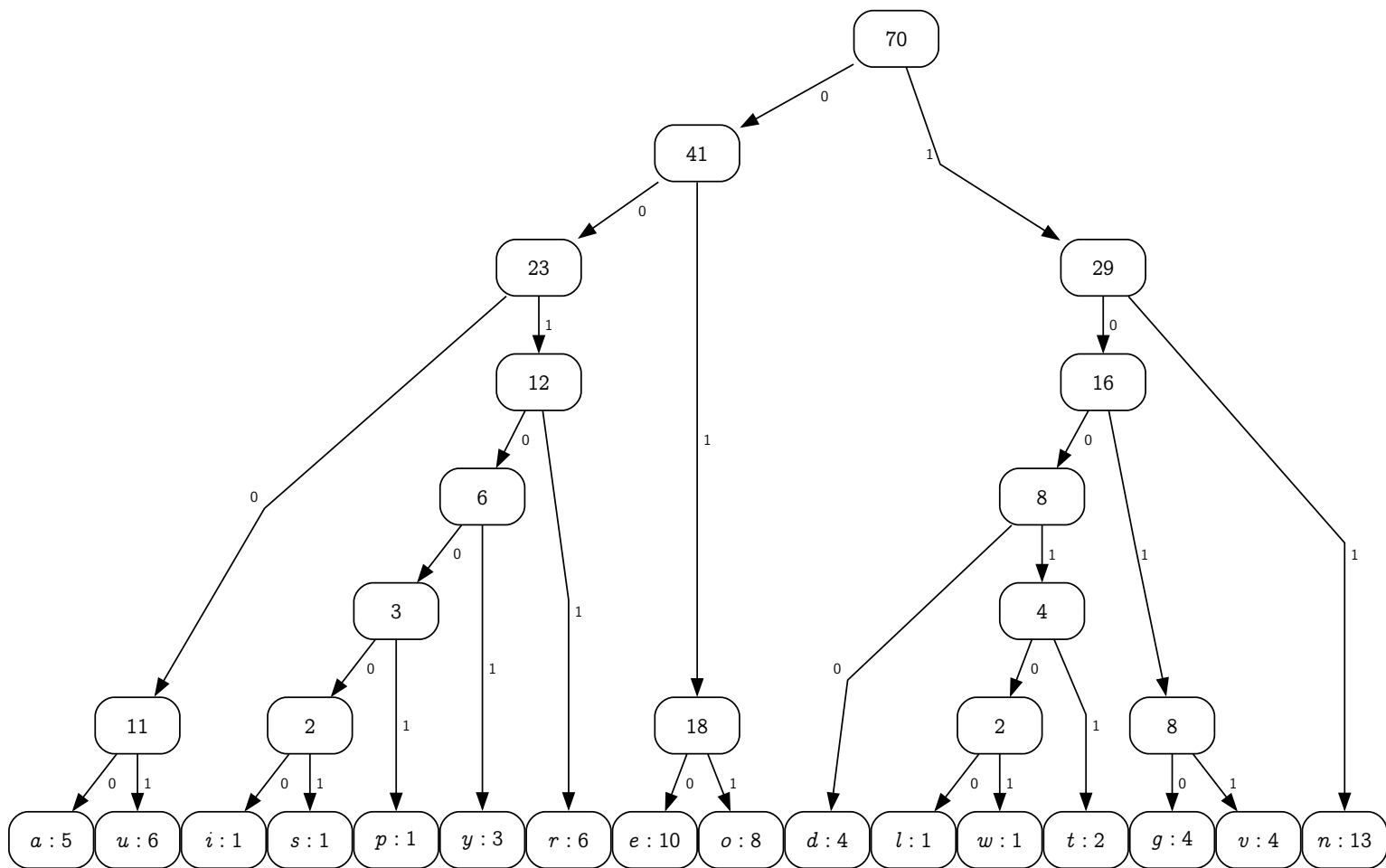
a = 010

b = 011

c = 1

**Huffman Tree**

- Find 2 min nodes
- Create a new node with those two nodes as children, set freq equal to summation
- Repeat until only one node left



## Huffman Tree

### Huffman Tree Node

- Instead of data, we have both character and frequency
- Since we have to pick two nodes with minimum freq, we overload operator< to do so and use priority\_queue

### Huffman Code: Node

```
class huffman_tree {
protected:
    class huffman_node {
    public:
        char letter;
        int freq;
        huffman_node *left, *right;
        huffman_node() : letter('*'), freq(0), left(NULL), right(NULL) {}
        huffman_node(char letter, int freq, huffman_node *left, huffman_node *right) :
            letter(letter), freq(freq), left(left), right(right) {}

        bool is_leaf() { return left == NULL && right == NULL; }
    };
};
```

```

class node_comparator {
public:
    bool operator()(const huffman_node *a, huffman_node *b) {
        return a->freq > b->freq;
    }
};

```

### Huffman Code: Build Tree

```

class huffman_tree {
protected:
    huffman_node *root;
    void build_tree(vector<huffman_node*> data) {
        priority_queue<huffman_node*, vector<huffman_node*>, node_comparator> pq;
        for (auto &x : data) pq.push(x);
        while (pq.size > 1) {
            huffman_node *right = pq.top(); pq.pop();
            huffman_node *left = pq.top(); pq.pop();
            pq.push(new huffman_node('*', left->freq+right->freq, left, right));
        }
        root = pq.top();
    }
public:
    huffman_tree(string s) {
        map<char, int> count;
        for (auto &c : s) {
            count[c]++; // word count
        }
        vector<huffman_node*> nodes;
        for (auto &x : count)
            nodes.push_back(new huffman_node(x.first, x.second, NULL, NULL));
        build_tree(nodes);
    }
};

```

### Recursive Programming

Calling itself

#### Recursive

- A function that call itself
- Must have some input, usually via function argument
- The function must check a condition for execution
  - Result in either **terminating case** where the function won't call itself
  - or **recursion case** where the function will call itself **with different parameters**

```

// calculate sum 0..n
int recur1(int n) {
    if (n <= 0) {
        // terminating case
        return 0;
    } else {
        // recursion case
        return recur1(n-1) + n;
    }
}

```

### Why recursion?

- Much simpler code
  - When the task is right
  - Recursion is natural for several mathematical model that is recursi
- Comparing to a normal loop, recursion has the same growth rate but recursion might take more time because function call is costlier than a loop

### More Example

```
void print_range1(int step, int goal) {
    if (step < goal) {
        std::cout << step << " ";
        print_range1(step+1, goal);
    }
}

void print_range2(int step, int goal) {
    if (step < goal) {
        print_range2(step+1, goal);
        std::cout << step << " ";
    }
}
```

- Terminating Case do nothing
- Which is the output of print\_range1(0,5) and print\_range2(0,5)

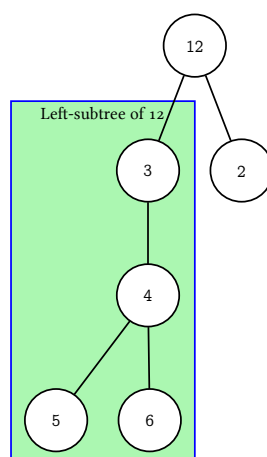
```
1. 0 1 2 3 4 5
2. 0 1 2 3 4
3. 5 4 3 2 1 0
4. 4 3 2 1 0
```

### Binary Tree Recursive Definition

- A Binary Tree is
  - A tree with no nodes (root is NULL)
  - A tree with a root
    - Both children of the root must be a binary tree
    - Each child is called left-subtree and right-subtree
- Since binary tree can be defined recursively, operation on a binary tree can be naturally written as a recursion

### Subtree

- For any node
  - its left (or right) child and all of the child's descendants is called left-subtree (or right-subtree)



### Recursion on Binary Tree

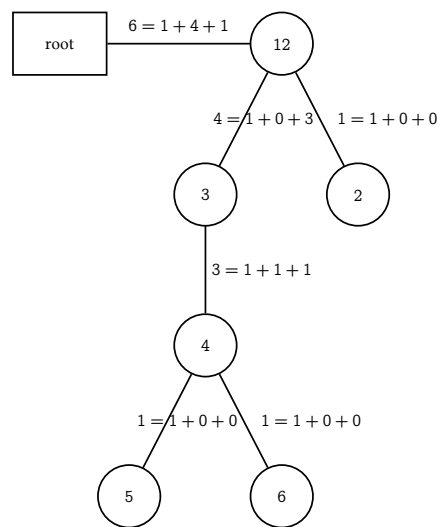
#### Tree Size by Recursion

- An empty tree has 0 nodes

- A tree with a root has 1 node (the root)
  - Plus the size of its two subtrees
- Easily written as recursive

```
class node {
public:
    int data;
    node *left, *right;
};

int get_size(node* n) {
    if (n == NULL) return 0;
    return 1 + get_size(n->left) + get_size(n->right);
}
```



### Tree Height

- Height of a tree is the number of link we have to go to reach it deepest children
- Empty tree has height -1
- Height of a tree is 1 + max of height of its children

```
class node {
public:
    int data;
    node *left, *right;
};

int get_height(node* n) {
    if (n == NULL) return 0;
    return 1 + std::max(get_height(n->left),
                        get_height(n->right));
}
```

### Tree Copy

```
class node {
public:
    int data;
    node *left, *right;
}
```

```

    node() : data(0), left(NULL), right(NULL);
    node(int data, node *left, node *right)
        : data(data), left(left), right(right);
};

node* copy(node *n) {
    if (n == NULL) return NULL;
    node *lc = copy(n->left);
    node *rc = copy(n->right);
    node *result = new node(n->data, lc, rc);
}

```

### Walk over a tree

- Visiting all nodes (and maybe do something)

```

// Preorder traversal
void preorder(node *n) {
    if (n == NULL) return;
    std::cout << n->data << " ";
    preorder(n->left);
    preorder(n->right);
}

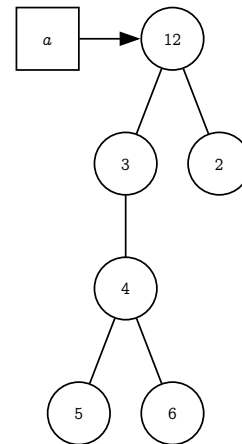
// Inorder traversal
void inorder(node *n) {
    if (n == NULL) return;
    inorder(n->left);
    std::cout << n->data << " ";
    inorder(n->right);
}

```

```

// Postorder traversal
void postorder(node *n) {
    if (n == NULL) return;
    postorder(n->left);
    postorder(n->right);
    std::cout << n->data << " ";
}

```



What is the result of

- preorder(a)
  - 12 3 4 5 6 2
- inorder(a)
  - 5 4 6 3 12 2
- postorder(a)
  - 5 6 4 3 2 12

### Huffman Tree: Encoding

```

class huffman_tree {
protected:
    class huffman_node { };
    class node_comparator { };
    huffman_node *root;
public:
    void print(huffman_node *n, string s) {
        if (n->is_leaf()) {
            cout << n->letter << ": " << s << endl;
        } else {
            print(n->left, s+"0");
            print(n->right, s+"1");
        }
    }

    void print() {

```

```

    print(root, "");
}
};

```

- Recursive printing
- Use `s` to store path

```

class huffman_tree {
protected:
    class huffman_node { };
    class node_comparator { };
    huffman_node *root;

    void delete_node(huffman_node *n) {
        if (n == NULL) return;
        delete_node(n->left);
        delete_node(n->right);
        delete n;
    }

public:
    ~huffman_tree() {
        delete_node(root);
    }
};

```

- Recursive delete node
- Use postorder traversal
- Can we use inorder or preorder?

## Lecture 16

### Binary Search Tree

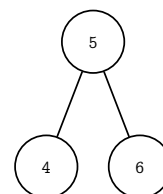
Binary Tree with value condition

#### Overview

- We add additional **value constraint** to a Binary Tree
- The constraint make finding data in the tree much faster
  - $\mathcal{O}(h)$  where  $h$  is the height of the tree
  - The tree is expected to have  $h$  be in  $\mathcal{O}(\lg n)$ , but this is not always true
  - The next tree (AVL tree) will add more constraint so that we can guarantee that  $h = \mathcal{O}(\log n)$
- Using the same approach as a binary heap, maintain the constraint during modification

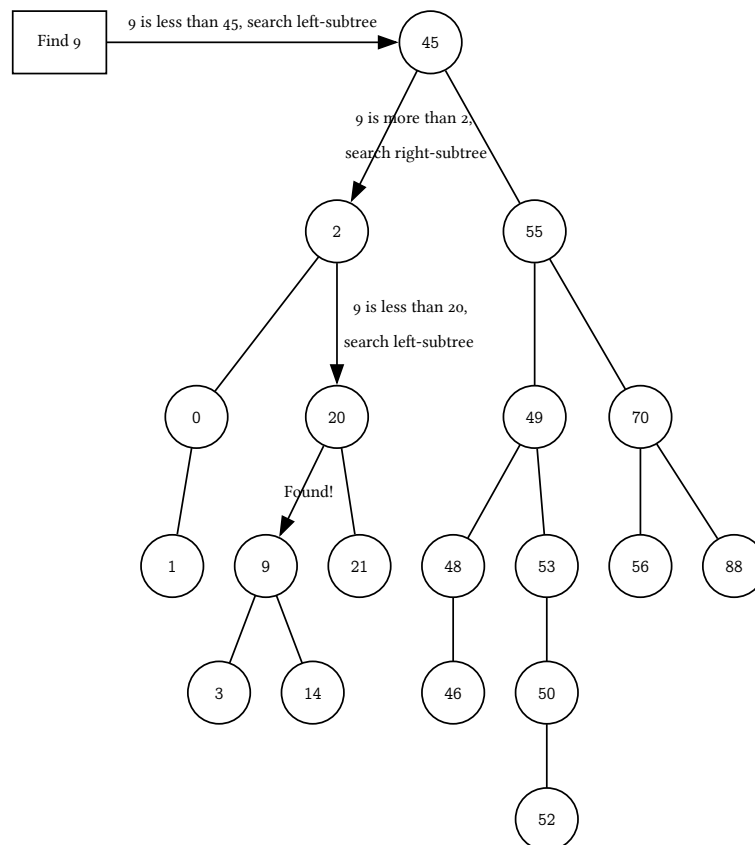
#### Binary Search Tree

- Structure rule: must be a Binary Tree
- Value rule: for any node `x`
  - data in **left-subtree** must be **less than** the data in `x`
  - data in **right-subtree** must be **more than** the data in `x`
- Recursive Definition
  - An **empty tree** is a Binary Search Tree (BST)
  - A node `X` is a BST when
    - Its subtrees (if any) must be BST and
    - If **left-subtree** exists, `X->data` must be more than `X->left->data`
    - If **right-subtree** exists, `X->data` must be less than `X->right->data`



### Finding Value in BST

- Value rules make finding fast
- To find **e**, start from root
  - If the current node is not **e**,
    - search in left-subtree if **e** is **less** than the **current node**
    - search in right-subtree if **e** is **more** than the **current node**
  - Keep going until we find **e** or reach **NULL**
- Other operation is also depends on **find**



### Insert & Erase with 0, 1 child

#### Find Node

```
node* find_node(const ValueT& k, node* r, node* &parent) {
    node *ptr = r;
    while (ptr != NULL) {
        int cmp == compare(k, ptr->data); // return -1 if a < b, 1 if a > b, 0 if equal
        if (cmp == 0) return ptr;
        parent = ptr;
        ptr = cmp < 0 ? ptr->left : ptr->right;
    }
    return NULL;
}
```

#### Insert

- Assumption: Data in BST is unique
- **insert(e)** by find **e**
  - If **e** is found, don't add any node

- If **e** is not in BST, find must reach **NULL** somewhere, that **NULL** is where to put **e**
- Both structure and value constraints are satisfied

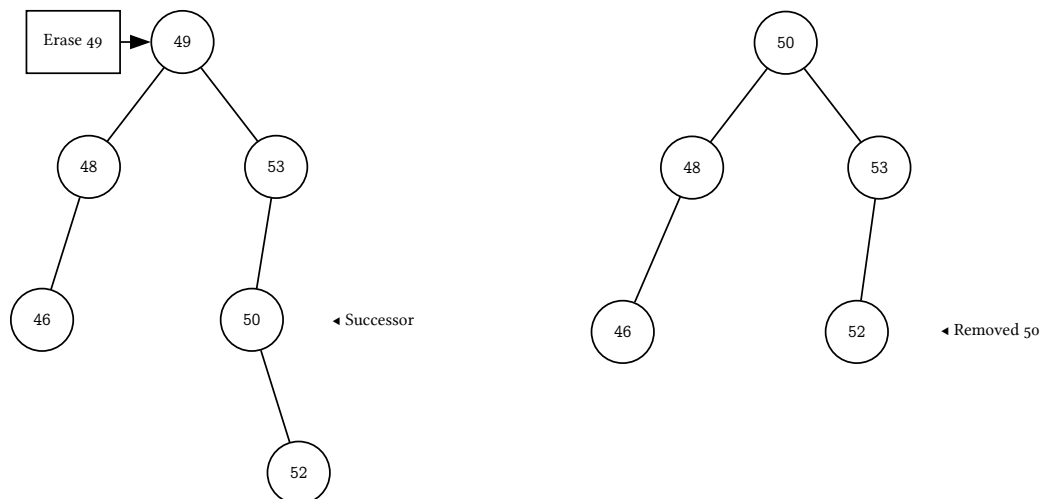
### Erase

- **erase(e)** first have to find **e** as well
- If not found, do nothing
- If found at node **X**, there are 3 cases depend on the number of children of **e**
  - If has **no child**, just simply delete **X**
  - If has **one child**, have parent of **X** points (using the same link) to the child of **X** instead
  - If has **two children**, pick either **successor** or **predecessor** of **e**
    - Assume we choose **successor p** (must be in right-subtree), replace **X** with **p** and **erase(p)** from right subtree

### Erase with 2 children

#### Erase node with 2 children

- Replace by **successor** (or **predecessor**) preserves value rules
  - **Successor** is the minimum in the **right-subtree**
  - **Predecessor** is the maximum in the **left-subtree**
- Both exists (because the node has both subtrees)



### Find min max

#### Finding Successor and Predecessor

- If a tree has **left-subtree**, min is the min of **left-subtree**
  - If not, min is the root

```
node* find_min_node(node* r) {
    // r must not be NULL
    node *min = r;
    while (min->left != NULL) {
        min = min->left;
    }
    return min;
}
```

- If a tree has **right-subtree**, max is the max of **right-subtree**
  - If not, max is the root

```
node* find_max_node(node* r) {
    // r must not be NULL
    node *max = r;
    while (max->right != NULL) {
        max = max->right;
    }
    return max;
}
```

**Finding Successor and Predecessor (recursive)**

```

node* find_min_node(node* r) {
    // r must not be NULL
    if (r->left == NULL) return r;
    return find_min_node(r->left);
}

node* find_max_node(node* r) {
    // r must not be NULL
    if (r->right == NULL) return r;
    return find_max_node(r->right);
}

```

**Complexity Analysis**

- insert, erase depends on `find`, `find_min` (or `find_max`)
- All finds start from root and in the worst case reach the leaf
  - Hence,  $\mathcal{O}(h)$
- Height of the tree can be in the range from  $\lg n$  to  $n$
- For 1,000,000 nodes, it's in the range of [20, 999999]
  - $\mathcal{O}(h)$  is, right now,  $\mathcal{O}(n)$
  - Will be fixed by AVL tree

**CP::map\_bst (insert)**

Using Binary Search Tree to create associated data structure

**Layout**

- Need node class
- Also need iterator class
- Template has two types
  - Key Type and Mapped Type
  - ValueType is `pair<KeyType, MappedType>`
- Also need custom comparator

```

template <typename KeyT,
          typename MappedT,
          typename CompareT = std::less<KeyT> >
class map_bst {
protected:
    typedef std::pair<KeyT, MappedT> ValueT;
    class node {
        friend class map_bst;
    protected:
        ValueT data;
        node *left;
        node *right;
        node *parent;
    };
    class tree_iterator {
    protected:
        node* ptr;
    public:
    };
    node *mRoot;
    CompareT mLess;
    size_t mSize;
public:
    typedef tree_iterator iterator;
};

```

**Node class**

- Data stores both the key type and mapped type (as a pair)

- Map finds by key

```
class node {
    friend class map_bst;
protected:
    ValueT data;
    node *left;
    node *right;
    node *parent;

    node() :
        data( ValueT() ), left( NULL ), right( NULL ), parent( NULL ) { }

    node(const ValueT& data, node* left, node* right, node* parent) :
        data( data ), left( left ), right( right ), parent( parent ) { }
};
```

#### Ctors, Dtor

```
map_bst(const map_bst<KeyT, MappedT, CompareT> & other) :
    mLess(other.mLess), mSize(other.mSize)
{ mRoot = copy(other.mRoot, NULL); } // recursive copy

map_bst(const CompareT& c = CompareT() ) :
    mRoot(NULL), mLess(c), mSize(0)
{ }
```

```
map_bst<KeyT, MappedT, CompareT>& operator=(map_bst<KeyT, MappedT, CompareT> other) {
    using std::swap;
    swap(this->mRoot, other.mRoot);
    swap(this->mLess, other.mLess);
    swap(this->mSize, other.mSize);
    return *this;
}
```

```
~map_bst() {
    clear(); // recursive delete
}
```

#### Actual Find

```
iterator find(const KeyT &key) {
    node *parent;
    node *ptr = find_node(key, mRoot, mParent);
    return ptr == NULL ? end() : iterator(ptr);
}
```

- Find by Key

```
int compare(const KeyT& k1, const KeyT& k2) {
    if (mLess(k1, k2)) return -1;
    if (mLess(k2, k1)) return 1;
    return 0;
}
```

```
node* find_node(const KeyT& k, node* r, node* &parent) { // modify parent pointer
    node *ptr = r;
    while (ptr != NULL) {
        int cmp = compare(k, ptr->data.first);
```

```

    if (cmp == 0) return ptr;
    parent = ptr;
    ptr = cmp < 0 ? ptr->left : ptr->right;
}
return NULL;
}

```

### Insert

- insert returns pair of iterator and insert result

```

std::pair<iterator, bool> insert(const ValueT& val) {
    node *parent = NULL;
    node *ptr = find_node(val.first, mRoot, parent);
    bool not_found = (ptr == NULL);
    if (not_found) {
        ptr = new node(val, NULL, NULL, parent);
        child_link(parent, val.first) = ptr;
        mSize++;
    }
    return std::make_pair(iterator(ptr), not_found);
}

```

child\_link returns a reference (the variable) to the pointer of the appropriate child of the parent with respect to `k`

```

node* &child_link(node* parent, const KeyT& k) {
    if (parent == NULL) return mRoot;
    return mLess(k, parent->data.first) ?
        parent->left : parent->right;
}

```

### CP::map\_bst (erase and iterator)

#### Erase

- Handle multiple cases

```

size_t erase(const KeyT &key) {
    if (mRoot == NULL) return 0; // empty tree
    node *parent = NULL;
    node *ptr = find_node(key, mRoot, parent);
    if (ptr == NULL) return 0; // not found
    if (ptr->left != NULL && ptr->right != NULL) {
        // have two children => erase min
        node *min = find_min_node(ptr->right); // successor
        // remove min like case 0 or 1 child
        node * &link = child_link(min->parent, min->data.first);
        link = (min->left == NULL) ? min->right : min->left;
        if (link != NULL) link->parent = min->parent;
        // swap min with ptr
        std::swap(ptr->data.first, min->data.first);
        std::swap(ptr->data.second, min->data.second);
        ptr = min; // we are going to delete this node instead
    } else {
        // case 0 or 1 child => erase ptr
        node * &link = child_link(ptr->parent, key); // find the pointer that holds ptr
        link = (ptr->left == NULL) ? ptr->right : ptr->left; // replace it with ptr child
        if (link != NULL) link->parent = ptr->parent; // if child exists, update parent
    }
}

```

```

    delete ptr;
    mSize--;
    return 1;
}

```

### Operator[]

```

MappedT& operator[](const KeyT& key) {
    node *parent = NULL;
    node *ptr = find_node(key, mRoot, parent);
    if (ptr == NULL) {
        ptr = new node(std::make_pair(key, MappedT()), NULL, NULL, parent);
        child_link(parent, key) = ptr;
        mSize++;
    }
    return ptr->data.second;
}

```

- Find node
- If not exists, create one with default MappedTypeReturn MappedType of the node

### Iterator

- Just like linked list, we need a class for iterator
  - Because we need custom operator++, -- (and some more)
- Iterator class just store a **pointer to a node**

```

class tree_iterator {
protected:
    node* ptr;

public:
    tree_iterator() : ptr( NULL ) { }
    tree_iterator(node *a) : ptr(a) { }
    // more functions below
};

```

### Other Functions

```

tree_iterator operator++(int) { // ++it
    tree_iterator tmp(*this);
    operator++;
    return tmp;
}

```

```

tree_iterator operator--(int) { // --it
    tree_iterator tmp(*this);
    operator--;
    return tmp;
}

```

```

ValueT& operator*() { return ptr->data; }
ValueT* operator->() { return &(ptr->data); }
bool operator==(const tree_iterator& other)
{ return other.ptr == ptr; }
bool operator!=(const tree_iterator& other)
{ return other.ptr != other }

```

### Operator++

- Find **successor** of **x**, easy if **x** have **right-subtree**

- ▶ Just find **min** of **right-subtree**
- If not, we have to go up (go toward root) until we find one that is **more** than **x**
  - ▶ This is always the closest **ancestor** of **x** that has **x** in its **left-subtree**

```
tree_iterator& operator++() {
    if (ptr->right == NULL) {
        node *parent = ptr->parent;
        while (parent != NULL &&
            parent->right == ptr) {
            ptr = parent;
            parent = ptr->parent;
        }
        ptr = parent;
    } else {
        ptr = ptr->right;
        while (ptr->left != NULL)
            ptr = ptr->left;
    }
    return (*this);
}
```

### Summary

- Binary Search Tree relies on **Value Constraint** to make find fast
  - ▶ Possible to be slow (will be fixed later)
- Erase requires `find_min`, `find_max`
- `CP::map_bst` uses pair to store **KeyT** and **MappedT**
  - ▶ `find` uses key
- **Iterator** is just a pointer
  - ▶ Have a problem of **operator--** at `end()` (will be fixed later)

## Lecture 17

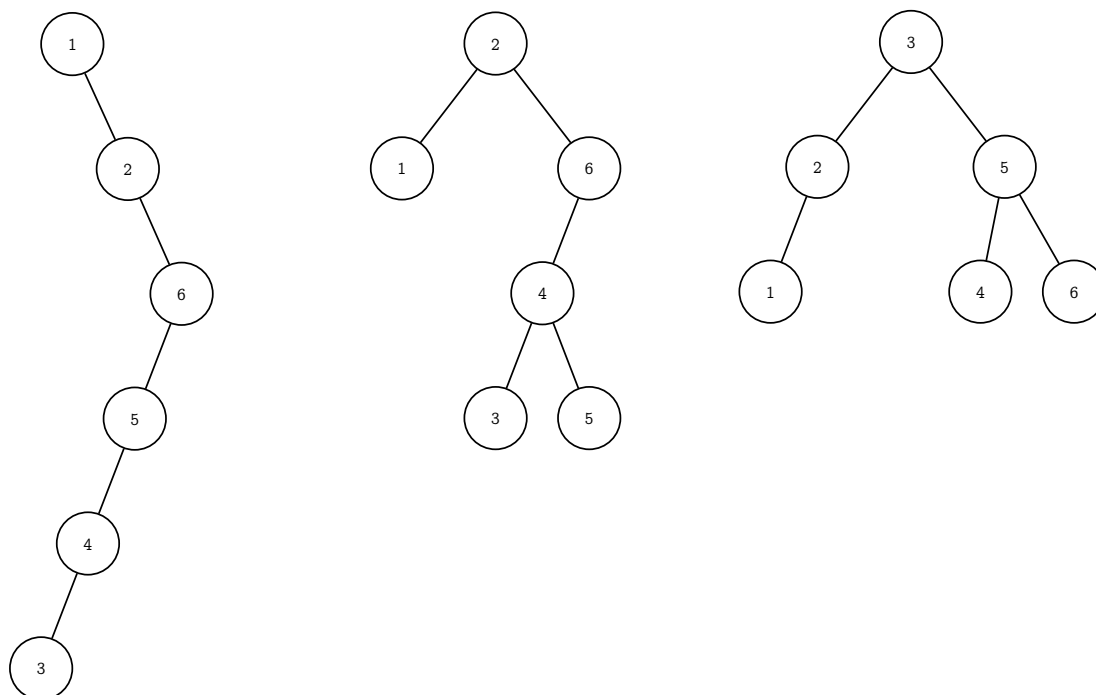
### AVL Tree

#### หัวข้อ

- นิยามต้นไม้เอวีแอล
- การวิเคราะห์ความสูงของต้นไม้เอวีแอล
- การปรับต้นไม้เอวีแอลให้สูงสมดุล
- กระบวนการหมุนปม

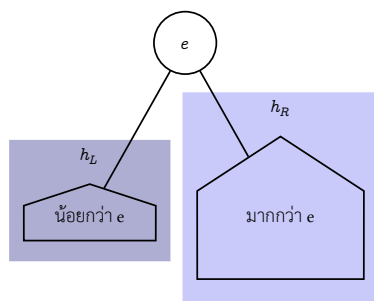
#### ต้นไม้ค้นหาแบบทวิภาค

- เวลาการทำงานเป็น  $\mathcal{O}(h)$
- $\lfloor \log_2 n \rfloor \leq h \leq n - 1$
- โชคดีทำงานเร็ว  $\mathcal{O}(\log n)$ , โชคร้ายทำงานเป็น  $\mathcal{O}(n)$



### ต้นไม้เอวีแอล

- AVL = Binary Search Tree + กฎความสูงสมดุล

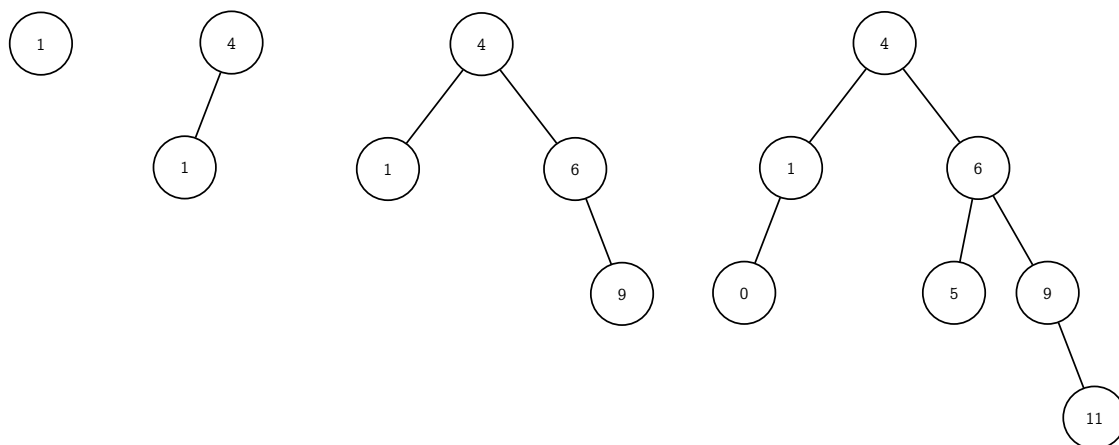


$$|h_L - h_R| \leq 1$$

ต้นไม้ย่อยทุกต้นต้องเป็นไปตามกฎ

AVL: Adelson-Velskii and Landis

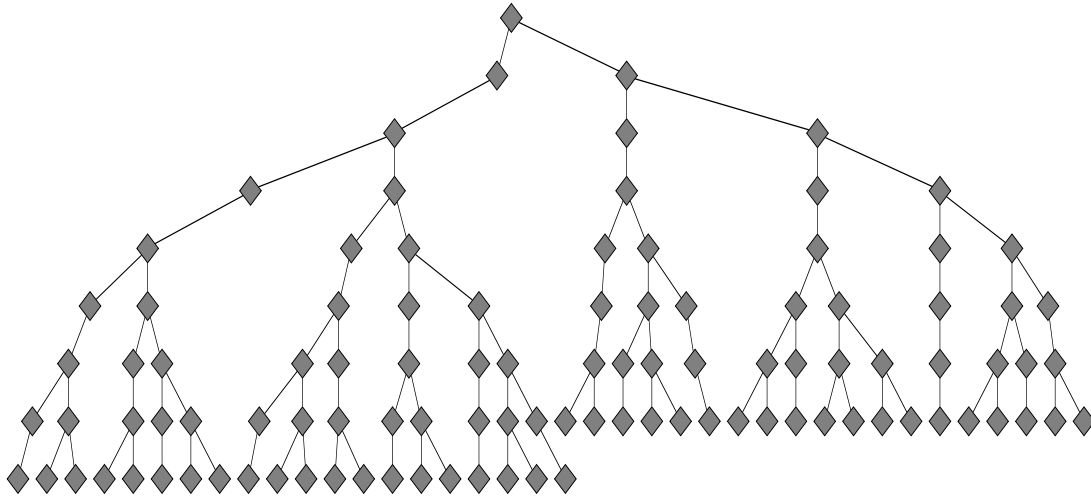
## ตัวอย่างต้นไม้ไอบีแอล



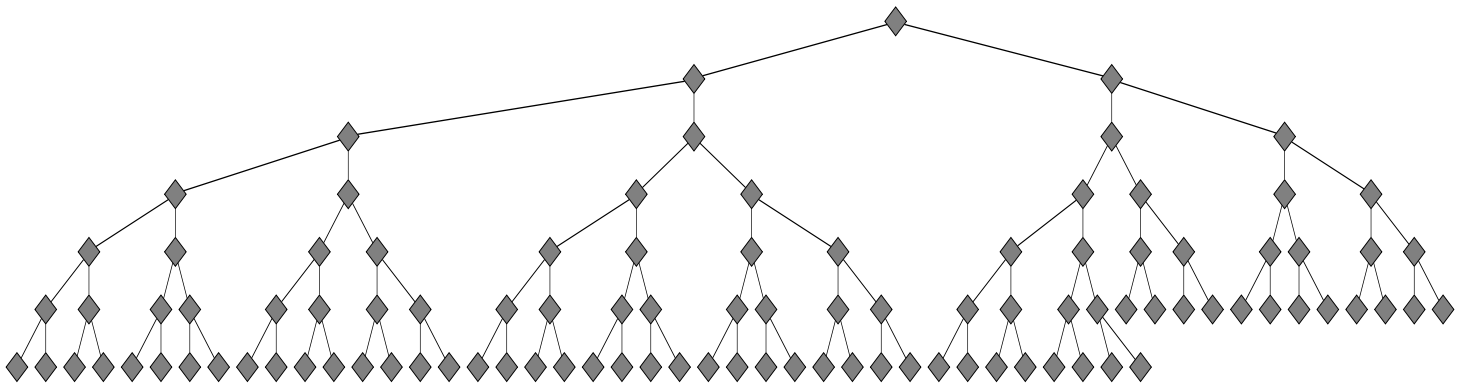
ต้นไม้ว่าง (null) สูง -1

## AVL Balance Rule

ต้นไม้ค้นหาแบบทวิภาคเอวีแอล



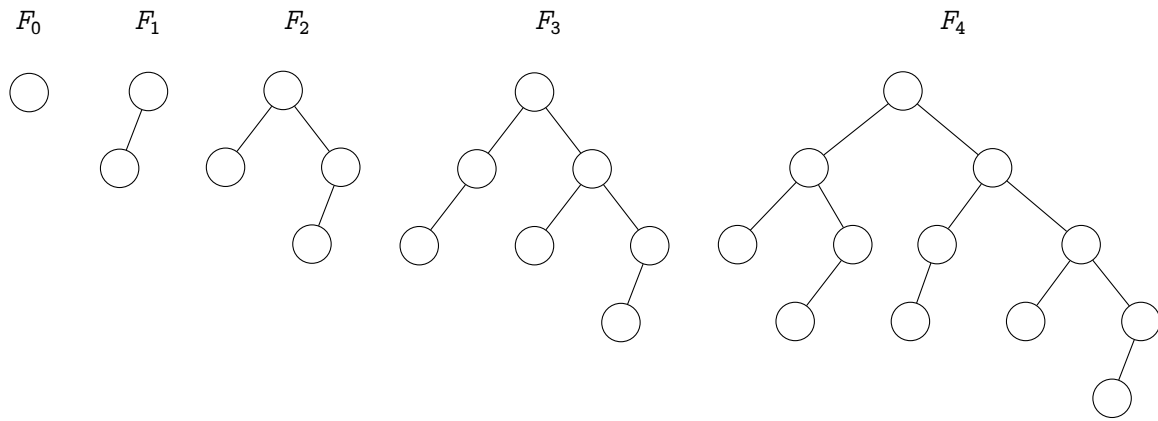
ต้นไม้ค้นหาแบบทวิภาคที่สร้างจากข้อมูลสุ่ม 100 ตัว



ต้นไม้เอวีแอลที่สร้างจากข้อมูลสุ่ม 100 ตัว

ต้นไม้เอวีแอลสูงเท่าใด ?

- ให้  $H_n$  คือต้นไม้เอวีแอลซึ่งสูง  $h$  ที่มีจำนวนปมน้อยที่สุด



$$|F_h| = 1 + |F_{h-1}| + |F_{h-2}|$$

Fibonacci Tree

ความสูงของต้นไม้ฟีโบนัชชี

$$|F_h| = 1 + |F_{h-1}| + |F_{h-2}|$$

$$n_h = 1 + n_{h-1} + n_{h-2} \quad h \geq 2, \quad n_0 = 1, \quad n_1 = 2$$

$$n_h = \alpha_1 \phi^h + \alpha_2 \hat{\phi}^h - 1, \quad \phi = 1.618..., \quad \hat{\phi} = -0.618$$

$$n_h \approx \alpha_1 \phi^h$$

$$h \approx \frac{1}{\log_2 \phi} (\log_2 n_h)$$

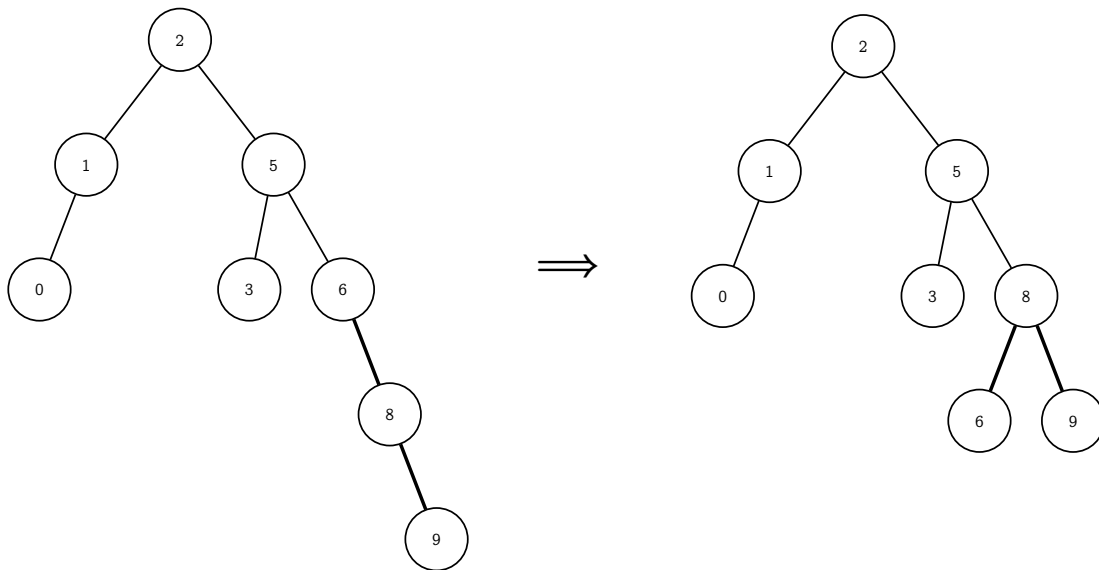
$$h \approx 1.44 (\log_2 n_h)$$

สรุป: ต้นไม้เวิแอลที่มี  $n$  ปม สูงไม่เกิน  $1.44 \log_2 n$

## AVL Rotation

ทำอย่างไรให้เป็นไปตามกฎของ AVL

- การเพิ่ม/ลบข้อมูลทำเหมือน BSTree
- แต่หลังการเพิ่ม/ลบ อาจทำให้ผิดกฎสูงสมดุล
- ถ้าผิดกฎต้องปรับต้นไม้



```

map_avl
template <typename KeyT,
          typename MappedT,
          typename CompareT = std::less<KeyT> >
class map_avl {
protected:
    class node {
        friend class map_bst;
        ...
    };

    class tree_iterator { // เหมือน map_bst
        ...
    };
public:
    ...
};

node
class node {
    friend class map_avl;
protected:
    ValueT data;
    node *left;
    node *right;
    node *parent;
    int height; // แต่ละบมมีความสูงกำกับ

    node() :
        data( ValueT() ), left( NULL ), right( NULL ),
        parent( NULL ), height(0) { }

    node(const ValueT& data, node* left,
         node* right, node* parent) : data(data),
         left(left), right(right), parent(parent) {
        set_height();
    }
};

```

```

}

int get_height(node *n) { // ไม้ ๐๐ ?
    return (n == NULL ? -1 : n->height);
}

void set_height() {
    int hL = get_height(this->left);
    int hR = get_height(this->right);
    height = 1 + (hL > hR ? hL : hR);
}

int balance_value() {
    return get_height(this->right) -
           get_height(this->left);
}

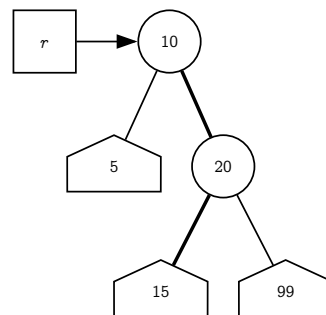
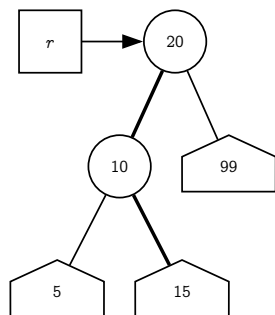
void set_left(node *n) {
    this->left = n;
    if (n != NULL) this->left->parent = this;
}

void set_right(node *n) {
    this->right = n;
    if (n != NULL) this->right->parent = this;
}
};

```

### การหมุนปม

- การปรับต้นไม้อาศัยการหมุน (rotation)



`rotate_left_child(r)`  
 $h_r = \max(2 + 5, 2 + 15, 1 + 99)$

`rotate_right_child(r)`  
 $h_r = \max(1 + 5, 2 + 15, 2 + 99)$

### CP::map\_avl

#### rotate\_left\_child(r)

```

node *rotate_left_child(node *r) {
    node *new_root = r->left;
    r->set_left(new_root->right);
    new_root->set_right(r);

    new_root->right->set_height();
    new_root->set_height();
}

```

```

    return new_root;
}

```

### rotate\_right\_child(r)

```

node *rotate_right_child(node *r) {
    node *new_root = r->right;
    r->set_right(new_root->left);
    new_root->set_left(r);

    new_root->left->set_height();
    new_root->set_height();
    return new_root;
}

```

### insert และ erase ใช้ rebalance

```

node* insert(const ValueT& val, node *r, node * &ptr) {
    ... // same as insert in map_bst

    r = rebalance(r); // เพิ่มตามปกติ แล้วค่อยปรับ
    return r;
}

```

```

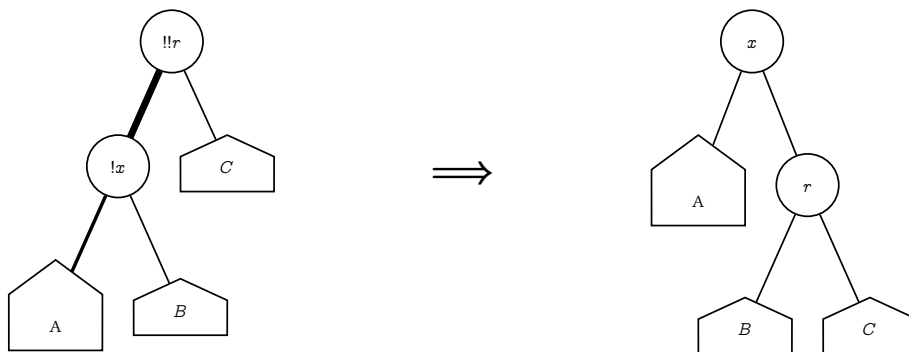
node *erase(const KeyT &key, node *r) {
    ... // same as erase in map_bst

    r = rebalance(r); // ลบตามปกติ แล้วค่อยปรับ
    return r;
}

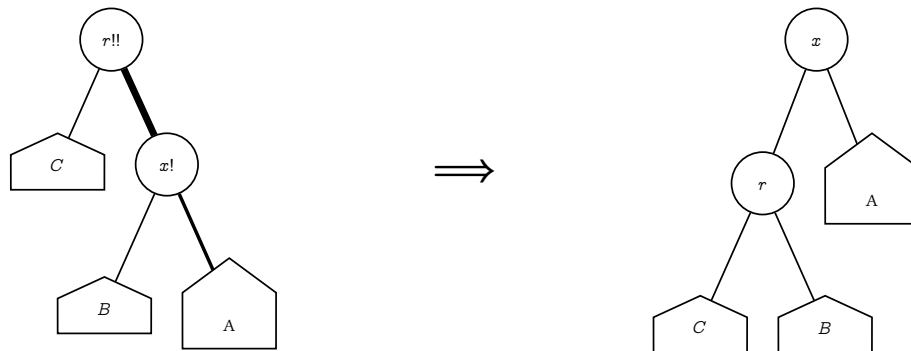
```

### rebalance

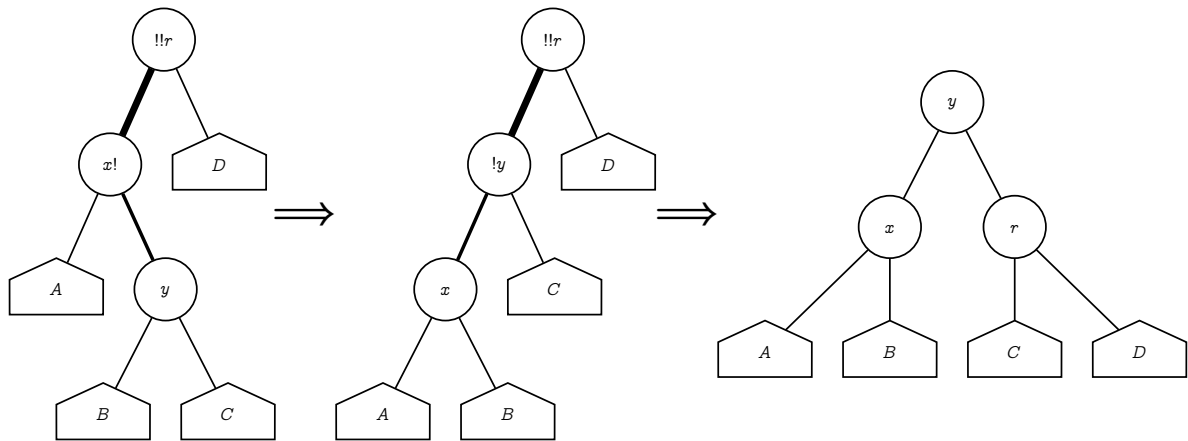
1. r เอียงซ้ายมากเกินไป และ x เอียงซ้าย  $\Rightarrow$  rotate\_left\_child(r);



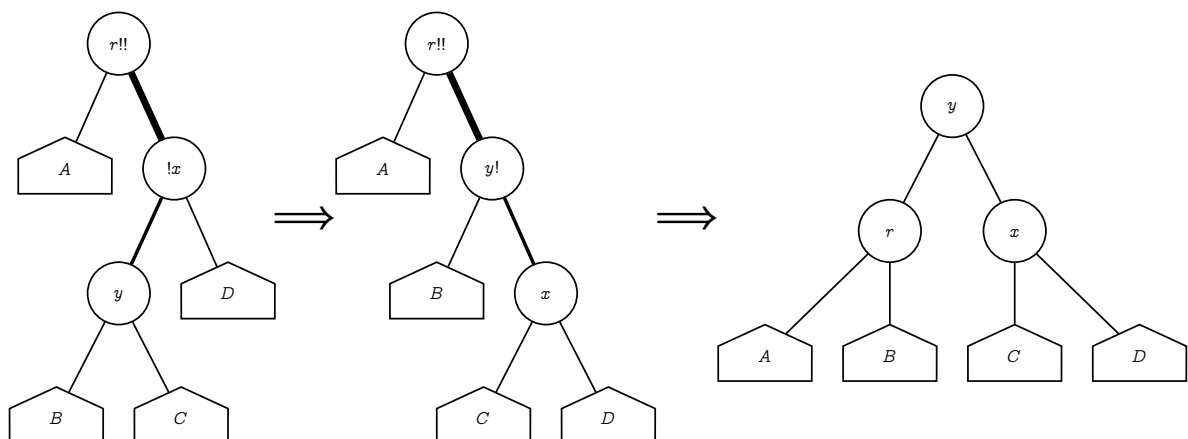
2. r เอียงขวาเกินไป และ x เอียงขวา  $\Rightarrow$  `rotate_right_child(r);`



3. r เอียงซ้ายมากเกินไป และ x เอียงขวา  $\Rightarrow$  `rotate_right_child(r->left); rotate_left_child(r);`



4. r เอียงขวาเกินไป และ x เอียงซ้าย  $\Rightarrow$  `rotate_left_child(r->right); rotate_right_child(r);`



```
node *rebalance(node *r) {
    if (r == NULL) return r;
    int balance = r->balance_value();
```

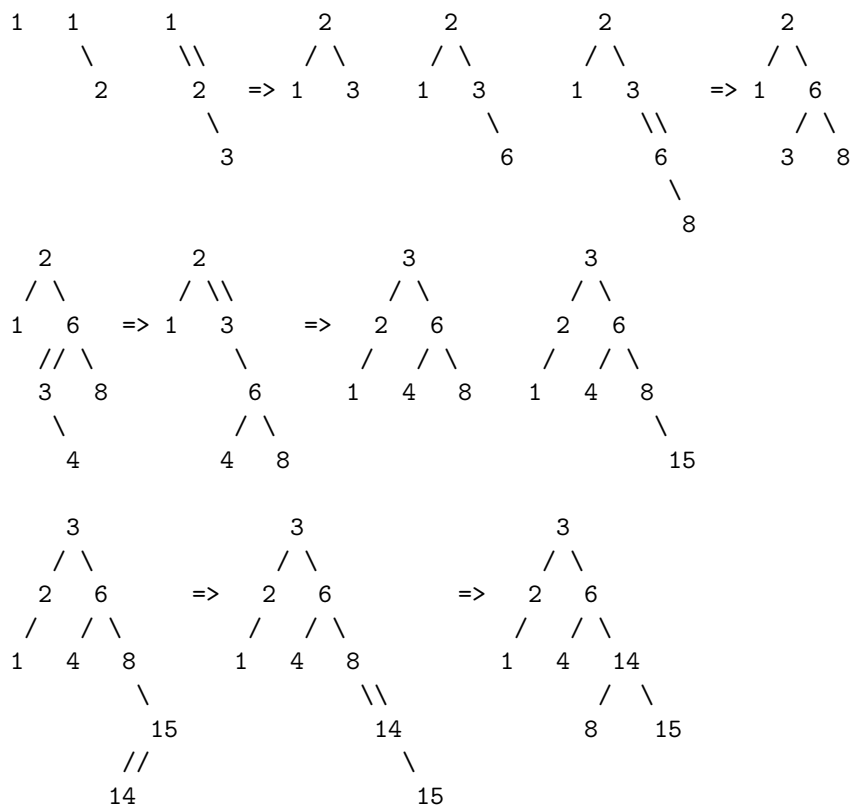
```

if (balance == -2) {
    if (r->left->balance_value() == 1) // case 3
        r->set_left(rotate_right_child(r->left));
    r = rotate_left_child(r);
} else if (balance == 2) {
    if (r->right->balance_value() == -1) // case 4
        r->set_right(rotate_left_child(r->right));
    r = rotate_right_child(r);
}
r->set_height();
return r;
}

```

ตัวอย่าง

1, 2, 3, 6, 8, 4, 15, 14



สรุป

- ต้นไม้เอวีแอลคือต้นไม้ค้นหาที่ถูกรักษาความสูง
- ผลต่างความสูงของลูกสองข้างห้ามเกินหนึ่ง
- พิสูจน์ได้ว่า  $\lfloor \log_2 n \rfloor \leq h < 1.44 \log_2 n$
- แต่ละปมเก็บความสูงไว้ตรวจสอบ
- ถ้าหลังเพิ่ม/ลดข้อมูลแล้วผิดกฎให้ปรับต้นไม้
- การปรับต้นไม้อาศัยการหมุนปม
- เวลาการทำงานของเพิ่ม ลง และค้นหาเป็น  $\mathcal{O}(\log n)$

ความสูงของต้นไม้ AVL

```

class node {
    friend class map_avl;
protected:
    ValueT data; // int, int 8

```

```

class node {
    friend class map_avl;
protected:
    ValueT data;

```

```

node *left;    // 4
node *right;   // 4
node *parent;  // 4
int height;    // 4
}

node *left;
node *right;
node *parent;
unsigned char height; // 1
}

```

### นอกเรื่อง

- std::map ใช้ Red Black Tree
- self-balanced tree มี AVL, Red Black, Splay

## Lecture 18

### Hash Table

#### หัวข้อ

- การใช้ตารางเก็บข้อมูลด้วยฟังก์ชันดัชนี
- การเก็บข้อมูลแบบแยกกันโยง
- ฟังก์ชันแฮช
- กลวิธีการเขียนฟังก์ชันแฮช
- การแฮชใน C++
- การกำหนดเลขที่อยู่เปิด
- การเกาะกลุ่มของข้อมูล

#### ใช้ฟังก์ชันดัชนีคำนวณตำแหน่ง

- key ของข้อมูลคือส่วนของข้อมูลที่ใช้ในการค้น
- มีตารางซึ่งแต่ละช่องเป็นที่เก็บข้อมูล
- หา  $f(\text{key})$  เพื่อแปลง key ไปเป็น index ของตาราง
- ฟังก์ชันดัชนีหาไม่ยาก ถ้าองตารางขนาดใหญ่

#### ฟังก์ชันดัชนีนั้นหายาก

- เมื่อต้องเก็บอย่างประหยัด
- เมื่อต้องประกันว่าไม่เกิดการ “ชน”

#### เปลี่ยนกลยุทธ์: อนุญาตให้ชนกันได้

- จะได้เก็บข้อมูลในตารางที่ไม่ใหญ่มาก
- แต่ต้องหาวิธีแก้ไขปัญหาคollision ที่ทำงานได้เร็ว ๆ

#### Separate Chaining

- จัดเก็บข้อมูลที่ชนกันไว้ในรายการเดียวกัน

```

0 [ ] --> < 14, 7 >
1 [ ] --> < 22, 1 >
2 [ ] --||
3 [ ] --> < 3, 38 >      h(x) = x % 7
4 [ ] --> < 46 >
5 [ ] --> < 40 >
6 [ ] --> < 6, 13, 27, 20 >

```

#### การกระจายข้อมูล

- ถ้าข้อมูลกระจายทั่วตาราง
  - แต่ละช่องเก็บรายการยาว  $\approx \lambda$
  - ถ้า  $\lambda$  น้อย ค้นหาได้เร็ว
- ถ้าไม่กระจาย

load factor

$$\lambda = n/m$$

- มีบางรายการยาวเกิน  $\lambda$  มาก
- การค้นหาซ้ำเหมือนเก็บด้วย list

```
0 [ ] --||
1 [ ] --> < 401, 321, 105, 185, 129 >
2 [ ] --||
3 [ ] --||
4 [ ] --||
5 [ ] --> < 180, 260, 60, 252, 92, 764,
6 [ ] --||           4, 76, 28, 108 >
7 [ ] --||
           h(x) = x % 8
```

$n$  = ปริมาณข้อมูล  
 $m$  = ขนาดของตาราง

```
           h(x) = x % 10
0 [ ] --> < 180, 260, 80 >
1 [ ] --> < 401, 321 >
2 [ ] --> < 252, 92 >
3 [ ] --||
4 [ ] --> < 764, 4 >
5 [ ] --> < 105, 185 >
6 [ ] --> < 76 >
7 [ ] --||
8 [ ] --> < 28, 108 >
9 [ ] --> < 129 >
```

## Hash Function

### ฟังก์ชันแฮช (Hash Function)

- [www.webster.com](http://www.webster.com)
  - **hash**: to chop (as meat and potatoes) into small pieces
- สอ เสถบุตร
  - สับ แหวก นำมาโขลกเข้าด้วยกัน

### ตัวอย่างฟังก์ชันแฮช

```
size_t h1(size_t x) {
    return (2654435769U * x) >> 22;
}
```

```
size_t h2(size_t x) {
    x = ~x + (x << 15);
    x ^= (x >> 11);
    x += (x << 3);
    x ^= (x >> 5);
    x += (x << 10);
    x ^= (x >> 16);
    return x & 0x3FF;
}
```

| x     | 1   | 2    | 3   | 4   | 5   | 6    | 7   | 8   |
|-------|-----|------|-----|-----|-----|------|-----|-----|
| h1(x) | 632 | 241  | 874 | 483 | 92  | 725  | 334 | 966 |
| h2(x) | 500 | 1001 | 507 | 978 | 486 | 1014 | 403 | 933 |

### การวิเคราะห์เลขโดด (Digit Analysis)

- คัดเลือกเลขโดดบางหลักของคีย์มาพิจารณา
- มั่นใจว่าที่ตัดไปไม่ทำให้เกิดความเอนเอียงในการกระจายของคีย์
- เช่น
  - รหัสนิติศึกษา ป.ตรี มีรูปแบบ: xx3xxxxx21
  - ก็ตัดเลข 3 และ 21 ออกจากการพิจารณา
  - $k = 4830109521$
  - $k1 = \lfloor k / 100 \rfloor$  //  $k1 = 48301095$
  - $k2 = \lfloor k1 / 100 \rfloor$  //  $k2 = 48$
  - $k3 = k2 * 10^5 + k1 \% 10^5$  //  $k3 = 4801095$

**การคูณ (Multiplicative Hashing)**

- คูณคีย์ด้วยจำนวนจริง  $A$  ที่มีค่าระหว่าง  $(0, 1)$
- นำเศษมาคูณกับขนาดของตาราง ( $m = 2^p$ )

$$h(x) = \lfloor m((xA - \lfloor xA \rfloor)) \rfloor$$

**การคูณ: Fibonacci Hashing**

- ถ้า  $A$  = golden ratio 0.6180339887 จะแยกคีย์ที่มีค่าใกล้เคียงกันออกจากกันได้ดี

```
size_t multHash(size_t x, size_t p) {
    size_t s = 2654435769U; // 0.6180339887*2^32
    size_t hash = (s * x);
    return (hash >> (32-p));
}
```

```
for (size_t i = 0; i < 10; i++) {
    cout << multHash(i, 10) << ", ";
}
```

0, 632, 241, 874, 483, 92, 725, 334, 966

**การพับ (Folding)**

- แบ่งคีย์ออกเป็นส่วน ๆ แล้วนำมา “รวม” กัน
- “รวม”  $\equiv$  บวก, xor, ...

**การหาร (Modulus Hashing)**

- $h(x) = x \% p$
- ไม่ควรเลือก
  - ▶  $p = 10^q$  เพราะเลือกเฉพาะ  $q$  หลักขวา ถ้าคีย์เป็นหลักสิบ
  - ▶  $p = 2^q$  เพราะเลือกเฉพาะ  $q$  บิตขวา
  - ▶  $p$  ที่มีค่าน้อย ๆ เป็นตัวประกอบ
    - ถ้า  $c$  คือตัวประกอบร่วมของ  $p$  และ  $x$
    - ค่า  $x\%p$  จะเป็นจำนวนเท่าของ  $c$
    - ถ้า  $c$  มีค่าน้อย ๆ จะมีคีย์จำนวนมากที่ได้  $x\%p$  มีค่าเป็นจำนวนเท่าของตัวประกอบนั้น ซึ่งไม่กระจาย
  - ▶ โดยทั่วไปเลือก  $p$  ที่เป็นจำนวนเฉพาะ

**c++11 std::unordered\_map**

```
#include <iostream>
#include <unordered_map>

using namespace std;

int main() {
    unordered_map<string, int> facultyCode;
    facultyCode["engineering"] = 21;
    facultyCode["accounting"] = 26;
    facultyCode["science"] = 23;

    cout << facultyCode["engineering"] << endl;
    cout << facultyCode["science"] << endl;
    cout << facultyCode["communication"] << endl;

    return 0;
}
```

**c++11 std::hash**

```
#include <functional>
using namespace std;
int main() {
    hash<string> hStr;
    hash<float> hFloat;
    hash<int> hInt;

    cout << hStr("C++") << endl; // 2262514926
    cout << hFloat(1.2f) << endl; // 2462087341
    cout << hInt(123) << endl; // 123

    return 0;
}

cout << hash<string>()("C++") << endl;
cout << hash<float>()(1.2f) << endl;
cout << hash<int>()(123) << endl;
```

**อยากใช้ Book เป็นคีย์**

```
class Book {
public:
    string title;
    int edition;
    double price;

    Book(string title, int ed = 1, double price = 199.0) :
        title(title), edition(ed), price(price)
    {}
    ...
};
```

**เขียน Hasher class, Equal class**

```
class BookHasher {
public:
    size_t operator()(const Book& b) const {
        return hash<string>()(b.title) ^ hash<int>()(b.edition);
    }
};

class BookEqual {
public:
    bool operator()(const Book& b1, const Book b2) const {
        return b1.title==b2.title && b1.edition==b2.edition;
    }
};

unordered_map<Book,string,BookHasher,BookEqual> m;
m[Book("Data Structures", 1, 200)] = "reserved";
m[Book("Algorithm", 5, 200)] = "available";

Book b1("Data Structures, 1");
Book b2("algorithm", 5);
```

```
cout << m[b1] << endl; // "reserved"
cout << m[b2] << endl; // ""
```

### ปฏิกิริยาวันเกิด (Birthday Paradox)

- ต้องมีคนในห้องกี่คนขึ้นไป จึงจะเกิดโอกาสเกินครึ่งที่จะมีคนเกิดวันเดียวกันสองคนขึ้นไป

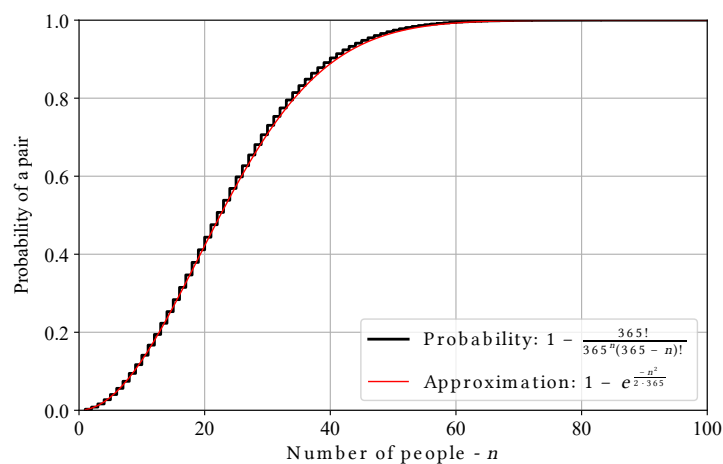
1 คน โอกาสที่มีวันเกิดไม่ซ้ำกัน =  $\left(\frac{366}{366}\right)$

2 คน โอกาสที่มีวันเกิดไม่ซ้ำกัน =  $\left(\frac{366}{366}\right)\left(\frac{365}{366}\right)$

3 คน โอกาสที่มีวันเกิดไม่ซ้ำกัน =  $\left(\frac{366}{366}\right)\left(\frac{365}{366}\right)\left(\frac{364}{366}\right)$

k คน โอกาสที่มีวันเกิดไม่ซ้ำกัน =  $\left(\frac{366}{366}\right)\left(\frac{365}{366}\right)\left(\frac{364}{366}\right)\dots\left(\frac{366-k+1}{366}\right)$

$$1 - \left(\frac{366}{366}\right)\left(\frac{365}{366}\right)\left(\frac{364}{366}\right)\dots\left(\frac{366-k+1}{366}\right) > 0.5$$



### CP::unordered\_map

```
CP::unordered_map<KeyT, MappedT>
bucket count = 10
[ ] -> [ * , * ] <-- bucket size = 2
[ ] -> [ * , * ]
[ ] -> [ * , * ]
[ ] -> [ * , (*) ] -> pair<key, mappedValue>
[ ] -> [ * , * ]
[ ] -> [ * , * ]
[ ] -> [ ] <-- bucket size = 0
[ ] -> [ ]
[ ] -> [ * , * , * ]
[ ] -> [ * , * , * ]
  ▲          ▲ vector< ValueT > = BucketT
  |
vector< vector< ValueT > > = vector< BucketT >

typedef pair<KeyT, MappedT> ValueT;
typedef vector< ValueT > BucketT;
```

### Constructors

```
template <typename KeyT,
          typename MappedT,
```

```

        typename HasherT = std::hash<KeyT>,
        typename EqualT = std::equal_to<KeyT> >
class unordered_map {
protected:
    typedef std::pair<KeyT, MappedT> ValueT;
    typedef std::vector<ValueT> BucketT;
    ...

    std::vector<BucketT> mBuckets;
    size_t mSize;
    HasherT mHasher; // ใช้ในการเปรียบเทียบ ระหว่างการค้น Key
    EqualT mEqual; // hash function เพื่อคำนวณตำแหน่ง bucket
    float mMaxLoadFactor;
    ...

    unordered_map() :
        mBuckets( std::vector<BucketT>(11) ), mSize(0),
        mHasher( HasherT() ), mEqual( EqualT() ),
        mMaxLoadFactor(1.0)
    { }

    unordered_map(const
        unordered_map<KeyT,MappedT,HasherT,EqualT> &other) :
        mBuckets(other.mBuckets), mSize(other.mSize),
        mHasher(other.mHasher), mEqual(other.mEqual),
        mMaxLoadFactor(other.mMaxLoadFactor)
    { }

    unordered_map<KeyT,MappedT,HasherT,EqualT>&
    operator=(unordered_map<KeyT,MappedT,HasherT,EqualT> other) {
        using std::swap;
        swap(this->mBuckets, other.mBuckets);
        swap(this->mSize, other.mSize);
        swap(this->mHasher, other.mHasher);
        swap(this->mEqual, other.mEqual);
        swap(this->mMaxLoadFactor, other.mMaxLoadFactor);
        return *this;
    }
}

```

### Functions

```

template < ... >
class unordered_map {
public:
    bool empty() { return mSize == 0; }
    size_t size() { return mSize; }
    size_t bucket_count() { return mBuckets.size(); }
    size_t bucket_size(size_t n) { return mBuckets[n].size(); }
    float load_factor() { return (float)mSize/mBuckets.size(); }
    float max_load_factor() { return mMaxLoadFactor; }
    void max_load_factor(float z) {...}

    iterator begin() {...}
    iterator end() {...}

```

```

MappedT& operator[](const KeyT& key) {...}
void      clear()          {...}
void      rehash(size_t n)  {...}
size_t    erase(const KeyT &key) {...}
pair<iterator,bool> insert(const ValueT& val) {...}
}

operator[]
size_t hash_to_bucket(const KeyT& key) {
    return mHasher(key) % mBuckets.size();
}

ValueIterator find_in_bucket(BucketT& b, const KeyT& key) {
    for (ValueIterator it = begin(); it != b.end(); it++) {
        if (mEqual(it->first, key)) return it;
    }
    return b.end();
}

ValueIterator insert_to_bucket(const ValueT& val, size_t& bi) {
    if ( ตารางแน่นเกินไป ) { ปรับตาราง }
    ++mSize;
    return mBuckets[bi].insert(mBuckets[bi].end(), val);
} // ผลการ insert ใน vector คือ iterator ไปยังข้อมูลที่ถูกเพิ่มใหม่

MappedT& operator[](const KeyT& key) {
    size_t      bi = hash_to_bucket(key);
    ValueIterator it = find_in_bucket(mBuckets[bi], key);

    // ถ้าหาไม่พบ ต้องเพิ่ม pair(key, ค่า default ของ mapped value)
    if (it == mBuckets[bi].end()) {
        it = insert_to_bucket(make_pair(key, MappedT()), bi);
    }

    return it->second;
}

erase
size_t erase(const KeyT & key) {
    size_t      bi = hash_to_bucket(key);
    ValueIterator it = find_in_bucket(mBuckets[bi], key);
    if (it == mBuckets[bi].end()) {
        return 0; // erase 0 element
    } else {
        mBuckets[bi].erase(it);
        mSize--;
        return 1; // erase 1 element
    }
}

```

ผลของการ erase คือ จำนวนข้อมูลที่ถูกลบ

- 0 คือ ไม่พบ ไม่มีการลบเกิดขึ้น
- 1 คือ พบ key มีการลบ pair ของ key และ mapped value ของ key นั้น

**insert**

```

pair<iterator, bool> insert(const ValueT& val) {
    pair<iterator, bool> result;
    const KeyT& key = val.first;
    size_t bi = hash_to_bucket(key);
    ValueIterator it = find_in_bucket(mBuckets[bi], key);
    result.second = false;
    if (it == mBuckets[bi].end()) {
        it = insert_to_bucket(val, bi); // bi อาจเปลี่ยน ถ้ามีการปรับตาราง
        result.second = true;
    }
    result.first = iterator(it, // iterator ของ unordered_map
                           mBuckets.begin() + bi,
                           mBuckets.end());
    return result;
}

```

val คือ pair(key, mappedValue)

**clear**

```

void clear() {
    for (vector<BucketT>::iterator it = mBuckets.begin();
         it != mBuckets.end();
         ++it) {
        (*it).clear();
    }
    mSize = 0;
}

void clear() {
    for (BucketT & bucket : mBuckets) { // for each bucket in mBuckets
        bucket.clear();
    }
    mSize = 0;
}

```

**destructor**

```

class unordered_map {
    ...
    ~unordered_map() {
        clear();
    }
    ...
};

```

**Rehashing**

ถ้าฟังก์ชันแฮชกระจายดี การลบและค้นใช้เวลา  $\mathcal{O}(1)$

ถ้าควบคุม  $\lambda$  ไม่ให้เกินค่าคงตัว  $k$  การลบและค้นใช้เวลาคงตัว

ถ้า “แน่น” เกิน ต้อง rehash

```

void rehash(size_t m) {
    if ( m <= mBuckets.size() &&
         load_factor() <= max_load_factor() ) return;
    m = std::max(m, (size_t)(0.5*mSize/mMaxLoadFactor));
    m = *std::lower_bound(PRIMES, PRIMES+N_PRIMES, m); 1
}

```

```

vector<ValueT> tmp;
for (auto& val : *this) tmp.push_back(val);
this->clear();
mBuckets.resize(m);
for (auto& val : tmp) this->insert(val);
}

ValueIterator insert_to_bucket(const ValueT& val, size_t& bi) {
    if (load_factor() > max_load_factor()) {
        rehash(2*bucket_count());
        bi = hash_to_bucket(val.first);
    }
    ++mSize;
    return mBuckets[bi].insert(mBuckets[bi].end(), val);
}

```

ขอปรับขนาดตารางเป็นจำนวนเฉพาะ

```

const size_t      N_PRIMES      = 256;
const unsigned long PRIMES[256] = {
    2ul, 3ul, 5ul, 7ul, 11ul, 13ul, 17ul, 19ul, 23ul, 29ul,
    ...
};

```

## Separate Chaining Iterator

iterator

```

class unordered_map {
protected:
    ...
    class hashtable_iterator {
        ...
    public:
        hashtable_iterator() { ... }
        hashtable_iterator& operator++() { ... } // ++it
        hashtable_iterator operator++(int) { ... } // it++
        ValueT & operator*() { ... } // *it
        ValueT * operator->() { ... } // it->first
        bool operator!=(const hashtable_iterator &other) { ... }
        bool operator==(const hashtable_iterator &other) { ... }
    };
public:
    typedef hashtable_iterator iterator;
    ...
};

```

++iterator

```

vector<ValueT>::iterator
                        ValueIterator
vector<BucketT>::iterator ▼
BucketIterator ▶ [ ] -> [ * , * , * ]
                  [ ] -> [ ]
                  [ ] -> [ * , * ]
                  [ ] -> [ ]
                  [ ] -> [ * ]

```

```

class hashtable_iterator{
protected:
    ValueIterator  mCurValueItr;  // ▼
    BucketIterator mCurBucketItr;  // ►
    BucketIterator mEndBucketsItr;  // เก็บ mBuckets.end() ตอนสร้าง iterator

    void to_next_data() {
        while(mCurBucketItr != mEndBucketsItr &&
              mCurValueItr == mCurBucketItr->end())
            mCurBucketItr++;
        if (mCurBucketItr == mEndBucketsItr) break;
        mCurValueItr = mCurBucketItr->begin();
    }

public:
    hashtable_iterator& operator++() {  // it++
        mCurValueItr++;
        to_next_data();
        return (*this);  // ++(++it) it เลื่อนสองครั้ง
    }
    hashtable_iterator operator++(int) {  // ++it
        hashtable_iterator tmp(*this);
        operator++;
        return tmp;  // (it++)++ it เลื่อนครั้งเดียว!
    }

    *it กับ it->
    typedef ValueT & reference;
    typedef ValueT * pointer;

    reference operator*() {
        return *mCurValueItr;  // (*it).first
    }

    pointer operator->() {
        return &(*mCurValueItr);  // it->first
    }

    it1 == it2 กับ it1 != it2
    bool operator==(const hashtable_iterator &other) {
        return mCurValueItr == other.mCurValueItr;
    }
    bool operator!=(const hashtable_iterator &other) {
        return mCurValueItr != other.mCurValueItr;
    }

    constructor
    hashtable_iterator(ValueIterator valueItr,
                      BucketIterator bucketItr,
                      BucketIterator endBucketItr) :
        mCurValueItr(valueItr);
        mCurBucketItr(bucketItr);
        mEndBucketItr(endBucketItr);
    {
        to_next_data();
    }

```

```

    }
};

begin() && end()
iterator begin() {
    return iterator( mBuckets.begin()->begin(),
                    mBuckets.begin(),
                    mBuckets.end() );
}

iterator end() {
    return iterator( mBuckets[mBuckets.size()-1].end(),
                    mBuckets.end(),
                    mBuckets.end() );
}

```

### อย่าลืม default constructor

```

class unordered_map {
    ...
    class hashtable_iterator {
        ...
        hashtable_iterator() { }
        ...
    };
    ...
    pair<iterator, bool> insert(const ValueT& val) {
        pair<iterator, bool> result;
        ...
        return result;
    }
    ...
};

```

## Open Addressing

### การแก้ปัญหาการชนแบบอื่น

- แบบแยกกันโยง (separate addressing)
  - ▶ แต่ละช่องในตารางเก็บรายการโยงของข้อมูล
  - ▶ ข้อมูลที่ชนกันเก็บอยู่ด้วยกัน ไม่กระทบข้อมูลอื่น
- แบบเลขที่อยู่เปิด (open addressing)
  - ▶ แต่ละช่องในตารางเก็บข้อมูล
  - ▶ ถ้าชน ก็หาช่องว่างใหม่ในตารางเพื่อเก็บข้อมูล
  - ▶  $\lambda = n/m \leq 1$  เสมอ ต้องคุมไม่ให้เกินเกณฑ์ ( $\lambda \leq 0.5$ )
  - ▶ มีหลายวิธีในการหาช่องว่างใหม่ในตาราง เมื่อเกิดการชน
    - การตรวจเชิงเส้น (linear probing)
    - การตรวจกำลังสอง (quadratic probing)
    - การตรวจสองชั้น (double hashing)

### การตรวจเชิงเส้น (Linear Probing)

- เมื่อชน หาช่องว่างถัดไปด้วยวิธีดูตัวถัดไปเรื่อย ๆ

ใช้  $h(x) = x \% 13$  แล้วเพิ่มข้อมูลที่มีคีย์ตามลำดับดังนี้

|    |    |    |   |   |    |    |    |    |  |  |  |  |
|----|----|----|---|---|----|----|----|----|--|--|--|--|
| 17 | 32 | 26 | 7 | 4 | 43 | 12 | 11 | 24 |  |  |  |  |
|----|----|----|---|---|----|----|----|----|--|--|--|--|

|   |   |   |   |   |   |   |   |   |   |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|----|----|----|

```

[26, , , , 17, , 32, 7, , , , ]
[26, , , , 17, 4, 32, 7, , , , ] // 4 ไม่ว่าง ลง 5 แทน
[26, , , , 17, 4, 32, 7, 43, , , , ] // 4 ไม่ว่าง ลง 8 แทน
[26, , , , 17, 4, 32, 7, 43, , , 11, 12]
[26, 24, , , , 17, 4, 32, 7, 43, , , 11, 12] // 11 ไม่ว่าง วนไปลง 1 แทน

```

- ให้  $h_j(x)$  คือช่องที่ probe หลังจากชนครั้งที่  $j$
- $h_0(x) = h(x)$  คือช่องที่ hash เริ่มต้น (home address)

$$h_j(x) = (h(x) + j) \% m$$

$$h_j(x) = (h_{j-1}(x) + 1) \% m$$

- ค้น - หาไม่พบเมื่อเจอช่องว่าง
- ลบ - ถ้าเจอช่องว่างที่เคยมีข้อมูล ให้ค้นต่อไป

### สถานะของช่องเก็บข้อมูล (bucket)

- แต่ละช่องมี 3 สถานะ
  - 0: empty : ช่องว่าง ๆ ไม่เคยมีข้อมูลมาก่อนเลย
  - 1: deleted: ช่องที่เก็บข้อมูลที่ถูกลบไปแล้ว
  - 2: data : ช่องที่มีข้อมูลเก็บอยู่

### Linear Probing Iterator

#### ข้อมูลที่เก็บในตาราง

```

template <...>
class unordered_map {
protected:
    typedef pair<KeyT, MappedT> ValueT;
    class BucketT {
    public:
        ValueT      value;
        unsigned char status;

        bool available() { return status < 2; }
        bool empty()     { return status == 0; }
        bool has_data()  { return status == 2; }
        void mark_deleted() { status = 1; }
        void mark_empty()  { status = 0; }
        void mark_data()   { status = 2; }
    };
    vector<BucketT> mBuckets;

```

#### การเปลี่ยนสถานะของ bucket

- constructor -> empty
- m.insert(val) -> mark\_data
- m["X"] = 2 -> mark\_data
- m.erase("X") -> mark\_deleted
- m.clear() -> mark\_empty
- m.rehash(...)
- clear -> mark\_empty
- insert -> mark\_data

#### มาดู iterator กันก่อน

```

class hashtable_iterator {
protected:
    BucketIterator mCurBucketItr;
    BucketIterator mEndBucketItr;

```

```

    void to_next_data() {
        while ( mCurBucketItr != mEndBucketItr &&
                !mCurBucketItr->has_data() ) {
            mCurBucketItr++;
        }
    }
public:
    hashtable_iterator(BucketIterator bucket,
                      BucketIterator endBucket) :
        mCurBucketItr(bucket), mEndBucketItr(endBucket) {
        to_next_data();
    }
};

public:
    iterator begin() {
        return iterator( mBuckets.begin(), mBuckets.end() );
    }
};

++it ãŸŸ it++
class hashtable_iterator {
protected:
    BucketIterator mCurBucketItr;
    BucketIterator mEndBucketItr;
    void to_next_data() {...}

public:
    ...
    hashtable_iterator& operator++() { // ++it
        mCurBucketItr++;
        to_next_data();
        return (*this);
    }
    hashtable_iterator operator++(int) { // it++
        hashtable_iterator tmp(*this);
        operator++();
        return tmp;
    }
}

*it ãŸŸ it->
    ValueT & operator*() {
        return mCurBucketItr->value;
    }

    ValueT * operator->() {
        return &(mCurBucketItr->value);
    }
};

```

## Linear Probing Hash Table

**unordered\_map (linear probing)**

```

template <...>
class unordered_map {
protected:

```

```

typedef pair<KeyT, MappedT> ValueT;
class BucketT {...};
class hashtable_iterator {...};

vector<BucketT> mBuckets;
size_t         mSize;
HasherT        mHasher;           // Will hash_to_bucket
EqualT         mEqual;            // Will find_position
float          mMaxLoadFactor;    // Will insert_to_position < 1
size_t         mUsed;             // # data + # deleted

size_t hash_to_bucket(const KeyT& key) {
    return mHasher(key) % mBuckets.size();
}

size_t find_position(const KeyT& key) {...}
BucketIterator insert_to_position(const ValueT& val, size_t& pos) {...}

```

### find\_position

```

vector<BucketT> mBuckets;
size_t find_position(const KeyT& key) {
    size_t homePos = hash_to_bucket(key);
    size_t pos     = homePos;
    while ( !mBuckets[pos].empty() &&
            !mEqual(mBuckets[pos].value.first, key) ) {
        pos = (pos + 1) % mBuckets.size();
    }
    return pos; // position of the target or the next empty bucket
}

```

### insert

```

BucketIterator insert_to_position(const ValueT& val, size_t& pos) {
    if (load_factor() > max_load_factor()) {
        rehash(2*bucket_count());
        pos = find_position(val.first);
    }
    mSize++;
    mBuckets[pos].value = val;
    if (mBuckets[pos].empty()) mUsed++;
    mBuckets[pos].mark_data();
    return mBuckets.begin()+pos;
}

pair<iterator, bool> insert(const ValueT& val) {
    size_t pos = find_position(val.first);
    if (mBuckets[pos].available()) {
        BucketIterator it = insert_to_position(val, pos);
        result.first = iterator(it, mBuckets.end());
        result.second = true;
    } else {
        result.first = iterator(mBuckets.begin()+pos, mBuckets.end());
        result.second = false;
    }
    return result;
}

```

**คำนวณ load factor**

```

float load_factor() {
    return (float)mUsed/mBuckets.size();
}

float max_load_factor() {
    return mMaxLoadFactor;
}

void max_load_factor(float z) {
    mMaxLoadFactor = z;
    rehash(bucket_count());
}

```

**operator[]**

```

MappedT& operator[](const KeyT& key) {
    size_t pos = find_position(key);
    if (mBuckets[pos].available()) { // ไม่มีข้อมูล
        insert_to_position(make_pair(key, MappedT()), pos);
    }
    return mBuckets[pos].value.second;
}

```

**erase**

```

size_t erase(const KeyT & key) {
    size_t pos = find_position(key);
    if (mBuckets[pos].has_data()) {
        mBuckets[pos].mark_deleted();
        mSize--;
        return 1;
    } else {
        return 0;
    }
}

```

**clear**

```

void clear() {
    for (auto& bucket : mBuckets) {
        bucket.mark_empty();
    }
    mSize = 0;
    mUsed = 0;
}

```

**rehash**

```

void rehash(size_t m) {
    if (load_factor() <= max_load_factor() &&
        m <= mBuckets.size()) return;
    m = max(m, (size_t)(0.5+mSize/mMaxLoadFactor));
    m = *lower_bound(PRIMES, PRIMES+N_PRIMES, m);
    vector<ValueT> tmp;
    for (auto& val : this) {
        tmp.push_back(val);
    }
    this->clear();
    mBuckets.resize(m);
    for (auto& val : tmp) {

```



คลาสเพื่อการคำนวณตำแหน่งถัดไป

```
class LinearProbing {
public:
    size_t operator()(size_t homePos,
                      size_t col_count,
                      size_t bucket_count) {
        return (homePos + col_count) % bucket_count;
    }
};

class QuadraticProbing {
public:
    size_t operator()(size_t homePos,
                      size_t col_count,
                      size_t bucket_count) {
        return (homePos + col_count*col_count) % bucket_count;
    }
};
```

**NextAddressT**

```
template <typename KeyT,
          typename MappedT,
          typename HasherT = std::hash<KeyT>,
          typename EqualT = std::equal_to<KeyT>,
          typename NextAddressT = LinearProbing >
class unordered_map {
    ...
    NextAddressT    mNextAddress;
```

**default & copy constructor**

```
unordered_map() :
    mBuckets(vector<BucketT>(11), mSize(0)),
    mHasher(HasherT()), mEqual(EqualT()),
    mMaxLoadFactor(0.5), mUsed(0),
    mNextAddress( NextAddressT() )
{ }

unordered_map(const
    unordered_map<KeyT,MappedT,HasherT,EqualT> &other) :
    mBuckets(other.mBuckets), mSize(other.mSize),
    mHasher(other.mHasher), mEqual(other.mEqual),
    mMaxLoadFactor(other.mMaxLoadFactor), mUsed(other.mUsed),
    mNextAddress( other.mNextAddress )
{ }
};
```

การตรวจกำลังสองไม่ตรวจทุกช่อง

- ลองเพิ่ม 30 อีกตัว ( $h(x) = x \% 13$ )

$$\begin{array}{ll}
 h(x) = 4 & (4 + 7^2) \% 13 = 1 \\
 (4 + 1^2) \% 13 = 5 & (4 + 8^2) \% 13 = 3 \\
 (4 + 2^2) \% 13 = 8 & (4 + 9^2) \% 13 = 7 \\
 (4 + 3^2) \% 13 = 0 & (4 + 10^2) \% 13 = 0 \\
 (4 + 4^2) \% 13 = 7 & (4 + 11^2) \% 13 = 8 \\
 (4 + 5^2) \% 13 = 3 & (4 + 12^2) \% 13 = 5 \\
 (4 + 6^2) \% 13 = 1 & (4 + 13^2) \% 13 = 4 \\
 & \dots
 \end{array}$$

- มีช่องว่างอาจหาไม่พบ

#### เมื่อตารางมีขนาดเป็นจำนวนเฉพาะ

- การตรวจกำลังสองจะดูอย่างน้อยครึ่งหนึ่งของตาราง
- ดังนั้น ถ้า **load factor**  $\leq 1/2$  ก็สบายใจว่า จะหาช่องว่างพบ เมื่อเพิ่มข้อมูล
- พิสูจน์ : ให้  $0 \leq i \leq j \leq \lfloor m/2 \rfloor$  ถ้าข้างบนไม่จริง ต้องมีการ probe ครั้งที่  $i$  และ  $j$  ที่ดูช่องซ้ำกัน

$$\begin{aligned}
 h(x) + j^2 &\equiv h(x) + i^2 \pmod{m} \\
 j^2 &\equiv i^2 \pmod{m} \\
 (j^2 - i^2) &\equiv 0 \pmod{m} \\
 (j - i)(j + i) &\equiv 0 \pmod{m}
 \end{aligned}$$

- เป็นไปไม่ได้:  $(j - i)$  ไม่เป็น 0,  $(j + i)$  ก็ไม่เป็น  $m$   
อีกทั้ง  $(j - i)(j + i) \% m \neq 0$  เพราะทั้งสองพจน์  $< m$  และ  $m$  เป็นจำนวนเฉพาะ

### Double Hashing

#### การเกาะกลุ่ม

- การเกาะกลุ่มทุติยภูมิ (secondary clustering)
  - ▶ ข้อมูลที่มี  $h(x)$  เดียวกัน จะตรวจช่องในตารางเหมือนกัน
  - ▶ ระยะกระโดดของการตรวจแปรตามหมายเลขครั้งที่ชน
  - ▶  $h_j(x) = (h(x) + j) \% m$ ,  $h_j(x) = (h(x) + j^2) \% m$
  - ▶ แก้ปัญหาได้ โดยให้ข้อมูลที่มี  $h(x)$  เดียวกัน ไม่จำเป็นต้องมีระยะโดดของการตรวจเหมือนกัน
  - ▶ ให้ระยะกระโดดคำนวณจากค่าของข้อมูล

#### การแฮชสองชั้น (Double Hashing)

- ใช้ฟังก์ชันแฮชอีกตัวเพื่อคำนวณระยะกระโดด
- ทำให้ชุดข้อมูลที่แฮชไปที่ช่องเดียวกัน อาจมีระยะกระโดดต่างกัน

$$h_j(x) = (h(x) + j \cdot g(x)) \% m \qquad h_j(x) = (h_{j-1}(x) + g(x)) \% m$$

- โดยที่  $g(x) \% m \neq 0$  (เพื่อไม่ให้ย่ำอยู่กับที่) เช่น
  - ▶  $g(x) = R - (x \% R)$ ,  $R$  เป็นจำนวนเฉพาะ และ  $R < m$
- และ ตัวหารร่วมมากของ  $g(x)$  และ  $m$  ต้องเป็น 1 จะได้ตรวจทุกช่องในตาราง
  - ▶ ประกันเงื่อนไขนี้ได้โดยให้  $m$  เป็นจำนวนเฉพาะ
  - ▶  $h(x) = 0, g(x) = 4, m = 8$  จะตรวจช่อง 0 และ 4 เท่านั้น
  - ▶  $h(x) = 0, g(x) = 4, m = 7$  จะตรวจช่อง 0, 4, 1, 5, 2, 6, 3

#### เปรียบเทียบจำนวนการตรวจเฉลี่ย

|  | Linear Probing |       | Quadratic Probing |       | Double Hashing |       |
|--|----------------|-------|-------------------|-------|----------------|-------|
|  | พบ             | ไม่พบ | พบ                | ไม่พบ | พบ             | ไม่พบ |

|                 |      |       |      |       |      |       |
|-----------------|------|-------|------|-------|------|-------|
| $\lambda = 0.3$ | 1.21 | 1.52  | 1.21 | 1.47  | 1.19 | 1.43  |
| $\lambda = 0.4$ | 1.33 | 1.89  | 1.31 | 1.75  | 1.28 | 1.67  |
| $\lambda = 0.5$ | 1.50 | 2.50  | 1.43 | 2.14  | 1.39 | 2.02  |
| $\lambda = 0.6$ | 1.75 | 3.63  | 1.59 | 2.72  | 1.53 | 2.54  |
| $\lambda = 0.7$ | 2.16 | 6.02  | 1.82 | 3.70  | 1.74 | 3.44  |
| $\lambda = 0.8$ | 3.00 | 12.84 | 2.16 | 5.64  | 2.05 | 5.32  |
| $\lambda = 0.9$ | 5.44 | 49.70 | 2.79 | 11.37 | 2.67 | 11.63 |

|                                             | จำนวนการตรวจเฉลี่ย                                   |                                                          |
|---------------------------------------------|------------------------------------------------------|----------------------------------------------------------|
|                                             | หาพบ                                                 | หาไม่พบ                                                  |
| แบบแยกกันโยง ( $\lambda \geq 0$ )           | $1 + \frac{\lambda}{2}$                              | $1 + \lambda$                                            |
| การตรวจเชิงเส้น ( $0 \leq \lambda \leq 1$ ) | $\frac{1}{2} \left( 1 + \frac{1}{1-\lambda} \right)$ | $\frac{1}{2} \left( 1 + \frac{1}{(1-\lambda)^2} \right)$ |
| การแฮชสองชั้น ( $0 \leq \lambda \leq 1$ )   | $\frac{1}{\lambda} \ln \frac{1}{1-\lambda}$          | $\frac{1}{1-\lambda}$                                    |

ถาม: เก็บข้อมูลโดยใช้การตรวจเชิงเส้น ถ้าต้องการตรวจโดยเฉลี่ยไม่เกิน 5 ครั้ง ต้องควบคุมให้ตารางแฮชมี  $\lambda$  เป็นเท่าใด

ตอบ:

$$\begin{aligned}
 5 &\geq \frac{1}{2} \left( 1 + \frac{1}{(1-\lambda)^2} \right) \\
 9 &\geq \frac{1}{(1-\lambda)^2} \\
 1-\lambda &\geq \sqrt{\frac{1}{9}} \\
 \lambda &\leq \frac{2}{3}
 \end{aligned}$$

### ข้อควรระวัง

- ไม่เหมาะกับบริการที่
  - ▶ แจกแจงข้อมูลด้วย iterator
  - ▶ เกี่ยวข้องกับอันดับของข้อมูล `getMin`, `getMax`, ...
  - ▶ ต้องค้นทั้งตาราง  $\Theta(m+n)$
- ต้องระวังเรื่องฟังก์ชันแฮช
  - ▶ ฟังก์ชันแฮชไม่ดี ก็ใช้งานได้ แต่ถ้า  $n$  มาก อาจช้าเป็น  $\mathcal{O}(n)$

```

class BookHasher {
public:
    size_t operator()(const Book& b) const {
        return hash<string>()(b.title) ^
               hash<int>()(b.edition);
    }
};

```

### สรุป

- การค้น เพิ่ม ลบข้อมูลในตารางแฮชทำได้รวดเร็ว
- สามารถปรับเวลาการทำงานให้เร็วขึ้นด้วยการใช้เนื้อที่เข้าแลก เพื่อให้ได้  $\lambda$  ที่เหมาะสม
- ฟังก์ชันแฮชมีผลต่อประสิทธิภาพการทำงาน